

PRODUCTS OF PRIME IDEALS IN RAY CLASS GROUPS

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ABSTRACT. We prove that every class in the narrow ray class group modulo an integral ideal $\mathfrak{q}$ of a fixed number field is represented by a product of three prime ideals of norm at most $(N\mathfrak{q})^{\max(1, 3\alpha, 4\alpha_0) + \kappa}$ for any $\kappa > 0$, where α is the exponent in short character sum bounds for general non-principal ray class characters and α_0 comes from a bounded-order subconvexity input for Hecke L -functions. Wu's subconvexity bound gives the admissible choice $\alpha = \alpha_0 = 103/256$, hence the explicit bound $(N\mathfrak{q})^{103/64 + \kappa}$. This improves the previous $O_K((N\mathfrak{q})^3)$ -scale bound of Deshouillers, Gun, Ramaré, and Sivaraman. We also prove that a positive proportion of ray classes are represented by products of two prime ideals. The proof extends the multiplicative dense-model and transference framework of Matomäki–Teräväinen to narrow ray class groups.

1. INTRODUCTION

A conjecture of Erdős states that, for every sufficiently large prime modulus q , every reduced residue class modulo q is represented by a product $p_1 p_2$ of two primes with $p_1, p_2 \leq q$; see [6, §2]. Matomäki and Teräväinen [16] proved a ternary version of this problem. They showed that, for cubefree q , every reduced residue class modulo q is represented by a product of three primes, each at most q , and that for general q the same is true with the bound $q^{1+\varepsilon}$. They also obtained lower bounds for the proportion of classes represented by products of two primes. Their main tool is a multiplicative dense-model theorem for primes, which transfers the problem to a product-set problem in a finite abelian group, up to certain coset obstructions. This approach is rooted in the dense-model transference method introduced by Green [8] and further developed by Green and Tao [9].

There are also function-field analogues of Erdős's conjecture, where geometric methods become available; see for instance the work of Sawin [22], and also [26]. This paper considers the number-field setting, where the approach is closer in spirit to the integer case. Our purpose is to extend the dense-model and transference framework of Matomäki and Teräväinen [16] to narrow ray class groups over number fields. Let K be a fixed number field with ring of integers $\mathcal{O}_K$. Let $\mathfrak{q} \subseteq \mathcal{O}_K$ be a non-zero integral ideal, put $Q := N\mathfrak{q}$, and let $G := \text{Cl}_{\mathfrak{q}}^{(\infty)}$ be the narrow ray class group modulo $\mathfrak{q}$, where all real places are included in the modulus. For $X \geq 2$, define

$$P_X(\mathfrak{q}) := \{[\mathfrak{p}] \in G : \mathfrak{p} \nmid \mathfrak{q}, N\mathfrak{p} \leq X\},$$

and

$$E_k(X; \mathfrak{q}) := \underbrace{P_X(\mathfrak{q}) \cdot \dots \cdot P_X(\mathfrak{q})}_{k \text{ times}} \subseteq G.$$

Thus $c \in E_k(X; \mathfrak{q})$ precisely when c is represented by a product of k prime ideal classes, with each prime ideal having norm at most X .

The bounds obtained below depend on two analytic inputs. Let

$$0 < \alpha_0 \leq \alpha < 1.$$

The first input is a short character-sum estimate for arbitrary non-principal ray class characters. For every $\varepsilon > 0$ there exists $\eta = \eta(K, \varepsilon) > 0$ such that for every finite order non-principal Hecke character χ with module of definition $\mathfrak{q}$,

$$(\text{CS}(\alpha)) \quad \sum_{N\mathfrak{a} \leq T} \chi(\mathfrak{a}) \ll_{K, \varepsilon} T^{1-\eta} \quad (T \geq Q^{\alpha+\varepsilon}).$$

2020 *Mathematics Subject Classification.* Primary 11R44; Secondary 11N36.

Key words and phrases. Ray class groups, products of prime ideals, dense model.

The second input is a bounded-order subconvexity estimate. For every fixed integer $\ell \geq 2$ and every $\varepsilon > 0$, every primitive non-principal Hecke character χ^* of order at most ℓ satisfies

$$(L^b(\alpha_0)) \quad L\left(\frac{1}{2} + it, \chi^*\right) \ll_{K, \ell, \varepsilon} C(\chi^*, t)^{\alpha_0/2 + \varepsilon} \quad (t \in \mathbb{R}),$$

where $C(\chi^*, t)$ denotes the analytic conductor. The first exponent α controls the general dense-model estimates, while α_0 controls only the bounded-order obstruction estimates.

In particular, $L^b(\alpha_0)$ implies the following short character-sum estimate for non-principal bounded order ray class characters: for every fixed integer $\ell \geq 2$ and every $\varepsilon > 0$, there exists $\eta = \eta(K, \ell, \varepsilon) > 0$ such that every non-principal Hecke character χ of order at most ℓ with module of definition $\mathfrak{q}$ satisfies

$$(CS^b(\alpha_0)) \quad \sum_{N\mathfrak{a} \leq T} \chi(\mathfrak{a}) \ll_{K, \ell, \varepsilon} T^{1-\eta} \quad (T \geq Q^{\alpha_0 + \varepsilon}).$$

The exponent α and α_0 control the general dense-model estimates. Our main result is the following ternary representation theorem for number fields.

Theorem 1.1. *Assume $CS(\alpha)$ and $L^b(\alpha_0)$. Let $\kappa > 0$ be fixed. For Q sufficiently large, if $X \geq Q^{\max(1, 3\alpha, 4\alpha_0) + \kappa}$, then $E_3(X; \mathfrak{q}) = G$.*

We also prove the following two-prime density result.

Theorem 1.2. *Assume $CS(\alpha)$ and $L^b(\alpha_0)$. Let $\kappa > 0$ be fixed, and let $\varepsilon > 0$ be sufficiently small. For Q sufficiently large in terms of ε , the following hold.*

- (i) *If $X \geq Q^{\max(1, 3\alpha, 4\alpha_0) + \kappa}$, then $|E_2(X; \mathfrak{q})| \geq (\frac{2}{3} - \varepsilon)|G|$.*
- (ii) *If $X \geq Q^{\max(1, 4\alpha, 4\alpha_0) + \kappa}$, then $|E_2(X; \mathfrak{q})| \geq (\frac{11}{16} - \varepsilon)|G|$.*

For a general fixed number field K , Wu's subconvexity theorem for Hecke L -functions [25], together with the admissible Ramanujan exponent $\theta = 7/64$ of Blomer–Brumley [3], gives

$$\alpha = \alpha_0 = \frac{103}{256};$$

see Proposition 2.4. Hence Theorem 1.1 gives

$$E_3(X; \mathfrak{q}) = G \quad \text{whenever} \quad X \geq Q^{103/64 + \kappa}, \quad 103/64 \approx 1.609.$$

This improves the exponent in the three-prime result of Deshouillers–Gun–Ramaré–Sivaraman [5], where the scale is $O_K(Q^3)$. They show that every narrow ray class modulo $\mathfrak{q}$ can be represented by a product of three degree-one unramified prime ideals of norm $\ll_K Q^3$. We state our theorem for arbitrary prime ideals. The restriction to unramified degree-one prime ideals may also be imposed by removing the negligible contribution of prime ideals of higher residue degree from the prime-counting arguments, since the number of prime ideals of residue degree > 1 and norm at most T is $O_K(T^{1/2})$.

The separation between α and α_0 is useful when stronger bounds are available for bounded-order characters. In particular, one may expect a bounded-order input $L^b(1/3)$ from the Weyl-type cube-free conductor subconvexity results of Balkanova, Frolenkov, and Wu [2]. If a Hecke character has order dividing a fixed integer ℓ , then its conductor is cube-free away from the finitely many primes lying above the rational primes dividing ℓ . Thus one may speculate that a fixed-local-ramification variant of their method yields $L^b(1/3)$. We do not pursue this refinement here. Assuming such a bounded-order input $L^b(1/3)$, while using the general character-sum exponent $\alpha = 103/256$ from Wu's subconvexity bound, Theorems 1.1 and 1.2(i) would hold at the scale $X \geq Q^{4/3 + \kappa}$. While Theorem 1.2(ii) depends on the term 4α , and therefore improving its scale would require an improvement in the general character-sum exponent α .

We now summarize the proof. The first step is to construct a dense model for the prime ideal classes in G . This uses the linear sieve over ideals, mean-value estimates in ray classes, and the character-sum input $CS(\alpha)$. The dense model reduces the representation problem to product-set estimates for dense subsets of the finite abelian group G .

The second step is finite-group theoretic. Kneser's theorem shows that either the relevant product sets are already large enough, or the dense model is concentrated in a small number of cosets of a subgroup $H \leq G$. The coset obstructions are then treated analytically. The obstructions

of indices 5 and 8 are ruled out by weighted prime-sum estimates for bounded-order characters, which is where $L^b(\alpha_0)$ enters. The index 2 obstruction is the exceptional quadratic case and requires a separate ideal-theoretic version of the argument of Matomäki–Teräväinen. The proof of the binary theorem uses the same dense model and Kneser analysis, together with a multiplicative energy estimate for products of prime ideals in ray classes.

The paper is organized as follows. Section 2 collects the linear sieve, ray-class mean-value estimates, and the subconvexity-to-character-sums input. Section 3 proves the dense model theorem over ray class groups and derives the main transference criteria. Section 4 contains the finite abelian group product-set arguments. Section 5 proves the prime-escape estimates for quotient indices 5 and 8. Section 6 proves the ternary Theorem 1.1, Section 7 treats the exceptional quadratic case, and Section 8 proves the binary density Theorem 1.2.

1.1. Notations. Let K be a number field of degree $n_K := [K : \mathbb{Q}]$, with ring of integers $\mathcal{O}_K$. Denote by $\zeta_K(s)$ the Dedekind zeta function of K , and write $\rho_K := \text{Res}_{s=1} \zeta_K(s)$ for its residue at $s = 1$. For a nonzero integral ideal $\mathfrak{q} \subset \mathcal{O}_K$, we write $N\mathfrak{q} = [\mathcal{O}_K : \mathfrak{q}]$ for its absolute norm.

Let $J(\mathfrak{q})$ denote the group of fractional ideals of K coprime to $\mathfrak{q}$, and let $P^+(\mathfrak{q})$ denote the subgroup of $J(\mathfrak{q})$ consisting of principal fractional ideals (α) , $\alpha \in K^\times$, $\alpha \equiv 1 \pmod{\mathfrak{q}}$, and $\sigma(\alpha) > 0$ for every real embedding $\sigma : K \hookrightarrow \mathbb{R}$.

We write

$$G := \text{Cl}_{\mathfrak{q}}^{(\infty)} = J(\mathfrak{q})/P^+(\mathfrak{q})$$

for the narrow ray class group modulo $\mathfrak{q}$.

Throughout this paper, by a Hecke character modulo $\mathfrak{q}$ we mean a finite-order Hecke character for which $\mathfrak{q}$ is a module of definition in the terminology of Neukirch [19, Def. VII.6.11]. Equivalently, this is a character of the narrow ray class group G ; or a Dirichlet character modulo $\mathfrak{q}$ in the sense of [19, Def. VII.6.8].

We denote by $\widehat{G}$ the character group of G . If $\chi \in \widehat{G}$, we also regard χ as a function on integral ideals by setting

$$\chi(\mathfrak{a}) = \begin{cases} \chi([\mathfrak{a}]), & \text{if } (\mathfrak{a}, \mathfrak{q}) = 1, \\ 0, & \text{otherwise,} \end{cases}$$

where $[\mathfrak{a}]$ denotes the class of $\mathfrak{a}$ in G . We shall use the usual orthogonality relations for the finite abelian group G . In particular, if $(\mathfrak{a}\mathfrak{b}, \mathfrak{q}) = 1$, then

$$\frac{1}{|G|} \sum_{\chi \in \widehat{G}} \chi(\mathfrak{a}) \overline{\chi(\mathfrak{b})} = \mathbf{1}_{[\mathfrak{a}] = [\mathfrak{b}]}.$$

2. PRELIMINARY RESULTS

2.1. Linear sieve in number fields. We record a form of the fundamental lemma of the linear sieve for ideals. It follows from Coleman’s extension of the Rosser–Iwaniec sieve to number fields [4], together with the classical construction of the linear sieve weights [7, Ch. 3].

Fix once and for all a total order $\leq_K$ on the non-zero integral ideals of K , such that

$$N\mathfrak{a} < N\mathfrak{b} \implies \mathfrak{a} <_K \mathfrak{b}.$$

For $z \geq 2$, put

$$\mathcal{P}(z) := \prod_{N\mathfrak{p} < z} \mathfrak{p}.$$

Lemma 2.1. *Let $z \geq 2$ and $D = z^s$ with $s \geq 1$. There exist coefficients $\lambda_{\mathfrak{d}}^{\pm} = \lambda_{\mathfrak{d}}^{\pm}(D, z)$, supported on squarefree ideals $\mathfrak{d} \mid \mathcal{P}(z)$ with $N\mathfrak{d} \leq D$, such that $|\lambda_{\mathfrak{d}}^{\pm}| \leq 1$ and the following hold.*

For every integral ideal $\mathfrak{n}$,

$$\sum_{\mathfrak{d} \mid \mathfrak{n}} \lambda_{\mathfrak{d}}^{-} \leq \mathbf{1}_{(\mathfrak{n}, \mathcal{P}(z))=1} \leq \sum_{\mathfrak{d} \mid \mathfrak{n}} \lambda_{\mathfrak{d}}^{+}.$$

Let h be a multiplicative function on ideals satisfying $0 \leq h(\mathfrak{p}) < 1$ for every prime ideal $\mathfrak{p}$. Assume that h has sieve dimension 1, in the following sense: there exists a constant $C_0 \geq 1$ such

that, for all ideals $\mathfrak{h}, \mathfrak{j}$ with $\mathfrak{h} <_K \mathfrak{j}$ and $N\mathfrak{h} \geq 2$,

$$\prod_{\mathfrak{h} \leq_K \mathfrak{p} <_K \mathfrak{j}} (1 - h(\mathfrak{p}))^{-1} \leq \frac{\log N\mathfrak{j}}{\log N\mathfrak{h}} \left(1 + \frac{C_0}{\log N\mathfrak{h}}\right).$$

If $1 \leq s \leq 3$, then

$$\sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_{\mathfrak{d}}^+ h(\mathfrak{d}) \leq (F_0(s) + o_K(1)) \prod_{N\mathfrak{p} < z} (1 - h(\mathfrak{p})),$$

where

$$F_0(s) = \frac{2e^\gamma}{s}.$$

If $2 \leq s \leq 4$, then

$$\sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_{\mathfrak{d}}^- h(\mathfrak{d}) \geq (f_0(s) + o_K(1)) \prod_{N\mathfrak{p} < z} (1 - h(\mathfrak{p})),$$

where

$$f_0(s) = \frac{2e^\gamma \log(s-1)}{s}.$$

Here the $o_K(1)$ -terms are as $z \rightarrow \infty$.

2.2. Character sums from subconvexity. We record a standard consequence of conductor-aspect subconvexity for Hecke L -functions. This will provide a concrete admissible value of the character-sum exponent α in $\text{CS}(\alpha)$.

We use the following subconvexity theorem of Wu.

Theorem 2.2 ([25], Thm. 1.1). *Let χ be a Hecke character of K , with analytic conductor $C(\chi)$. Then*

$$\left| L\left(\frac{1}{2}, \chi\right) \right| \ll_{K,\varepsilon} C(\chi)^{\frac{1}{4} - \frac{1-2\theta}{16} + \varepsilon},$$

where θ is any admissible exponent toward the Ramanujan–Petersson conjecture for GL_2 over K .

Remark 2.3. The best currently known value is $\theta = 7/64$, due to Blomer–Brumley [3, Thm. 1].

Proposition 2.4. *Let K be a fixed number field, and let $\mathfrak{q} \subseteq \mathcal{O}_K$ be a non-zero integral ideal. Let χ be a non-principal Hecke character modulo $\mathfrak{q}$. Then, for every $\varepsilon > 0$, there exists $\delta = \delta(K, \varepsilon) > 0$ such that*

$$\sum_{N\mathfrak{a} \leq X} \chi(\mathfrak{a}) \ll_{K,\varepsilon} X^{1-\delta}$$

whenever

$$X \geq (N\mathfrak{q})^{\frac{1}{2} - \frac{1-2\theta}{8} + \varepsilon}.$$

Remark 2.5. In particular, $\text{CS}(\alpha)$ holds with $\alpha = \frac{1}{2} - \frac{1-2\theta}{8}$. Using $\theta = 7/64$, this gives $\alpha = \frac{103}{256}$. Since $\text{CS}(\alpha)$ applies to all non-principal Hecke characters, we may also take $\alpha_0 = \alpha = \frac{103}{256}$ in $\text{CS}^b(\alpha_0)$.

Proof. Put

$$\sigma := \frac{1-2\theta}{16}, \quad \rho := \frac{1}{2} - 2\sigma = \frac{1}{2} - \frac{1-2\theta}{8}.$$

Let χ^* be the primitive character inducing χ , and let $\mathfrak{f} \mid \mathfrak{q}$ be the conductor of χ^* . Since χ is extended by zero on ideals not coprime to $\mathfrak{q}$, its Dirichlet series is

$$D(s, \chi) := \sum_{\mathfrak{a}} \frac{\chi(\mathfrak{a})}{(N\mathfrak{a})^s} = L(s, \chi^*) \prod_{\substack{\mathfrak{p}|\mathfrak{q} \\ \mathfrak{p} \nmid \mathfrak{f}}} \left(1 - \frac{\chi^*(\mathfrak{p})}{(N\mathfrak{p})^s}\right).$$

The finite Euler product satisfies, on $\Re s = 1/2$,

$$\left| \prod_{\substack{\mathfrak{p}|\mathfrak{q} \\ \mathfrak{p} \nmid \mathfrak{f}}} \left(1 - \frac{\chi^*(\mathfrak{p})}{(N\mathfrak{p})^s}\right) \right| \leq \prod_{\mathfrak{p}|\mathfrak{q}} \left(1 + \frac{1}{(N\mathfrak{p})^{1/2}}\right) \ll_{K,\varepsilon} Q^\varepsilon.$$

Hence, after replacing ε by a smaller value if necessary, Theorem 2.2 applied to the twist $\chi_t^* := \chi^*|\cdot|_{\mathbb{A}}^{it}$ gives

$$(2.1) \quad D\left(\frac{1}{2} + it, \chi\right) \ll_{K,\varepsilon} Q^{1/4-\sigma+\varepsilon} (1+|t|)^{B_K},$$

for some constant $B_K > 0$ depending only on K . Here we have used the standard conductor estimate

$$C(\chi_t^*) \ll_K N\mathfrak{f}(1+|t|)^{B_K} \leq Q(1+|t|)^{B_K}.$$

Let $0 < \eta < 1/10$, and choose $W_\eta \in C_c^\infty([0, \infty))$ such that

$$0 \leq W_\eta \leq 1, \quad W_\eta(y) = 1 \quad (0 \leq y \leq 1), \quad W_\eta(y) = 0 \quad (y \geq 1 + \eta),$$

and

$$\left| \left(y \frac{d}{dy}\right)^j W_\eta(y) \right| \ll_j \eta^{-j}.$$

Let

$$\widehat{W}_\eta(s) := \int_0^\infty W_\eta(y) y^{s-1} dy.$$

Repeated integration by parts gives, for every $A \geq 1$,

$$(2.2) \quad \widehat{W}_\eta(\sigma + iT) \ll_{A,c} \eta^{-A} (1+T)^{-A} \quad \left(\frac{1}{2} \leq \sigma \leq c\right),$$

Define the smoothed sum

$$S_\eta(X) := \sum_{\mathfrak{a}} \chi(\mathfrak{a}) W_\eta\left(\frac{N\mathfrak{a}}{X}\right).$$

Since $W_\eta = 1$ on $[0, 1]$ and is supported in $[0, 1 + \eta]$, the ideal-counting estimate [14, Thm. VI.3.3]

$$(2.3) \quad \#\{\mathfrak{a} : N\mathfrak{a} \leq T\} = \rho_K T + O_K(T^{1-1/n_K}),$$

gives

$$(2.4) \quad \sum_{N\mathfrak{a} \leq X} \chi(\mathfrak{a}) = S_\eta(X) + O_K(\eta X + X^{1-1/n_K}),$$

where $\rho_K = \text{Res}_{s=1} \zeta_K(s)$ and $n_K = [K : \mathbb{Q}]$.

By Mellin inversion,

$$S_\eta(X) = \frac{1}{2\pi i} \int_{(c)} D(s, \chi) \widehat{W}_\eta(s) X^s ds \quad (c > 1).$$

Since χ^* is non-principal, $L(s, \chi^*)$ hence $D(s, \chi)$ is entire [12, Sect. 6] and of finite order by the functional equation.

Moreover, by the Phragmen–Lindelöf principle for Hecke L -functions, $D(s, \chi)$ has polynomial growth in every fixed vertical strip; see (5.4) in the proof of Lemma 5.1. Thus, for fixed $c > 1$, there is $C_K > 0$ such that

$$D(\sigma + it, \chi) \ll_{K,c,\varepsilon} Q^{O(1)} (1+|t|)^{C_K} \quad \left(\frac{1}{2} \leq \sigma \leq c\right).$$

For $T \geq 1$, integrate over the rectangle with vertical sides $\Re s = c$ and $\Re s = 1/2$, and horizontal sides $\Im s = \pm T$. No pole is crossed. On the horizontal sides, using (2.2), we get

$$\int_{1/2}^c D(\sigma + iT, \chi) \widehat{W}_\eta(\sigma + iT) X^{\sigma+iT} d\sigma \ll_{K,c,A,\varepsilon} Q^{O(1)} X^c \eta^{-A} (1+T)^{C_K-A}.$$

Choosing $A > C_K + 1$, this tends to 0 as $T \rightarrow \infty$, and the same argument applies to the lower horizontal side. Hence the contour may be shifted to $\Re s = 1/2$, using (2.1) and (2.2), and choosing $A > \max\{C_K + 1, B_K + 2\}$, we obtain

$$\begin{aligned} S_\eta(X) &= \frac{1}{2\pi} \int_{-\infty}^\infty D\left(\frac{1}{2} + it, \chi\right) \widehat{W}_\eta\left(\frac{1}{2} + it\right) X^{1/2+it} dt \\ &\ll_{K,\varepsilon,A} X^{1/2} Q^{1/4-\sigma+\varepsilon} \eta^{-A} \int_{-\infty}^\infty (1+|t|)^{B_K-A} dt \\ &\ll_{K,\varepsilon,A} X^{1/2} Q^{1/4-\sigma+\varepsilon} \eta^{-A}. \end{aligned}$$

Therefore,

$$(2.5) \quad \sum_{N\mathfrak{a} \leq X} \chi(\mathfrak{a}) \ll_{K, \varepsilon, A} X^{1/2} Q^{1/4 - \sigma + \varepsilon} \eta^{-A} + \eta X + X^{1-1/n_K}.$$

Now suppose $X \geq Q^{\rho + \varepsilon_0}$ for some fixed $\varepsilon_0 > 0$. Choose $\varepsilon > 0$ so small that $\varepsilon < \varepsilon_0/2$. Since $\frac{1}{4} - \sigma = \frac{\rho}{2}$, we have

$$X^{1/2} Q^{1/4 - \sigma + \varepsilon} \leq X^{1/2 + \frac{\rho/2 + \varepsilon}{\rho + \varepsilon_0}} = X^{1-c_0}$$

for some $c_0 = c_0(\varepsilon_0) > 0$.

Choose $\eta = X^{-\lambda}$ with $0 < \lambda < c_0/(2A)$. Then

$$X^{1/2} Q^{1/4 - \sigma + \varepsilon} \eta^{-A} \ll X^{1-c_0/2}, \quad \eta X = X^{1-\lambda}.$$

Substituting into (2.5), we get

$$\sum_{N\mathfrak{a} \leq X} \chi(\mathfrak{a}) \ll_{K, \varepsilon_0} X^{1-c_0/2} + X^{1-\lambda} + X^{1-1/n_K}.$$

Thus

$$\sum_{N\mathfrak{a} \leq X} \chi(\mathfrak{a}) \ll_{K, \varepsilon_0} X^{1-\delta},$$

where $\delta := \min\left\{\frac{c_0}{2}, \lambda, \frac{1}{n_K}\right\} > 0$. This proves the proposition. $\square$

2.3. Mean values of character sums.

Lemma 2.6. *Let K be a number field, let $\mathfrak{q} \subseteq \mathcal{O}_K$ be a non-zero integral ideal, and put $G = \text{Cl}_{\mathfrak{q}}^{(\infty)}$. Let $X \geq 2$, and let $a_{\mathfrak{a}}$ be complex coefficients supported on integral ideals $\mathfrak{a}$ satisfying $(\mathfrak{a}, \mathfrak{q}) = 1$ and $N\mathfrak{a} \leq X$.*

Define

$$M_K(X; \mathfrak{q}) := \max_{c \in G} \#\{\mathfrak{a} \subseteq \mathcal{O}_K : (\mathfrak{a}, \mathfrak{q}) = 1, N\mathfrak{a} \leq X, [\mathfrak{a}] = c\}.$$

Then

$$(2.6) \quad \sum_{\chi \in \widehat{G}} \left| \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathfrak{q}) = 1}} a_{\mathfrak{a}} \chi(\mathfrak{a}) \right|^2 \leq |G| M_K(X; \mathfrak{q}) \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathfrak{q}) = 1}} |a_{\mathfrak{a}}|^2.$$

Proof. Squaring out and using the orthogonality of the characters of G , we obtain

$$\sum_{\chi \in \widehat{G}} \left| \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathfrak{q}) = 1}} a_{\mathfrak{a}} \chi(\mathfrak{a}) \right|^2 = \sum_{\chi \in \widehat{G}} \sum_{\substack{N\mathfrak{a}, N\mathfrak{b} \leq X \\ (\mathfrak{a}, \mathfrak{q}) = 1 \\ (\mathfrak{b}, \mathfrak{q}) = 1}} a_{\mathfrak{a}} \overline{a_{\mathfrak{b}}} \chi(\mathfrak{a}) \overline{\chi(\mathfrak{b})} = |G| \sum_{\substack{N\mathfrak{a}, N\mathfrak{b} \leq X \\ (\mathfrak{a}, \mathfrak{q}) = 1, [\mathfrak{a}] = [\mathfrak{b}]}} a_{\mathfrak{a}} \overline{a_{\mathfrak{b}}}.$$

Using $|a_{\mathfrak{a}} a_{\mathfrak{b}}| \leq \frac{|a_{\mathfrak{a}}|^2 + |a_{\mathfrak{b}}|^2}{2}$, we get

$$\begin{aligned} \sum_{\chi \in \widehat{G}} \left| \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathfrak{q}) = 1}} a_{\mathfrak{a}} \chi(\mathfrak{a}) \right|^2 &\leq |G| \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathfrak{q}) = 1}} |a_{\mathfrak{a}}|^2 \#\{\mathfrak{b} : (\mathfrak{b}, \mathfrak{q}) = 1, N\mathfrak{b} \leq X, [\mathfrak{b}] = [\mathfrak{a}]\} \\ &\leq |G| M_K(X; \mathfrak{q}) \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathfrak{q}) = 1}} |a_{\mathfrak{a}}|^2. \end{aligned}$$

This proves the lemma. $\square$

Lemma 2.7 (Mean value theorem for ray class characters). *Let K be a number field, let $\mathfrak{q} \subseteq \mathcal{O}_K$ be a non-zero integral ideal, and put $G = \text{Cl}_{\mathfrak{q}}^{(\infty)}$. Let $X \geq 2$, and let $a_{\mathfrak{a}}$ be complex coefficients supported on integral ideals $\mathfrak{a}$ satisfying $(\mathfrak{a}, \mathfrak{q}) = 1$ and $N\mathfrak{a} \leq X$. Then, we have*

$$\sum_{\chi \in \widehat{G}} \left| \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathfrak{q}) = 1}} a_{\mathfrak{a}} \chi(\mathfrak{a}) \right|^2 \ll_K (X + Q)(\log(3Q))^{n_K} \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathfrak{q}) = 1}} |a_{\mathfrak{a}}|^2.$$

In particular, for every $\varepsilon > 0$,

$$\sum_{\chi \in \widehat{G}} \left| \sum_{\substack{Na \leq X \\ (a, q) = 1}} a_a \chi(a) \right|^2 \ll_{K, \varepsilon} Q^\varepsilon (X + Q) \sum_{\substack{Na \leq X \\ (a, q) = 1}} |a_a|^2.$$

Proof. By Lemma 2.6, it is enough to bound $|G| M_K(X; \mathfrak{q})$.

By [10, Thm. 1], we have the following uniform bound for $M_K(X; \mathfrak{q})$:

$$M_K(X; \mathfrak{q}) \ll_K F(\mathfrak{q}) (\log(3F(\mathfrak{q})))^{n_K} \left(1 + \frac{X}{Q}\right),$$

where

$$F(\mathfrak{q}) = \frac{2^{r_1} \varphi_K(\mathfrak{q}) h_K}{h_{K, \mathfrak{q}}}, \quad h_{K, \mathfrak{q}} = |G|,$$

$n_K = r_1 + 2r_2 = [K : \mathbb{Q}]$, and h_K is the class number of K .

Since $h_{K, \mathfrak{q}} = |G|$, we have $|G| F(\mathfrak{q}) = 2^{r_1} h_K \varphi_K(\mathfrak{q}) \ll_K Q$. In particular, $F(\mathfrak{q}) \ll_K Q$, hence $\log(3F(\mathfrak{q})) \ll_K \log(3Q)$. It follows that

$$|G| M_K(X; \mathfrak{q}) \ll_K (X + Q) (\log(3Q))^{n_K} \ll_{K, \varepsilon} Q^\varepsilon (X + Q).$$

Substituting this into Lemma 2.6 gives the stated bounds. $\square$

Lemma 2.8. *Let $Q = N\mathfrak{q}$. For $T \geq 1$,*

$$\#\{\mathfrak{a} : Na \leq T, (a, \mathfrak{q}) = 1\} = \rho_K \frac{\varphi_K(\mathfrak{q})}{N\mathfrak{q}} T + O_{K, \lambda} \left(Q^\lambda T^{1-1/n_K} \right)$$

for every fixed $\lambda > 0$. In particular, uniformly for $T \geq Q^\sigma$, with $\sigma > 0$ fixed,

$$\#\{\mathfrak{a} : Na \leq T, (a, \mathfrak{q}) = 1\} = \rho_K \frac{\varphi_K(\mathfrak{q})}{N\mathfrak{q}} T (1 + o_{K, \sigma}(1)).$$

Proof. By Möbius inversion and the ideal-counting asymptotic (2.3) from [14, Thm. VI.3.3], we have

$$\begin{aligned} \#\{\mathfrak{m} : N\mathfrak{m} \leq T, (\mathfrak{m}, \mathfrak{q}) = 1\} &= \rho_K \frac{\varphi_K(\mathfrak{q})}{N\mathfrak{q}} T + O_K \left(T^{1-\frac{1}{[K:\mathbb{Q}]}} \sum_{\mathfrak{e}|\mathfrak{q}} \frac{1}{(N\mathfrak{e})^{1-\frac{1}{[K:\mathbb{Q}]}}} \right) \\ &= \rho_K \frac{\varphi_K(\mathfrak{q})}{N\mathfrak{q}} T + O_K \left(\tau_K(\mathfrak{q}) T^{1-\frac{1}{[K:\mathbb{Q}]}} \right), \end{aligned}$$

where $\rho_K = \text{Res}_{s=1} \zeta_K(s)$, and $\tau_K(\mathfrak{q})$ denotes the number of ideal divisors of $\mathfrak{q}$.

Grouping prime ideals above each rational prime p , we have $\tau_K(\mathfrak{q}) \leq \tau(Q)^{[K:\mathbb{Q}]}$, where $Q = N\mathfrak{q}$ and τ is the usual divisor function for integers. Hence, for every fixed $\lambda > 0$,

$$\tau_K(\mathfrak{q}) \ll_{K, \lambda} Q^\lambda.$$

Moreover,

$$\left(\frac{\varphi_K(\mathfrak{q})}{N\mathfrak{q}} \right)^{-1} = \prod_{\mathfrak{p}|\mathfrak{q}} \left(1 - \frac{1}{N\mathfrak{p}} \right)^{-1} \ll_{K, \lambda} Q^\lambda.$$

For every fixed $\sigma > 0$, choosing $\lambda > 0$ sufficiently small in terms of K and σ , the error term is $o_K(1)$ times the main term. Therefore,

$$\#\{\mathfrak{m} : N\mathfrak{m} \leq T, (\mathfrak{m}, \mathfrak{q}) = 1\} = \rho_K \frac{\varphi_K(\mathfrak{q})}{N\mathfrak{q}} T + O_{K, \lambda} \left(Q^\lambda T^{1-\frac{1}{[K:\mathbb{Q}]}} \right) = \rho_K \frac{\varphi_K(\mathfrak{q})}{N\mathfrak{q}} T (1 + o_{K, \sigma}(1)),$$

uniformly for $T \geq Q^\sigma$. $\square$

Let $0 < \alpha < 1$. We use the following fixed-parameter version of $\text{CS}(\alpha)$. This formulation records the dependence on ε and η , since the estimates below will depend explicitly on these parameters.

Let $\varepsilon > 0$ and $\eta > 0$ be fixed. We say that $\text{CS}(\alpha, \varepsilon, \eta)$ holds if, for every non-principal character ψ modulo $\mathfrak{q}$, whenever $T \geq (N\mathfrak{q})^{\alpha+\varepsilon}$, we have

$$(\text{CS}(\alpha, \varepsilon, \eta)) \quad \sum_{Na \leq T} \psi(a) \ll_{K, \varepsilon} T^{1-\eta}.$$

Lemma 2.9 (Halász–Montgomery-type estimate). *Let $0 < \alpha < 1$, $\varepsilon > 0$, $\eta > 0$, and $C \geq 1$ be fixed. Assume $(\mathbf{CS}(\alpha, \varepsilon, \eta))$. Let $\chi_1, \dots, \chi_R$ be distinct characters of $G = \text{Cl}_{\mathfrak{q}}^{(\infty)}$, and put $Q = N\mathfrak{q}$.*

(i) *Let $X \geq Q^{\alpha+2\varepsilon}$. Then, for any complex coefficients $a_{\mathfrak{a}}$,*

$$\sum_{j=1}^R \left| \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathcal{P}(Q^\varepsilon))=1}} a_{\mathfrak{a}} \chi_j(\mathfrak{a}) \right|^2 \ll_{K, \varepsilon, \eta} \left(\frac{X}{\log Q} + R X^{1-\eta} Q^{\varepsilon\eta} \right) \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathcal{P}(Q^\varepsilon))=1}} |a_{\mathfrak{a}}|^2.$$

(ii) *Let $Q^C \geq X_2 \geq X_1 \geq Q^{\alpha+2\varepsilon}$. Then, for any complex coefficients $a_{\mathfrak{a}}$,*

$$\sum_{j=1}^R \left| \sum_{\substack{X_1 < N\mathfrak{a} \leq X_2 \\ (\mathfrak{a}, \mathcal{P}(Q^\varepsilon))=1}} \frac{a_{\mathfrak{a}}}{N\mathfrak{a}} \chi_j(\mathfrak{a}) \right|^2 \ll_{K, C, \varepsilon, \eta} \left(1 + R \left(\frac{Q^\varepsilon}{X_1} \right)^\eta \right) \sum_{\substack{X_1 < N\mathfrak{a} \leq X_2 \\ (\mathfrak{a}, \mathcal{P}(Q^\varepsilon))=1}} \frac{|a_{\mathfrak{a}}|^2}{N\mathfrak{a}}.$$

Proof. We first prove (ii), since the proof of (i) is similar.

Proof of (ii). By the duality principle [13, Sect. 7.1, p. 170], it suffices to show that for any complex numbers $c_1, \dots, c_R$,

$$(2.7) \quad \sum_{\substack{X_1 < N\mathfrak{a} \leq X_2 \\ (\mathfrak{a}, \mathcal{P}(Q^\varepsilon))=1}} \frac{1}{N\mathfrak{a}} \left| \sum_{j=1}^R c_j \chi_j(\mathfrak{a}) \right|^2 \ll \left(1 + R \left(\frac{Q^\varepsilon}{X_1} \right)^\eta \right) \sum_{j=1}^R |c_j|^2.$$

Let $\lambda_{\mathfrak{d}}^+$ be the upper-bound linear sieve weights as in Lemma 2.1 with sifting range $z = Q^\varepsilon$ and level $D = z$, so that $1_{(\mathfrak{a}, \mathcal{P}(Q^\varepsilon))=1} \leq \sum_{\mathfrak{d}|\mathfrak{a}} \lambda_{\mathfrak{d}}^+$. Hence the left-hand side of (2.7) is at most

$$(2.8) \quad \sum_{X_1 < N\mathfrak{a} \leq X_2} \frac{1}{N\mathfrak{a}} \sum_{\mathfrak{d}|\mathfrak{a}} \lambda_{\mathfrak{d}}^+ \left| \sum_{j=1}^R c_j \chi_j(\mathfrak{a}) \right|^2 = \sum_{j,k=1}^R c_j \overline{c_k} \sum_{N\mathfrak{d} \leq D} \frac{\lambda_{\mathfrak{d}}^+}{N\mathfrak{d}} \chi_j(\mathfrak{d}) \overline{\chi_k(\mathfrak{d})} \sum_{X_1/N\mathfrak{d} < N\mathfrak{m} \leq X_2/N\mathfrak{d}} \frac{\chi_j(\mathfrak{m}) \overline{\chi_k(\mathfrak{m})}}{N\mathfrak{m}}.$$

We separate the diagonal and off-diagonal terms.

Diagonal terms. If $j = k$, then $\chi_j(\mathfrak{m}) \overline{\chi_k(\mathfrak{m})} = 1$ for ideals $\mathfrak{m}$ coprime to $\mathfrak{q}$. Hence the diagonal contribution is

$$\sum_{j=1}^R |c_j|^2 \sum_{\substack{N\mathfrak{d} \leq D \\ (\mathfrak{d}, \mathfrak{q})=1}} \frac{\lambda_{\mathfrak{d}}^+}{N\mathfrak{d}} \sum_{\substack{X_1/N\mathfrak{d} < N\mathfrak{m} \leq X_2/N\mathfrak{d} \\ (\mathfrak{m}, \mathfrak{q})=1}} \frac{1}{N\mathfrak{m}}.$$

For $N\mathfrak{d} \leq D = Q^\varepsilon$, we have $X_1/N\mathfrak{d} \geq Q^{\alpha+\varepsilon}$. By Lemma 2.8 and partial summation,

$$\sum_{\substack{X_1/N\mathfrak{d} < N\mathfrak{m} \leq X_2/N\mathfrak{d} \\ (\mathfrak{m}, \mathfrak{q})=1}} \frac{1}{N\mathfrak{m}} = \frac{\varphi_K(\mathfrak{q})}{N\mathfrak{q}} \rho_K \log \frac{X_2}{X_1} + O_{K, C, \varepsilon}(Q^{-\delta_0})$$

for some $\delta_0 > 0$. Therefore the diagonal contribution is

$$\sum_{j=1}^R |c_j|^2 \left(\frac{\varphi_K(\mathfrak{q})}{N\mathfrak{q}} \log Q \sum_{\substack{N\mathfrak{d} \leq D \\ (\mathfrak{d}, \mathfrak{q})=1}} \frac{\lambda_{\mathfrak{d}}^+}{N\mathfrak{d}} + o_{K, C, \varepsilon}(1) \right).$$

Applying the upper-bound linear sieve with

$$h(\mathfrak{p}) = \begin{cases} 1/N\mathfrak{p}, & \mathfrak{p} \nmid \mathfrak{q}, \\ 0, & \mathfrak{p} \mid \mathfrak{q}, \end{cases}$$

we get

$$\sum_{\substack{N\mathfrak{d} \leq D \\ (\mathfrak{d}, \mathfrak{q})=1}} \frac{\lambda_{\mathfrak{d}}^+}{N\mathfrak{d}} \ll_{K, \varepsilon} \frac{N\mathfrak{q}}{\varphi_K(\mathfrak{q}) \log Q}.$$

Thus the diagonal contribution is

$$\ll_{K, C, \varepsilon} \sum_{j=1}^R |c_j|^2.$$

Off-diagonal terms. Now suppose $j \neq k$. Then $\psi_{j,k} := \chi_j \overline{\chi_k}$ is a non-principal character of G . Since $N\mathfrak{d} \leq D = Q^\varepsilon$ and $X_1 \geq Q^{\alpha+2\varepsilon}$, we have $\frac{X_1}{N\mathfrak{d}} \geq Q^{\alpha+\varepsilon}$. Thus $(\mathbf{CS}(\alpha, \varepsilon, \eta))$ applies to $\psi_{j,k}$.

By partial summation and $\text{CS}(\alpha, \varepsilon, \eta)$,

$$\sum_{X_1/N\mathfrak{d} < N\mathfrak{m} \leq X_2/N\mathfrak{d}} \frac{\psi_{j,k}(\mathfrak{m})}{N\mathfrak{m}} \ll_{K,\varepsilon,\eta} \left(\frac{X_1}{N\mathfrak{d}}\right)^{-\eta}.$$

Therefore, using $\sum_{N\mathfrak{d} \leq D} (N\mathfrak{d})^{-1+\eta} \ll_{K,\eta} D^\eta$, the off-diagonal contribution in (2.8) is

$$\ll_{K,\varepsilon,\eta} X_1^{-\eta} \sum_{\substack{1 \leq j,k \leq R \\ j \neq k}} |c_j c_k| \sum_{N\mathfrak{d} \leq D} (N\mathfrak{d})^{-1+\eta} \ll_{K,\eta} \left(\frac{D}{X_1}\right)^\eta \sum_{\substack{1 \leq j,k \leq R \\ j \neq k}} |c_j c_k|.$$

Since $\sum_{j \neq k} |c_j c_k| \leq (R-1) \sum_{j=1}^R |c_j|^2$, the off-diagonal contribution is

$$\ll_{K,\varepsilon,\eta} R \left(\frac{Q^\varepsilon}{X_1}\right)^\eta \sum_{j=1}^R |c_j|^2.$$

Combining the diagonal and off-diagonal bounds proves (2.7), and hence part (ii).

Proof of (i). By duality, it suffices to prove that for any complex coefficients $c_1, \dots, c_R$,

$$(2.9) \quad \sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathcal{P}(Q^\varepsilon))=1}} \left| \sum_{j=1}^R c_j \chi_j(\mathfrak{a}) \right|^2 \ll \left(\frac{X}{\log Q} + R X^{1-\eta} Q^{\varepsilon\eta} \right) \sum_{j=1}^R |c_j|^2.$$

Using the upper-bound sieve weights with sifting range Q^ε and level $D = Q^\varepsilon$, the left-hand side of (2.9) is at most

$$\sum_{j,k=1}^R c_j \overline{c_k} \sum_{N\mathfrak{d} \leq D} \lambda_{\mathfrak{d}}^+ \chi_j(\mathfrak{d}) \overline{\chi_k(\mathfrak{d})} \sum_{N\mathfrak{m} \leq X/N\mathfrak{d}} \chi_j(\mathfrak{m}) \overline{\chi_k(\mathfrak{m})}.$$

As in the diagonal estimate in part (ii), but without the factor $1/N\mathfrak{a}$, the upper-bound linear sieve together with the coprime ideal-counting asymptotic gives the diagonal contribution

$$\ll_{K,\varepsilon} \frac{X}{\log Q} \sum_{j=1}^R |c_j|^2.$$

For the off-diagonal terms $j \neq k$, the character $\chi_j \overline{\chi_k}$ is non-principal. Since $\frac{X}{N\mathfrak{d}} \geq \frac{X}{Q^\varepsilon} \geq Q^{\alpha+\varepsilon}$, the hypothesis $(\text{CS}(\alpha, \varepsilon, \eta))$ yields

$$\sum_{N\mathfrak{m} \leq X/N\mathfrak{d}} \chi_j(\mathfrak{m}) \overline{\chi_k(\mathfrak{m})} \ll_{K,\varepsilon,\eta} \left(\frac{X}{N\mathfrak{d}}\right)^{1-\eta}.$$

Thus the off-diagonal contribution is

$$\ll_{K,\varepsilon,\eta} \sum_{j \neq k} |c_j c_k| \sum_{N\mathfrak{d} \leq D} \left(\frac{X}{N\mathfrak{d}}\right)^{1-\eta} \ll_{K,\eta} X^{1-\eta} D^\eta \sum_{j \neq k} |c_j c_k| \ll R X^{1-\eta} Q^{\varepsilon\eta} \sum_{j=1}^R |c_j|^2.$$

Combining the diagonal and off-diagonal contributions proves (2.9), and hence part (i). $\square$

3. DENSE MODEL IN THE RAY CLASS GROUP

In this section we adapt the multiplicative dense model argument of Matomäki–Teräväinen to the narrow ray class group $G = \text{Cl}_{\mathfrak{q}}^{(\infty)}$. Throughout this section we keep the notation $Q = N\mathfrak{q}$ and $G = \text{Cl}_{\mathfrak{q}}^{(\infty)}$. We begin by proving a sieve estimate for ideals whose classes lie in a given coset.

Lemma 3.1. *Let $H \leq G$ be a subgroup of fixed index Y , and let $b \in G$. Let $X \geq Q^{3\alpha_0+\varepsilon}$, and define $\vartheta_0 := 1 - \varepsilon - \alpha_0 \frac{\log Q}{\log X}$. Let $0 < \gamma < 1$ and $s := \frac{\vartheta_0}{\gamma}$, define*

$$\mathcal{N}_\gamma(X; b, H) := \#\left\{ \mathfrak{n} : \begin{array}{l} N\mathfrak{n} \leq X, [\mathfrak{n}] \in bH, \\ N\mathfrak{p} > X^\gamma \text{ for every prime ideal } \mathfrak{p} \mid \mathfrak{n} \end{array} \right\}.$$

Then, uniformly in b and in subgroups $H \leq G$ of fixed index Y , the following hold.

If $1 \leq s \leq 3$, then

$$\mathcal{N}_\gamma(X; b, H) \leq (1 + o_K(1)) \frac{2}{Y\vartheta_0} \frac{X}{\log X}.$$

If $2 < s \leq 4$, then

$$\mathcal{N}_\gamma(X; b, H) \geq (1 + o_K(1)) \frac{2 \log(s-1)}{Y \vartheta_0} \frac{X}{\log X}.$$

Proof. Let

$$z := X^\gamma, \quad D := X^{\vartheta_0}, \quad s := \frac{\vartheta_0}{\gamma}, \quad \mathcal{P}(z) := \prod_{N\mathfrak{p} < z} \mathfrak{p}.$$

Let $\lambda_\mathfrak{d}^\pm$ be the upper and lower linear sieve weights as in Lemma 2.1, with sifting range z and level $D = X^{\vartheta_0} = z^s$. Then

$$\sum_{\mathfrak{d} | (\mathfrak{n}, \mathcal{P}(z))} \lambda_\mathfrak{d}^- \leq 1_{(\mathfrak{n}, \mathcal{P}(z))=1} \leq \sum_{\mathfrak{d} | (\mathfrak{n}, \mathcal{P}(z))} \lambda_\mathfrak{d}^+.$$

Hence

$$(3.1) \quad \mathcal{N}_\gamma(X; b, H) \geq \sum_{\mathfrak{d} | \mathcal{P}(z)} \lambda_\mathfrak{d}^- \sum_{\substack{N\mathfrak{m} \leq X/N\mathfrak{d} \\ [\mathfrak{d}\mathfrak{m}] \in bH}} 1,$$

and the corresponding upper bound holds with $\lambda_\mathfrak{d}^+$.

We now estimate the inner sum. Let $\widehat{G/H}$ denote the characters of G that are trivial on H . By orthogonality,

$$1_{[\mathfrak{d}\mathfrak{m}] \in bH} = \frac{1}{Y} \sum_{\chi \in \widehat{G/H}} \chi(\mathfrak{d}) \chi(\mathfrak{m}) \overline{\chi(b)}.$$

If $(\mathfrak{d}, \mathfrak{q}) = 1$,

$$\sum_{\substack{N\mathfrak{m} \leq X/N\mathfrak{d} \\ [\mathfrak{d}\mathfrak{m}] \in bH}} 1 = \frac{1}{Y} \sum_{\chi \in \widehat{G/H}} \chi(\mathfrak{d}) \overline{\chi(b)} \sum_{N\mathfrak{m} \leq X/N\mathfrak{d}} \chi(\mathfrak{m}).$$

If $(\mathfrak{d}, \mathfrak{q}) \neq 1$, the sum is zero.

Since $N\mathfrak{d} \leq D = X^{\vartheta_0}$, we have $\frac{X}{N\mathfrak{d}} \geq \frac{X}{D} = X^{1-\vartheta_0} = X^\varepsilon Q^{\alpha_0} \geq Q^{\alpha_0+3\alpha_0\varepsilon+\varepsilon^2}$. By Lemma 2.8, the principal character contributes

$$\frac{1}{Y} \#\{\mathfrak{m} : N\mathfrak{m} \leq X/N\mathfrak{d}, (\mathfrak{m}, \mathfrak{q}) = 1\} = \frac{\rho_K}{Y} \prod_{\mathfrak{p} | \mathfrak{q}} \left(1 - \frac{1}{N\mathfrak{p}}\right) \frac{X}{N\mathfrak{d}} (1 + o_{K,\varepsilon}(1)).$$

For every non-principal $\chi \in \widehat{G/H}$, we use (CS^b(α_0)). Then, there exists $\delta > 0$ such that

$$\sum_{N\mathfrak{m} \leq X/N\mathfrak{d}} \chi(\mathfrak{m}) \ll_{K,\varepsilon} \left(\frac{X}{N\mathfrak{d}}\right)^{1-\delta}.$$

Since Y is fixed, the total non-principal contribution is $O_{K,\varepsilon}\left(\left(\frac{X}{N\mathfrak{d}}\right)^{1-\delta}\right)$.

Combining the principal and non-principal estimates gives, uniformly for $N\mathfrak{d} \leq D$,

$$\sum_{\substack{N\mathfrak{m} \leq X/N\mathfrak{d} \\ [\mathfrak{d}\mathfrak{m}] \in bH}} 1 = \frac{\rho_K}{Y} \prod_{\mathfrak{p} | \mathfrak{q}} \left(1 - \frac{1}{N\mathfrak{p}}\right) \frac{X}{N\mathfrak{d}} (1 + o_{K,\varepsilon}(1)).$$

Substituting this into the lower sieve inequality (3.1), we obtain

$$\mathcal{N}_\gamma(X; b, H) \geq \frac{\rho_K X}{Y} \prod_{\mathfrak{p} | \mathfrak{q}} \left(1 - \frac{1}{N\mathfrak{p}}\right) \sum_{\substack{\mathfrak{d} | \mathcal{P}(z) \\ (\mathfrak{d}, \mathfrak{q})=1}} \frac{\lambda_\mathfrak{d}^-}{N\mathfrak{d}} (1 + o_{K,\varepsilon}(1)).$$

It remains to evaluate the sieve sum. Define a multiplicative function h on ideals by

$$h(\mathfrak{d}) = \begin{cases} 1/N\mathfrak{d}, & (\mathfrak{d}, \mathfrak{q}) = 1, \\ 0, & (\mathfrak{d}, \mathfrak{q}) \neq 1. \end{cases}$$

The linear sieve for number fields Lemma 2.1 gives

$$\sum_{\mathfrak{d} | \mathcal{P}(z)} \lambda_\mathfrak{d}^- h(\mathfrak{d}) \geq (f_0(s) + o_K(1)) \prod_{\substack{N\mathfrak{p} < z \\ \mathfrak{p} \nmid \mathfrak{q}}} \left(1 - \frac{1}{N\mathfrak{p}}\right).$$

Multiplying by $\prod_{\mathfrak{p}|\mathfrak{q}}(1 - 1/N\mathfrak{p})$, we get

$$\prod_{\mathfrak{p}|\mathfrak{q}} \left(1 - \frac{1}{N\mathfrak{p}}\right) \prod_{\substack{N\mathfrak{p} < z \\ \mathfrak{p} \nmid \mathfrak{q}}} \left(1 - \frac{1}{N\mathfrak{p}}\right) = (1 + o_K(1)) \prod_{N\mathfrak{p} < z} \left(1 - \frac{1}{N\mathfrak{p}}\right).$$

Here the primes dividing $\mathfrak{q}$ with norm $\geq z$ contribute only $1 + O(\frac{\log Q}{z \log z}) = 1 + o(1)$. By Mertens' theorem [21, Thm. 2] for K ,

$$(3.2) \quad \prod_{N\mathfrak{p} < z} \left(1 - \frac{1}{N\mathfrak{p}}\right) = \frac{e^{-\gamma}}{\rho_K \log z} + O_K\left(\frac{1}{(\log z)^2}\right).$$

When $2 \leq s \leq 4$, $f_0(s) = \frac{2e^\gamma \log(s-1)}{s}$. Applying the lower bound sieve in Lemma 2.1 gives

$$\mathcal{N}_\gamma(X; b, H) \geq (1 + o_{K,\varepsilon}(1)) \frac{X}{Y} f_0(s) \frac{e^{-\gamma}}{\log z} = (1 + o_{K,\varepsilon}(1)) \frac{2 \log(s-1)}{Y \vartheta_0} \frac{X}{\log X}.$$

This is the stated lower bound.

The upper bound is identical, using the upper-bound sieve weights $\lambda_{\mathfrak{d}}^+$. The linear sieve gives

$$\sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_{\mathfrak{d}}^+ h(\mathfrak{d}) \leq (F_0(s) + o_K(1)) \prod_{\substack{N\mathfrak{p} < z \\ \mathfrak{p} \nmid \mathfrak{q}}} \left(1 - \frac{1}{N\mathfrak{p}}\right),$$

and for $1 \leq s \leq 3$, $F_0(s) = \frac{2e^\gamma}{s}$. Hence

$$\mathcal{N}_\gamma(X; b, H) \leq (1 + o_{K,\varepsilon}(1)) \frac{2}{Y \vartheta_0} \frac{X}{\log X}.$$

This proves the lemma. $\square$

For $X \geq 2$, define

$$\mathcal{A}(X; \mathfrak{q}) := \{\mathfrak{a} \subset \mathcal{O}_K : (\mathfrak{a}, \mathfrak{q}) = 1, N\mathfrak{a} \leq X\}.$$

For functions $f : \mathcal{A}(X; \mathfrak{q}) \rightarrow \mathbb{R}_{\geq 0}$ and $g : G \rightarrow \mathbb{R}_{\geq 0}$, write

$$\mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a}) := \frac{1}{|\mathcal{A}(X; \mathfrak{q})|} \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a}), \quad \mathbb{E}_{c \in G} g(c) := \frac{1}{|G|} \sum_{c \in G} g(c).$$

More generally, for a finite non-empty set S and a function $F : S^k \rightarrow \mathbb{C}$, write

$$\mathbb{E}_{x_1, \dots, x_k \in S} F(x_1, \dots, x_k) := \frac{1}{|S|^k} \sum_{x_1, \dots, x_k \in S} F(x_1, \dots, x_k).$$

For $\chi \in \widehat{G}$, define

$$\widehat{f}(\chi) := \mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a}) \chi([\mathfrak{a}]), \quad \widehat{g}(\chi) := \mathbb{E}_{c \in G} g(c) \chi(c).$$

Proposition 3.2 (Multiplicative dense model). *Let $r > 1$, $C \geq 1$, and $\eta, \theta \in (0, 1)$ be fixed. Let $\delta \in \left(\left(\frac{10Cr \log \log Q}{\theta \log Q}\right)^{1/r}, \frac{1}{10}\right)$, where this interval is assumed to be nonempty. Suppose $f : \mathcal{A}(X; \mathfrak{q}) \rightarrow \mathbb{R}_{\geq 0}$ satisfies the following two conditions.*

(A1) *There exists a function $\nu : \mathcal{A}(X; \mathfrak{q}) \rightarrow \mathbb{R}_{\geq 0}$ such that*

$$f(\mathfrak{a}) \leq \nu(\mathfrak{a}) \quad (\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})),$$

and for some fixed constant $C_1 > 0$,

$$\mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} \nu(\mathfrak{a}) \leq 1 + \eta, \quad \max_{\chi \neq \chi_0} |\widehat{\nu}(\chi)| \leq C_1 Q^{-\theta}.$$

(A2) *There are at most $C\delta^{-r}$ characters $\chi \in \widehat{G}$ satisfying*

$$|\widehat{f}(\chi)| \geq \delta.$$

Then there exists a function $g : G \rightarrow \mathbb{R}_{\geq 0}$ satisfying the following properties.

(i) *For every $c \in G$,*

$$0 \leq g(c) \leq 1 + \eta + C_1 Q^{-\theta/2}.$$

(ii) For every character $\chi \in \widehat{G}$,

$$|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq \delta.$$

(iii) For every character $\chi \in \widehat{G}$,

$$|\widehat{g}(\chi)| \leq |\widehat{f}(\chi)|, \quad |\widehat{f}(\chi) - \widehat{g}(\chi)| \leq |\widehat{f}(\chi)|.$$

(iv)

$$\mathbb{E}_{c \in G} g(c) = \mathbb{E}_{\mathbf{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathbf{a}).$$

(v) Let $H \leq G$ be a subgroup of index $Y < 1/(2\delta)$. Then, for every coset bH of H ,

$$|\mathbb{E}_{\mathbf{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathbf{a}) \mathbf{1}_{[\mathbf{a}] \in bH} - \mathbb{E}_{c \in G} g(c) \mathbf{1}_{c \in bH}| < \delta.$$

Proof. Define the large spectrum of f by

$$\mathcal{T} := \{\chi \in \widehat{G} : |\widehat{f}(\chi)| \geq \delta\}.$$

By assumption (A2), $|\mathcal{T}| \leq C\delta^{-r}$.

We define the multiplicative Bohr set

$$B := \left\{ b \in G : |\chi(b) - 1| \leq \frac{\delta}{5} \text{ for every } \chi \in \mathcal{T} \right\}.$$

We first give a lower bound for $|B|$. Write

$$\mathcal{T} = \{\chi_1, \dots, \chi_k\}, \quad k \leq C\delta^{-r}.$$

Choose $L = \lceil \frac{10\pi}{\delta} \rceil$ equally spaced points $\zeta_1, \dots, \zeta_L \in S^1$. Then every point of S^1 has distance at most $\pi/L \leq \delta/10$ from some ζ_ℓ .

For each $a \in G$, and for each $j = 1, \dots, k$, choose an index $\ell_j(a) \in \{1, \dots, L\}$ such that $|\chi_j(a) - \zeta_{\ell_j(a)}| \leq \frac{\delta}{10}$. Thus each $a \in G$ determines a k -tuple $(\ell_1(a), \dots, \ell_k(a)) \in \{1, \dots, L\}^k$. Since there are L^k possible such tuples, the pigeonhole principle gives a tuple $(\ell_1, \dots, \ell_k)$ which occurs for at least $L^{-k}|G|$ elements of G . Setting $\xi_j := \zeta_{\ell_j}$, $j = 1, \dots, k$, we obtain a set

$$A := \left\{ a \in G : |\chi_j(a) - \xi_j| \leq \frac{\delta}{10} \text{ for every } j = 1, \dots, k \right\}$$

satisfying

$$|A| \geq L^{-k}|G| = \left\lceil \frac{10\pi}{\delta} \right\rceil^{-k} |G|.$$

If $x, y \in A$, then for every $j = 1, \dots, k$, $|\chi_j(x) - \chi_j(y)| \leq |\chi_j(x) - \xi_j| + |\chi_j(y) - \xi_j| \leq \frac{\delta}{5}$. Since $|\chi_j(y)| = 1$, it follows that

$$|\chi_j(xy^{-1}) - 1| = \left| \frac{\chi_j(x)}{\chi_j(y)} - 1 \right| = |\chi_j(x) - \chi_j(y)| \leq \frac{\delta}{5}.$$

Thus $xy^{-1} \in B$. Fix $y_0 \in A$, then $Ay_0^{-1} \subseteq B$. Therefore

$$|B| \geq |Ay_0^{-1}| = |A| \geq |G| \left\lceil \frac{10\pi}{\delta} \right\rceil^{-C\delta^{-r}}.$$

Since $\delta < 1/10$, we have $\left\lceil \frac{10\pi}{\delta} \right\rceil \leq \frac{20\pi}{\delta}$. By the choice of δ , $C\delta^{-r} \leq \frac{\theta \log Q}{10r \log \log Q}$. Moreover, since Q is an integer and the interval defining δ is nonempty, we have $\log \log Q \geq 1$. It follows that $\log \frac{20\pi}{\delta} \leq 5r \log \log Q$. Consequently

$$\log \frac{|G|}{|B|} \leq C\delta^{-r} \log \frac{20\pi}{\delta} \leq \frac{\theta \log Q}{10r \log \log Q} \cdot 5r \log \log Q = \frac{\theta}{2} \log Q.$$

Hence

$$\frac{|G|}{|B|} \leq Q^{\theta/2}.$$

Now define

$$g(c) := |G| \mathbb{E}_{b_1, b_2 \in B} \mathbb{E}_{\mathbf{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathbf{a}) \mathbf{1}_{[\mathbf{a}] = cb_1 b_2^{-1}}.$$

Clearly $g(c) \geq 0$. We now verify the five claims.

Using that $f \leq \nu$ and the orthogonality of characters of G ,

$$1_{[\mathfrak{a}] = cb_1 b_2^{-1}} = \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \chi([\mathfrak{a}]) \overline{\chi(c)} \overline{\chi(b_1)} \chi(b_2),$$

we have

$$g(c) \leq |G| \mathbb{E}_{b_1, b_2 \in B} \mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} \nu(\mathfrak{a}) 1_{[\mathfrak{a}] = cb_1 b_2^{-1}} = \sum_{\chi \in \widehat{G}} \widehat{\nu}(\chi) \overline{\chi(c)} |\mathbb{E}_{b \in B} \chi(b)|^2.$$

Separating the principal character,

$$g(c) \leq \widehat{\nu}(\chi_0) + \max_{\chi \neq \chi_0} |\widehat{\nu}(\chi)| \sum_{\chi \in \widehat{G}} |\mathbb{E}_{b \in B} \chi(b)|^2.$$

By (A1),

$$\widehat{\nu}(\chi_0) = \mathbb{E}_{\mathfrak{a}} \nu(\mathfrak{a}) \leq 1 + \eta, \quad \text{and} \quad \max_{\chi \neq \chi_0} |\widehat{\nu}(\chi)| \leq C_1 Q^{-\theta}.$$

Moreover, by orthogonality of characters

$$\sum_{\chi \in \widehat{G}} |\mathbb{E}_{b \in B} \chi(b)|^2 = \frac{|G|}{|B|} \leq Q^{\theta/2}.$$

Hence

$$g(c) \leq 1 + \eta + C_1 Q^{-\theta/2},$$

which proves (i).

For any $\chi \in \widehat{G}$, we have

$$\begin{aligned} \mathbb{E}_{c \in G} g(c) \chi(c) &= \mathbb{E}_{b_1, b_2 \in B} \mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a}) \chi([\mathfrak{a}] b_1^{-1} b_2) \\ &= |\mathbb{E}_{b \in B} \chi(b)|^2 \mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a}) \chi([\mathfrak{a}]). \end{aligned}$$

Thus

$$(3.3) \quad \widehat{g}(\chi) = |\mathbb{E}_{b \in B} \chi(b)|^2 \widehat{f}(\chi).$$

It follows from (3.3) that

$$(3.4) \quad \widehat{f}(\chi) - \widehat{g}(\chi) = \widehat{f}(\chi) \left(1 - |\mathbb{E}_{b \in B} \chi(b)|^2\right).$$

If $\chi \notin \mathcal{T}$, then $|\widehat{f}(\chi)| < \delta$. Since $0 \leq |\mathbb{E}_{b \in B} \chi(b)|^2 \leq 1$, we obtain

$$|\widehat{f}(\chi) - \widehat{g}(\chi)| < \delta.$$

Now suppose that $\chi \in \mathcal{T}$. For every $b \in B$, we have $|\chi(b) - 1| \leq \frac{\delta}{5}$, and hence $|\mathbb{E}_{b \in B} \chi(b) - 1| \leq \frac{\delta}{5}$. Therefore

$$\left|1 - |\mathbb{E}_{b \in B} \chi(b)|^2\right| \leq \left(1 + \frac{\delta}{5}\right)^2 - 1.$$

Since $f \leq \nu$,

$$|\widehat{f}(\chi)| \leq \mathbb{E}_{\mathfrak{a}} \nu(\mathfrak{a}) \leq 1 + \eta.$$

Consequently, by (3.4),

$$|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq (1 + \eta) \left(\left(1 + \frac{\delta}{5}\right)^2 - 1 \right) \leq \delta,$$

since $\eta < 1$ and $\delta < 1/10$. This proves (ii).

(iii) follows immediately from (3.3) and (3.4), since $0 \leq |\mathbb{E}_{b \in B} \chi(b)|^2 \leq 1$.

(iv) follows by taking $\chi = \chi_0$ in (3.3), since $\mathbb{E}_{b \in B} \chi_0(b) = 1$.

Let $H \leq G$ with index $Y < 1/(2\delta)$, and define

$$\mathcal{D} := \{\chi \in \widehat{G} : \chi(h) = 1 \text{ for every } h \in H\}.$$

Then $|\mathcal{D}| = Y$, and the orthogonality of characters gives

$$1_{d \in bH} = \frac{1}{Y} \sum_{\chi \in \mathcal{D}} \chi(d) \overline{\chi(b)}.$$

Therefore

$$\mathbb{E}_{\mathbf{a}} f(\mathbf{a}) 1_{[\mathbf{a}] \in bH} - \mathbb{E}_{c \in G} g(c) 1_{c \in bH} = \frac{1}{Y} \sum_{\chi \in \mathcal{D}} \overline{\chi(b)} (\widehat{f}(\chi) - \widehat{g}(\chi)).$$

We now show that each summand is $O(\delta)$.

If $\chi \notin \mathcal{T}$, then by (iii) and the definition of $\mathcal{T}$,

$$|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq |\widehat{f}(\chi)| < \delta.$$

Now suppose that $\chi \in \mathcal{T} \cap \mathcal{D}$. Since χ is trivial on H , its order divides Y . Hence every value $\chi(b)$ is a Y -th root of unity. Since $Y < \frac{1}{2\delta}$, the only Y -th root of unity lying within distance $\delta/5$ of 1 is 1 itself. Since

$$B := \left\{ b \in G : |\chi(b) - 1| \leq \frac{\delta}{5} \text{ for every } \chi \in \mathcal{T} \right\},$$

it follows that $\chi(b) = 1$ for every $b \in B$. Hence $\mathbb{E}_{b \in B} \chi(b) = 1$. By (3.3), $\widehat{g}(\chi) = \widehat{f}(\chi)$. Thus the contribution of every $\chi \in \mathcal{T} \cap \mathcal{D}$ vanishes.

Combining the two cases, every summand is $< \delta$, therefore

$$|\mathbb{E}_{\mathbf{a}} f(\mathbf{a}) 1_{[\mathbf{a}] \in bH} - \mathbb{E}_{c \in G} g(c) 1_{c \in bH}| < \left(1 - \frac{|\mathcal{T} \cap \mathcal{D}|}{Y}\right) \delta \leq \delta.$$

This proves (v), and hence the proposition. $\square$

3.1. Applying the transference principle. For $k \geq 1$, $X \geq 2$, recall the notations

$$\mathcal{A}(X; \mathbf{q}) := \{ \mathbf{a} \in \mathcal{O}_K : (\mathbf{a}, \mathbf{q}) = 1, N\mathbf{a} \leq X \},$$

$$E_k(X; \mathbf{q}) := \{ [\mathbf{p}_1] \cdots [\mathbf{p}_k] \in G : \mathbf{p}_i \nmid \mathbf{q}, N\mathbf{p}_i \leq X \}.$$

For a subgroup $H \leq G$ and a coset $bH \subseteq G$, write

$$(3.5) \quad \pi(X; bH) := \# \{ \mathbf{p} \in \mathcal{O}_K : \mathbf{p} \nmid \mathbf{q}, N\mathbf{p} \leq X, [\mathbf{p}] \in bH \}.$$

We shall use the following two short character-sum bound assumptions. Let $0 < \alpha_0 \leq \alpha < 1$. We assume the following two character sum assumptions. For every $\varepsilon > 0$, there exists $\eta_1 = \eta_1(\varepsilon) > 0$ such that for every non-principal Hecke character χ modulo $\mathbf{q}$, whenever $T \geq Q^{\alpha+\varepsilon}$,

$$(CS(\alpha)) \quad \sum_{N\mathbf{a} \leq T} \chi(\mathbf{a}) \ll_{K, \varepsilon} T^{1-\eta_1};$$

For every fixed integer $\ell \geq 2$ and every $\varepsilon > 0$, there exists $\eta_2 = \eta_2(\ell, \varepsilon) > 0$ such that for every non-principal Hecke character χ modulo $\mathbf{q}$ of order at most ℓ , whenever $T \geq Q^{\alpha_0+\varepsilon}$,

$$(CS^b(\alpha_0)) \quad \sum_{N\mathbf{a} \leq T} \chi(\mathbf{a}) \ll_{K, \ell, \varepsilon} T^{1-\eta_2}.$$

We assume throughout that $CS(\alpha)$ and $CS^b(\alpha_0)$ hold for some fixed $0 < \alpha_0 \leq \alpha < 1$.

For $X \geq 2$, define

$$(3.6) \quad \vartheta := 1 - \varepsilon - \alpha \frac{\log Q}{\log X}, \quad \vartheta_0 := 1 - \varepsilon - \alpha_0 \frac{\log Q}{\log X}.$$

Proposition 3.3 (Conclusion of transference over ray class groups). *Let $\kappa, \varepsilon > 0$ and $C \geq 1$ be fixed, with $\varepsilon > 0$ sufficiently small. Let ϑ and ϑ_0 be as in (3.6). Assume that*

$$X \in [Q^{\frac{\alpha+3\varepsilon}{1-\varepsilon}}, Q^C].$$

Then, when Q is sufficiently large, there exists a set $A \subseteq G$ such that the following hold.

(i)

$$|A| \geq \left(\frac{\vartheta}{2} - \varepsilon \right) |G|.$$

(ii) *Assume that $X \geq Q^{1+\kappa}$. Then for all but $O_{K, \varepsilon}(|G|(\log Q)^{-\varepsilon/2})$ elements $c \in G$, we have*

$$(1_A * 1_A)(c) \gg \frac{|G|}{(\log Q)^{1/2-\varepsilon}} \implies c \in E_2(X; \mathbf{q}).$$

(iii) Assume that $X \geq Q^{1+\kappa}$. Then, for every $c \in G$,

$$(1_A * 1_A * 1_A)(c) \gg \frac{|G|^2}{(\log Q)^{1/2-\varepsilon}} \implies c \in E_3(X; \mathfrak{q}).$$

(iv) Assume that $B := \frac{\log X}{\log Q} \geq 2\alpha + \kappa$. Let $\beta \in (2\varepsilon, B]$ and $L = \lfloor \frac{B}{\beta} \rfloor$. Then, for every $c \in G$,

$$\sum_{\substack{[\mathfrak{p}]^{a_1} a_2 = c, \\ Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta}} \frac{1}{N\mathfrak{p}} \gg \frac{|G|}{(\log Q)^{(1-2\varepsilon)/(2L)}} \implies c \in E_3(X; \mathfrak{q}).$$

(v) Assume that $X \geq Q^{3\alpha_0+\kappa}$. Let $H \leq G$ be a subgroup of index $Y < \varepsilon^{-1/2}$. Then there exist at least $\left\lceil \left(\frac{\vartheta_0}{2} - 3\varepsilon^{1/2} \frac{\vartheta_0}{\vartheta} \right) Y \right\rceil$ distinct cosets bH of H such that

$$|A \cap bH| > \varepsilon |G|.$$

(vi) Let $H \leq G$ have index $Y < \varepsilon^{-1/2}$, and let bH be a coset. Let $\pi(X; bH)$ be as defined in (3.5). Then

$$|A \cap bH| \geq \left(\frac{\vartheta}{2} \frac{\pi(X; bH)}{X/\log X} - \frac{\varepsilon}{5Y} \right) |G|.$$

Proof. Let $V_{\mathfrak{q}} := \rho_K \prod_{\mathfrak{p}|\mathfrak{q}} \left(1 - \frac{1}{N\mathfrak{p}}\right)$, so that by Lemma 2.8,

$$(3.7) \quad |\mathcal{A}(X; \mathfrak{q})| = V_{\mathfrak{q}} X (1 + o_K(1)).$$

Put

$$D = X^\vartheta, \quad z = X^{\vartheta/3}.$$

Let $\lambda_{\mathfrak{d}}^+$ be the upper-bound linear sieve weights with level D and sifting range z as in Lemma 2.1.

We shall apply Proposition 3.2 with

$$r = 2, \quad \delta := \frac{1}{(\log Q)^{1/2-\varepsilon/2}}.$$

Define $f, \nu : \mathcal{A}(X; \mathfrak{q}) \rightarrow \mathbb{R}_{\geq 0}$ by

$$f(\mathfrak{a}) := \frac{\vartheta}{2} V_{\mathfrak{q}} \log X \, 1_{\mathfrak{a}=\mathfrak{p}} 1_{N\mathfrak{p} \geq z},$$

and

$$(3.8) \quad \nu(\mathfrak{a}) := \frac{\vartheta}{2} V_{\mathfrak{q}} \log X \sum_{\substack{\mathfrak{d}|\mathfrak{a} \\ N\mathfrak{d} \leq D}} \lambda_{\mathfrak{d}}^+.$$

By the upper-bound sieve inequality as in Lemma 2.1,

$$f(\mathfrak{a}) \leq \nu(\mathfrak{a}) \quad (\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})).$$

We now verify the assumptions of Proposition 3.2.

Verification of (A2). Let

$$\mathcal{T} := \{\chi \in \widehat{G} : |\widehat{f}(\chi)| \geq \delta\}.$$

We claim that

$$(3.9) \quad |\mathcal{T}| \ll_{K,\varepsilon} \delta^{-2}, \quad \sum_{\chi \in \mathcal{T}} |\widehat{f}(\chi)|^2 \ll_{K,\varepsilon} 1$$

First note that the hypothesis $B := \frac{\log X}{\log Q} \geq \frac{\alpha+3\varepsilon}{1-\varepsilon}$ implies $B\vartheta = B(1-\varepsilon) - \alpha \geq 3\varepsilon$. Hence

$$z = X^{\vartheta/3} = Q^{B\vartheta/3} \geq Q^\varepsilon.$$

Thus every ideal in the support of f is coprime to $\mathcal{P}(Q^\varepsilon)$. Moreover, the same lower bound for B implies $X \geq Q^{\alpha+2\varepsilon}$, so Lemma 2.9(i) is applicable.

Applying Lemma 2.9(i) to the set of characters $\mathcal{T}$, with coefficients $a_{\mathfrak{a}} = f(\mathfrak{a})$, gives

$$\sum_{\chi \in \mathcal{T}} \left| \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a}) \chi(\mathfrak{a}) \right|^2 \ll_{K,\varepsilon} \left(\frac{X}{\log Q} + |\mathcal{T}| X^{1-\eta} Q^{\varepsilon\eta} \right) \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a})^2$$

for some $\eta = \eta(\varepsilon) > 0$. Dividing by $|\mathcal{A}(X; \mathfrak{q})|^2$, we obtain

$$\sum_{\chi \in \mathcal{T}} |\hat{f}(\chi)|^2 \ll_{K, \varepsilon} \left(\frac{X}{\log Q} + |\mathcal{T}| X^{1-\eta} Q^{\varepsilon\eta} \right) \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a})^2.$$

Now $f(\mathfrak{a}) = \frac{\vartheta}{2} V_{\mathfrak{q}} \log X$ on prime ideals $\mathfrak{p}$ with $z \leq N\mathfrak{p} \leq X$, and is zero otherwise. Hence, by the prime ideal theorem and (3.7),

$$\frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a})^2 \ll_K \frac{\log X}{X}.$$

Therefore

$$(3.10) \quad |\mathcal{T}| \delta^2 \leq \sum_{\chi \in \mathcal{T}} |\hat{f}(\chi)|^2 \ll_{K, \varepsilon} \frac{\log X}{\log Q} + |\mathcal{T}| X^{-\eta} Q^{\varepsilon\eta} \log X.$$

Since $X \leq Q^C$, the first term is $O_{K, C}(1)$. Also, since $X \geq Q^{3\varepsilon}$ and $\delta^2 = (\log Q)^{-1+\varepsilon}$,

$$X^{-\eta} Q^{\varepsilon\eta} \log X \leq Q^{-2\varepsilon\eta} \log Q = o_{\varepsilon}(\delta^2).$$

Thus, for Q sufficiently large, $|\mathcal{T}| \delta^2 \ll_{K, \varepsilon} 1 + \frac{1}{2} |\mathcal{T}| \delta^2$. Absorbing the second term gives $|\mathcal{T}| \ll_{K, \varepsilon} \delta^{-2}$. Applying the bound for $|\mathcal{T}|$ back to (3.10) gives $\sum_{\chi \in \mathcal{T}} |\hat{f}(\chi)|^2 \ll_{K, \varepsilon} 1$. This proves (3.9).

Verification of (A1) for non-principal characters. Let $\chi \neq \chi_0$. Since ν is supported on $\mathcal{A}(X; \mathfrak{q})$, after interchanging the order of summation in (3.8) we get

$$\hat{\nu}(\chi) = \mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} \nu(\mathfrak{a}) \chi(\mathfrak{a}) = \frac{\vartheta}{2} V_{\mathfrak{q}} \log X \sum_{(\mathfrak{d}, \mathfrak{q})=1} \lambda_{\mathfrak{d}}^+ \chi(\mathfrak{d}) \frac{1}{|\mathcal{A}(X; \mathfrak{q})|} \sum_{Nm \leq X/N\mathfrak{d}} \chi(\mathfrak{m}).$$

Since $N\mathfrak{d} \leq D = X^{\vartheta}$, we have $\frac{X}{N\mathfrak{d}} \geq X^{1-\vartheta} = X^{\varepsilon} Q^{\alpha} \geq Q^{\alpha+\varepsilon_1}$ for some $\varepsilon_1 > 0$ depending only on α and ε . Thus, by (CS(α)), there exists $\eta_1 > 0$ such that

$$\sum_{Nm \leq X/N\mathfrak{d}} \chi(\mathfrak{m}) \ll_{K, \varepsilon} \left(\frac{X}{N\mathfrak{d}} \right)^{1-\eta_1}.$$

Using $|\lambda_{\mathfrak{d}}^+| \leq 1$ and (3.7), we obtain

$$|\hat{\nu}(\chi)| \ll_{K, \varepsilon} \log X \sum_{N\mathfrak{d} \leq D} \frac{1}{X} \left(\frac{X}{N\mathfrak{d}} \right)^{1-\eta_1} = \log X X^{-\eta_1} \sum_{N\mathfrak{d} \leq D} \frac{1}{(N\mathfrak{d})^{1-\eta_1}}.$$

The elementary estimate $\sum_{N\mathfrak{d} \leq D} (N\mathfrak{d})^{-1+\eta_1} \ll_{K, \eta_1} D^{\eta_1}$ gives

$$|\hat{\nu}(\chi)| \ll_{K, \varepsilon} (\log X) X^{-\eta_1} D^{\eta_1} \ll_{K, \varepsilon} (\log Q) Q^{-\eta_1(\alpha+\varepsilon_1)} \ll_{K, \varepsilon} Q^{-\eta_2}$$

for some $\eta_2 > 0$. Therefore

$$\max_{\chi \neq \chi_0} |\hat{\nu}(\chi)| \ll_{K, \varepsilon} Q^{-\eta_2}.$$

Verification of (A1) for the principal character. We now estimate the average of the majorant ν .

From (3.8), after interchanging the order of summation, we have

$$\mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} \nu(\mathfrak{a}) = \frac{\vartheta}{2} V_{\mathfrak{q}} \log X \frac{1}{|\mathcal{A}(X; \mathfrak{q})|} \sum_{(\mathfrak{d}, \mathfrak{q})=1} \lambda_{\mathfrak{d}}^+ \# \left\{ \mathfrak{m} : Nm \leq \frac{X}{N\mathfrak{d}}, (\mathfrak{m}, \mathfrak{q}) = 1 \right\}.$$

Applying Lemma 2.8, together with (3.7), gives

$$(3.11) \quad \mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} \nu(\mathfrak{a}) = \frac{\vartheta}{2} V_{\mathfrak{q}} \log X \sum_{(\mathfrak{d}, \mathfrak{q})=1} \frac{\lambda_{\mathfrak{d}}^+}{N\mathfrak{d}} + o_K(1).$$

The upper-bound linear sieve Lemma 2.1 with

$$h(\mathfrak{p}) = \begin{cases} 1/N\mathfrak{p}, & \mathfrak{p} \nmid \mathfrak{q}, \\ 0, & \mathfrak{p} \mid \mathfrak{q}, \end{cases}$$

together with the prime-ideal Mertens theorem (3.2) gives

$$(3.12) \quad V_{\mathfrak{q}} \sum_{(\mathfrak{d}, \mathfrak{q})=1} \frac{\lambda_{\mathfrak{d}}^+}{N\mathfrak{d}} \leq \left(\frac{2e^\gamma}{3} + o_K(1) \right) \rho_K \prod_{\mathfrak{p}|\mathfrak{q}} \left(1 - \frac{1}{N\mathfrak{p}} \right) \prod_{\substack{N\mathfrak{p} < z \\ \mathfrak{p} \nmid \mathfrak{q}}} \left(1 - \frac{1}{N\mathfrak{p}} \right) \leq (1 + o_K(1)) \frac{2}{\vartheta \log X}.$$

Substituting (3.12) into (3.11), we obtain

$$\mathbb{E}_{\mathfrak{a}} \nu(\mathfrak{a}) \leq 1 + o_K(1).$$

This verifies the principal-character part of (A1).

Therefore, the hypotheses of Proposition 3.2 are satisfied for Q sufficiently large. Hence we obtain a function $g : G \rightarrow [0, 1 + o_K(1)]$ satisfying the conclusions of Proposition 3.2.

By Proposition 3.2(iv),

$$\mathbb{E}_{c \in G} g(c) = \mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a}).$$

Using the prime ideal theorem and (3.7), we have

$$\mathbb{E}_{\mathfrak{a}} f(\mathfrak{a}) = \frac{\vartheta}{2} V_{\mathfrak{q}} \log X \cdot \frac{\#\{\mathfrak{p} \nmid \mathfrak{q} : z \leq N\mathfrak{p} \leq X\}}{|\mathcal{A}(X; \mathfrak{q})|} = \frac{\vartheta}{2} + o_K(1).$$

Therefore

$$(3.13) \quad \mathbb{E}_{c \in G} g(c) = \frac{\vartheta}{2} + o_K(1).$$

Moreover,

$$(3.14) \quad \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a})^2 = \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} \left(\frac{\vartheta}{2} V_{\mathfrak{q}} \log X \right)^2 \#\{\mathfrak{p} \nmid \mathfrak{q} : z \leq N\mathfrak{p} \leq X\} \ll_K \frac{\log X}{X}.$$

Define

$$(3.15) \quad A := \left\{ c \in G : g(c) \geq \frac{\varepsilon}{10} \right\}.$$

We now prove the six conclusions.

Proof of (i). Since $g(c) \leq 1 + o_K(1)$,

$$\mathbb{E}_{c \in G} g(c) \leq \frac{\varepsilon}{10} + \frac{|A|}{|G|} (1 + o_K(1)).$$

Combining this with (3.13), for Q sufficiently large,

$$|A| \geq \left(\frac{\vartheta}{2} - \varepsilon \right) |G|.$$

Proof of (iii). For a character χ of G , recall that

$$\widehat{f}(\chi) := \mathbb{E}_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a}) \chi([\mathfrak{a}]), \quad \widehat{g}(\chi) := \mathbb{E}_{c \in G} g(c) \chi(c).$$

For $c \in G$, define

$$T_3(c) := \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^3} \sum_{\substack{\mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3 \in \mathcal{A}(X; \mathfrak{q}) \\ [\mathfrak{a}_1 \mathfrak{a}_2 \mathfrak{a}_3] = c}} f(\mathfrak{a}_1) f(\mathfrak{a}_2) f(\mathfrak{a}_3).$$

If $T_3(c) > 0$, then $c \in E_3(X; \mathfrak{q})$.

By orthogonality of characters, we have

$$T_3(c) - \frac{1}{|G|^3} (g * g * g)(c) = \frac{1}{|G|} \sum_{\chi} (\widehat{f}(\chi)^3 - \widehat{g}(\chi)^3) \overline{\chi(c)}.$$

Using $|\widehat{g}(\chi)| \leq |\widehat{f}(\chi)|$, we have

$$(3.16) \quad \left| T_3(c) - \frac{1}{|G|^3} (g * g * g)(c) \right| \leq \frac{3}{|G|} \sum_{\chi} |\widehat{f}(\chi)|^2 |\widehat{f}(\chi) - \widehat{g}(\chi)|.$$

We show that the sum in (3.16) is $O_{K, \varepsilon}(\delta)$. Put

$$\tau := \frac{\delta}{(\log Q)^{n_K + 2}}.$$

Split the characters into

$$\mathcal{X}_1 := \{\chi : |\widehat{f}(\chi)| \leq \tau\}, \quad \mathcal{X}_2 := \{\chi : \tau < |\widehat{f}(\chi)| \leq \delta\}, \quad \mathcal{X}_3 := \{\chi : |\widehat{f}(\chi)| > \delta\}.$$

By the mean value theorem Lemma 2.7, together with (3.14), we have

$$\begin{aligned} \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 &= \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} \sum_{\chi \in \widehat{G}} \left| \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a}) \chi([\mathfrak{a}]) \right|^2 \\ &\ll_K (X + Q)(\log(3Q))^{n_K} \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a})^2 \\ &\ll_K \left(1 + \frac{Q}{X}\right) (\log(3Q))^{n_K} \log X. \end{aligned}$$

Since $X \leq Q^C$ and $X \geq Q^{1+\kappa}$, this implies

$$(3.17) \quad \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 \ll_K (\log Q)^{n_K+1}.$$

For $\mathcal{X}_1$, using $|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq |\widehat{f}(\chi)| \leq \tau$ and (3.17), we obtain

$$(3.18) \quad \sum_{\chi \in \mathcal{X}_1} |\widehat{f}(\chi)|^2 |\widehat{f}(\chi) - \widehat{g}(\chi)| \leq \tau \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 \ll_K \frac{\delta}{(\log Q)^{n_K+2}} (\log Q)^{n_K+1} \ll \delta.$$

For $\mathcal{X}_2$, decompose dyadically according to $T < |\widehat{f}(\chi)| \leq 2T$, $\tau \leq T \leq \delta$. Let

$$R(T) := \#\{\chi : T < |\widehat{f}(\chi)| \leq 2T\}.$$

Since τ is a negative power of $\log Q$, applying Lemma 2.9(i) as in the verification of (A2), we have $R(T) \ll_{K,\varepsilon} T^{-2}$ uniformly for $\tau \leq T \leq \delta$. Hence

$$(3.19) \quad \sum_{\chi \in \mathcal{X}_2} |\widehat{f}(\chi)|^2 |\widehat{f}(\chi) - \widehat{g}(\chi)| \leq \sum_{\tau \leq T \leq \delta} R(T) T^3 \ll_{K,\varepsilon} \sum_{\tau \leq T \leq \delta} T \ll \delta.$$

For $\mathcal{X}_3$, by Lemma 2.9(i), again applied as in the proof of (3.9) in the verification of (A2), we have $|\mathcal{X}_3| \ll_{K,\varepsilon} \delta^{-2}$ and $\sum_{\chi \in \mathcal{X}_3} |\widehat{f}(\chi)|^2 \ll_{K,\varepsilon} 1$. Moreover, Proposition 3.2(ii) gives $|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq \delta$. Therefore

$$(3.20) \quad \sum_{\chi \in \mathcal{X}_3} |\widehat{f}(\chi)|^2 |\widehat{f}(\chi) - \widehat{g}(\chi)| \ll_{K,\varepsilon} \delta.$$

Combining (3.18), (3.19), and (3.20), we obtain

$$\sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 |\widehat{f}(\chi) - \widehat{g}(\chi)| \ll_{K,\varepsilon} \delta.$$

Thus (3.16) gives

$$(3.21) \quad T_3(c) = \frac{1}{|G|^3} (g * g * g)(c) + O_{K,\varepsilon} \left(\frac{\delta}{|G|} \right).$$

If $(1_A * 1_A * 1_A)(c) \gg \frac{|G|^2}{(\log Q)^{1/2-\varepsilon}}$, then, as $g \geq \varepsilon/10$ on A , we have

$$\frac{1}{|G|^3} (g * g * g)(c) \geq \frac{1}{|G|^3} \left(\frac{\varepsilon}{10} \right)^3 (1_A * 1_A * 1_A)(c) \gg \frac{1}{|G|(\log Q)^{1/2-\varepsilon}},$$

which dominates the error term in (3.21) for Q sufficiently large, since $\frac{\delta}{|G|} = \frac{1}{|G|(\log Q)^{1/2-\varepsilon/2}}$.

Therefore $T_3(c) > 0$, and hence $c \in E_3(X; \mathfrak{q})$.

Proof of (ii). Define

$$T_2(c) := \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} \sum_{\substack{\mathfrak{a}_1, \mathfrak{a}_2 \in \mathcal{A}(X; \mathfrak{q}) \\ [\mathfrak{a}_1 \mathfrak{a}_2] = c}} f(\mathfrak{a}_1) f(\mathfrak{a}_2).$$

If $T_2(c) > 0$, then $c \in E_2(X; \mathfrak{q})$.

Again by orthogonality of characters,

$$T_2(c) - \frac{1}{|G|^2}(g * g)(c) = \frac{1}{|G|} \sum_{\chi \in \widehat{G}} (\widehat{f}(\chi)^2 - \widehat{g}(\chi)^2) \overline{\chi(c)}.$$

Using $|\widehat{g}(\chi)| \leq |\widehat{f}(\chi)|$, we have

$$(3.22) \quad \sum_{c \in G} \left| T_2(c) - \frac{1}{|G|^2}(g * g)(c) \right|^2 \leq \frac{4}{|G|} \sum_{\chi} |\widehat{f}(\chi)|^2 |\widehat{f}(\chi) - \widehat{g}(\chi)|^2.$$

We estimate the sum in (3.22) using the same decomposition $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$ as in the proof of (iii).

For $\mathcal{X}_1$, using $|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq |\widehat{f}(\chi)|$, and (3.17), we obtain

$$\sum_{\chi \in \mathcal{X}_1} |\widehat{f}(\chi)|^2 |\widehat{f}(\chi) - \widehat{g}(\chi)|^2 \leq \tau^2 \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 \ll_K \frac{\delta^2}{(\log Q)^{2n_K+4}} (\log Q)^{n_K+1} \ll \delta^2.$$

For $\mathcal{X}_2$, dyadically decompose according to $T < |\widehat{f}(\chi)| \leq 2T$, $\tau \leq T \leq \delta$. As in the proof of (iii), Lemma 2.9(i) gives

$$R(T) := \#\{\chi : T < |\widehat{f}(\chi)| \leq 2T\} \ll_{K,\varepsilon} T^{-2}.$$

Therefore

$$\sum_{\chi \in \mathcal{X}_2} |\widehat{f}(\chi)|^2 |\widehat{f}(\chi) - \widehat{g}(\chi)|^2 \leq \sum_{\tau \leq T \leq \delta} R(T) T^4 \ll_{K,\varepsilon} \sum_{\tau \leq T \leq \delta} T^2 \ll \delta^2.$$

For $\mathcal{X}_3$, similar as before, Lemma 2.9(i) gives $|\mathcal{X}_3| \ll_{K,\varepsilon} \delta^{-2}$ and $\sum_{\chi \in \mathcal{X}_3} |\widehat{f}(\chi)|^2 \ll_{K,\varepsilon} 1$. Hence

$$\sum_{\chi \in \mathcal{X}_3} |\widehat{f}(\chi)|^2 |\widehat{f}(\chi) - \widehat{g}(\chi)|^2 \ll_{K,\varepsilon} \delta^2.$$

Combining the three estimates, we get

$$\sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 |\widehat{f}(\chi) - \widehat{g}(\chi)|^2 \ll_{K,\varepsilon} \delta^2.$$

Therefore (3.22) gives

$$\sum_{c \in G} \left| T_2(c) - \frac{1}{|G|^2}(g * g)(c) \right|^2 \ll_{K,\varepsilon} \frac{\delta^2}{|G|}.$$

Let

$$\lambda := \frac{1}{|G|(\log Q)^{1/2-3\varepsilon/4}}.$$

By Chebyshev's inequality,

$$\#\left\{c \in G : \left| T_2(c) - \frac{1}{|G|^2}(g * g)(c) \right| > \lambda \right\} \leq \lambda^{-2} \sum_{c \in G} \left| T_2(c) - \frac{1}{|G|^2}(g * g)(c) \right|^2 \ll_{K,\varepsilon} |G|(\log Q)^{-\varepsilon/2}.$$

Now suppose that c is outside this exceptional set of size $O_{K,\varepsilon}(|G|(\log Q)^{-\varepsilon/2})$, and that $(1_A * 1_A)(c) \gg \frac{|G|}{(\log Q)^{1/2-\varepsilon}}$. Then

$$T_2(c) = \frac{1}{|G|^2}(g * g)(c) + O\left(\frac{1}{|G|(\log Q)^{1/2-3\varepsilon/4}}\right).$$

Since $g \geq \varepsilon/10$ on A , we have

$$(g * g)(c) \geq \left(\frac{\varepsilon}{10}\right)^2 (1_A * 1_A)(c) \gg \frac{|G|}{(\log Q)^{1/2-\varepsilon}},$$

which dominates the error term. Therefore, for Q sufficiently large, $T_2(c) > 0$, and hence $c \in E_2(X; \mathfrak{q})$.

Proof of (iv). Assume that $B := \frac{\log X}{\log Q} \geq 2\alpha + \kappa$. Let $\beta \in (2\varepsilon, B]$, $L = \lfloor \frac{B}{\beta} \rfloor$. Define

$$f_0(\mathfrak{a}) := \frac{1_{\mathfrak{a}=\mathfrak{p}} 1_{Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta}}{N\mathfrak{p}},$$

$$(f * f * f_0)(c) := \sum_{\substack{\mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3 \in \mathcal{A}(X; \mathfrak{q}) \\ [\mathfrak{a}_1 \mathfrak{a}_2 \mathfrak{a}_3] = c}} f(\mathfrak{a}_1) f(\mathfrak{a}_2) f_0(\mathfrak{a}_3),$$

$$(g * g * f_0)(c) := \sum_{\substack{\mathfrak{a}_1, \mathfrak{a}_2 \in G, \mathfrak{a}_3 \in \mathcal{A}(X; \mathfrak{q}) \\ \mathfrak{a}_1 \mathfrak{a}_2 [\mathfrak{a}_3] = c}} g(\mathfrak{a}_1) g(\mathfrak{a}_2) f_0(\mathfrak{a}_3).$$

If $(f * f * f_0)(c) > 0$, then $c \in E_3(X; \mathfrak{q})$.

By orthogonality of characters and that $|\widehat{g}(\chi)| \leq |\widehat{f}(\chi)|$, we have

$$(3.23) \quad \left| \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} (f * f * f_0)(c) - \frac{1}{|G|^2} (g * g * f_0)(c) \right| \leq \frac{2}{|G|} \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)| |\widehat{f}_0(\chi)| |\widehat{f}(\chi) - \widehat{g}(\chi)|.$$

We split the characters into

$$\mathcal{X}_1 := \left\{ \chi \in \widehat{G} : |\widehat{f}_0(\chi)| < (\log Q)^{-(n_K+2)} \right\}, \quad \mathcal{X}_2 := \widehat{G} \setminus \mathcal{X}_1.$$

For $\chi \in \mathcal{X}_1$, using $|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq |\widehat{f}(\chi)|$ and (3.17), we have

$$(3.24) \quad \frac{1}{|G|} \sum_{\chi \in \mathcal{X}_1} |\widehat{f}(\chi)| |\widehat{f}_0(\chi)| |\widehat{f}(\chi) - \widehat{g}(\chi)| \leq \frac{1}{|G| (\log Q)^{n_K+2}} \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 \ll_K \frac{1}{|G| \log Q}$$

We now treat $\mathcal{X}_2$. By Proposition 3.2(ii),(iii), $|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq |\widehat{f}(\chi)|$ and $|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq \delta$, together with Hölder's inequality, we have

$$(3.25) \quad \frac{1}{|G|} \sum_{\chi \in \mathcal{X}_2} |\widehat{f}(\chi)| |\widehat{f}_0(\chi)| |\widehat{f}(\chi) - \widehat{g}(\chi)| \leq \frac{\delta^{\frac{1}{2L}}}{|G|} \left(\sum_{\chi \in \mathcal{X}_2} |\widehat{f}_0(\chi)|^{2L} \right)^{\frac{1}{2L}} \left(\sum_{\chi \in \mathcal{X}_2} |\widehat{f}(\chi)|^2 \right)^{1-\frac{1}{2L}}.$$

We first estimate the $2L$ -th moment. Expanding the L -th power,

$$(\widehat{f}_0(\chi))^L = \left(\sum_{\mathfrak{a}} f_0(\mathfrak{a}) \chi(\mathfrak{a}) \right)^L = \sum_{\mathfrak{n}} b_{\mathfrak{n}} \chi(\mathfrak{n}),$$

where $b_{\mathfrak{n}}$ is supported on ideals $\mathfrak{n} = \mathfrak{p}_1 \cdots \mathfrak{p}_L$ with $Q^{\beta-\varepsilon} < N\mathfrak{p}_j \leq Q^{\beta}$, and satisfies $|b_{\mathfrak{n}}| \leq \frac{\tau_L(\mathfrak{n})}{N\mathfrak{n}}$. Thus

$$Q^{L(\beta-\varepsilon)} < N\mathfrak{n} \leq Q^{L\beta} \leq Q^B = X.$$

We next verify that the lower endpoint lies in the range required for Lemma 2.9(ii). We claim that

$$(3.26) \quad L(\beta - \varepsilon) > \frac{B}{2} - \varepsilon.$$

If $L = 1$, then $\beta > B/2$, and the claim is immediate. Suppose now that $L \geq 2$. Since $L = \lfloor B/\beta \rfloor$, we have

$$\frac{B}{L+1} < \beta \leq \frac{B}{L}.$$

If $B/(L+1) \geq 2\varepsilon$, then

$$L(\beta - \varepsilon) > \frac{LB}{L+1} - L\varepsilon \geq \frac{B}{2} - \varepsilon.$$

If $B/(L+1) < 2\varepsilon$, then, since $\beta > 2\varepsilon$,

$$L(\beta - \varepsilon) > L\varepsilon > \frac{LB}{2(L+1)} = \frac{B}{2} - \frac{B}{2(L+1)} > \frac{B}{2} - \varepsilon.$$

This proves (3.26).

Because $B \geq 2\alpha + \kappa$, we may choose $\varepsilon > 0$ sufficiently small so that $L(\beta - \varepsilon) \geq \alpha + 3\varepsilon$. Moreover, since $\beta > 2\varepsilon$, every prime factor of $\mathfrak{n}$ has norm $> Q^{\varepsilon}$. Hence $(\mathfrak{n}, P(Q^{\varepsilon})) = 1$. Therefore Lemma 2.9(ii) applies to the coefficients $a_{\mathfrak{n}} := N\mathfrak{n} b_{\mathfrak{n}}$, and gives that for some $c_0 = c_0(\alpha, \varepsilon, \kappa) > 0$,

$$\sum_{\chi \in \mathcal{X}_2} |\widehat{f}_0(\chi)|^{2L} \ll_{K, \varepsilon} (1 + |\mathcal{X}_2| Q^{-c_0}) \sum_{\mathfrak{n}} \frac{|a_{\mathfrak{n}}|^2}{N\mathfrak{n}}.$$

Since $|a_n| \leq \tau_L(n)$, by Mertens' theorem for prime ideals (3.2), we have

$$\sum_n \frac{|a_n|^2}{Nn} \ll \prod_{Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta} \left(1 + \frac{O_L(1)}{N\mathfrak{p}}\right) \ll_{K,\varepsilon} 1.$$

Hence

$$(3.27) \quad \sum_{\chi \in \mathcal{X}_2} |\hat{f}_0(\chi)|^{2L} \ll_{K,\varepsilon} 1 + |\mathcal{X}_2| Q^{-c_0}.$$

On the other hand, by the definition of $\mathcal{X}_2$,

$$(\log Q)^{-2L(n_K+2)} |\mathcal{X}_2| \leq \sum_{\chi \in \mathcal{X}_2} |\hat{f}_0(\chi)|^{2L}.$$

Combining this with (3.27) and taking Q sufficiently large, we obtain

$$(3.28) \quad |\mathcal{X}_2| \ll_{K,\varepsilon} (\log Q)^{2L(n_K+2)}.$$

Applying this bound (3.28) for $|\mathcal{X}_2|$ back to (3.27), we have

$$(3.29) \quad \sum_{\chi \in \mathcal{X}_2} |\hat{f}_0(\chi)|^{2L} \ll_{K,\varepsilon} 1.$$

On the other hand, Lemma 2.9(i) gives for some $\eta = \eta(\varepsilon) > 0$,

$$\sum_{\chi \in \mathcal{X}_2} |\hat{f}(\chi)|^2 \ll \left(\frac{X}{\log Q} + |\mathcal{X}_2| X^{1-\eta} Q^{\varepsilon\eta} \right) \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a})^2.$$

Using (3.14) and (3.28) and taking ε sufficiently small, we have

$$(3.30) \quad \sum_{\chi \in \mathcal{X}_2} |\hat{f}(\chi)|^2 \ll_{K,\varepsilon} \frac{\log X}{\log Q} + |\mathcal{X}_2| X^{-\eta} Q^{\varepsilon\eta} \log X \ll 1 + |\mathcal{X}_2| X^{-\eta} Q^{\varepsilon\eta} \log Q \ll_\varepsilon 1.$$

Substituting (3.29) and (3.30) into (3.25), we obtain

$$\frac{1}{|G|} \sum_{\chi \in \mathcal{X}_2} |\hat{f}(\chi)| |\hat{f}_0(\chi)| |\hat{f}(\chi) - \hat{g}(\chi)| \ll_{K,\varepsilon} \frac{\delta^{1/L}}{|G|}.$$

Together with (3.23) and (3.24),

$$(3.31) \quad \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} (f * f * f_0)(c) = \frac{1}{|G|^2} (g * g * f_0)(c) + O_K \left(\frac{\delta^{1/L}}{|G|} \right).$$

Recall that $\delta = (\log Q)^{-1/2+\varepsilon/2}$, so

$$\frac{\delta^{1/L}}{|G|} = \frac{1}{|G|(\log Q)^{(1-\varepsilon)/(2L)}}.$$

Since $g \geq \varepsilon/10$ on A ,

$$\frac{1}{|G|^2} (g * g * f_0)(c) \gg_\varepsilon \frac{1}{|G|^2} \sum_{\substack{[\mathfrak{p}]_{a_1 a_2 = c} \\ Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta, a_1, a_2 \in A}} \frac{1}{N\mathfrak{p}} \gg \frac{1}{|G|(\log Q)^{(1-2\varepsilon)/(2L)}}.$$

which dominates the error term in (3.31). Hence $(f * f * f_0)(c) > 0$, and therefore $c \in E_3(X; \mathfrak{q})$.

Proof of (v). Let $H \leq G$ have index $Y < \varepsilon^{-1/2}$. Let $b_1 H, \dots, b_K H$ be the cosets of H such that A contains more than $\varepsilon|G|$ elements from $b_j H$. Let the remaining cosets be $c_1 H, \dots, c_{Y-K} H$.

For $j \leq K$, we first estimate the f -mass in the coset $b_j H$. By the definition of f ,

$$(3.32) \quad \mathbb{E}_{\mathfrak{a}} f(\mathfrak{a}) 1_{[\mathfrak{a}] \in b_j H} = \frac{\vartheta}{2} V_{\mathfrak{q}} \log X \frac{\#\{\mathfrak{p} : X^{\vartheta/3} \leq N\mathfrak{p} \leq X, [\mathfrak{p}] \in b_j H\}}{|\mathcal{A}(X; \mathfrak{q})|}.$$

Since $\vartheta \leq \vartheta_0$, the prime ideal theorem gives $\#\{\mathfrak{p} : X^{\vartheta/3} \leq N\mathfrak{p} < X^{\vartheta_0/3}\} = o_K(X/\log X)$. Applying the upper bound in Lemma 3.1 with $\gamma = \vartheta_0/3$ gives, uniformly in the coset $b_j H$,

$$\#\{\mathfrak{p} : X^{\vartheta_0/3} \leq N\mathfrak{p} \leq X, [\mathfrak{p}] \in b_j H\} \leq \mathcal{N}_{\vartheta_0/3}(X; b_j, H) \leq (1 + o_{K,\varepsilon}(1)) \frac{2}{Y \vartheta_0} \frac{X}{\log X}.$$

Therefore, combining these together with (3.7) in (3.32), we conclude that

$$\mathbb{E}_{\mathbf{a}} f(\mathbf{a}) 1_{[\mathbf{a}] \in b_j H} \leq (1 + o_{K,\varepsilon}(1)) \frac{\vartheta}{Y \vartheta_0}.$$

By Proposition 3.2(v),

$$(3.33) \quad \mathbb{E}_{c \in G} g(c) 1_{c \in b_j H} = \mathbb{E}_{\mathbf{a}} f(\mathbf{a}) 1_{[\mathbf{a}] \in b_j H} + O(\delta),$$

therefore

$$(3.34) \quad \frac{1}{|G|} \sum_{a \in b_j H} g(a) = \mathbb{E}_{c \in G} g(c) 1_{c \in b_j H} \leq (1 + o_{K,\varepsilon}(1)) \frac{\vartheta}{Y \vartheta_0}.$$

For $j > K$, we use $g \leq 1 + o_K(1)$ on A and $g < \varepsilon/10$ on $G \setminus A$, obtaining

$$\frac{1}{|G|} \sum_{a \in c_j H} g(a) \leq (1 + o_K(1)) \frac{|A \cap c_j H|}{|G|} + \frac{\varepsilon}{10} \frac{|c_j H|}{|G|} \leq (1 + o_K(1)) \varepsilon + \frac{\varepsilon}{10Y}.$$

Since $Y < \varepsilon^{-1/2}$, we have $\varepsilon \leq \varepsilon^{1/2}/Y$. Hence, for Q sufficiently large,

$$(3.35) \quad \frac{1}{|G|} \sum_{a \in c_j H} g(a) \leq \frac{2\varepsilon^{1/2}}{Y}.$$

Summing (3.34) and (3.35) over all cosets and using (3.13), we get

$$\frac{\vartheta}{2} + o_K(1) = \mathbb{E}_{c \in G} g(c) \leq (1 + o_K(1)) K \frac{\vartheta}{Y \vartheta_0} + 2\varepsilon^{1/2}.$$

Rearranging, for Q sufficiently large,

$$K \geq \left(\frac{\vartheta_0}{2} - 3\varepsilon^{1/2} \frac{\vartheta_0}{\vartheta} \right) Y.$$

Proof of (vi). Since the primes with $N\mathfrak{p} < X^{\vartheta_0/3}$ contribute $o_K(X/\log X)$, by (3.32) and (3.7), we have

$$(3.36) \quad \mathbb{E}_{\mathbf{a}} f(\mathbf{a}) 1_{[\mathbf{a}] \in bH} \geq \frac{\vartheta}{2} \frac{\pi(X; bH)}{X/\log X} + o_K(1).$$

On the other hand, using $g \leq 1 + o_K(1)$ on A and $g < \varepsilon/10$ on $G \setminus A$, we have

$$(3.37) \quad \frac{1}{|G|} \sum_{a \in bH} g(a) \leq (1 + o_K(1)) \frac{|A \cap bH|}{|G|} + \frac{\varepsilon}{10Y}$$

Combining (3.33), (3.37), and (3.36), we obtain that for Q sufficiently large,

$$|A \cap bH| \geq \left(\frac{\vartheta}{2} \frac{\pi(X; bH)}{X/\log X} - \frac{\varepsilon}{5Y} \right) |G|.$$

This proves (vi), and completes the proof. $\square$

4. PRODUCT SETS

This section converts the dense-model frameworks from Proposition 3.3 into product-set information inside the ray class group. Apart from the transference inputs proved in the previous section, the argument is purely finite-abelian-group theoretic and closely follows the product-set argument of Matomäki–Teräväinen [16].

Throughout this section, let $G = \text{Cl}_{\mathfrak{q}}^{(\infty)}$, $Q = N\mathfrak{q}$. For subsets $A, B \subseteq G$, we write

$$A \cdot B = \{ab : a \in A, b \in B\},$$

and for functions $f, g : G \rightarrow \mathbb{C}$, we write

$$(f * g)(c) := \sum_{\substack{a, b \in G \\ ab=c}} f(a)g(b).$$

We shall use the following elementary lower bound for convolutions of characteristic functions on finite abelian groups, which appears as [16, Lem. 3.4]. It may be viewed as a simple consequence of inclusion–exclusion and is useful for obtaining lower bounds on product sets inside cosets.

Lemma 4.1. [16, Lem. 3.4] *Let G be a finite abelian group.*

(i) *Let $A, B \subseteq G$ be nonempty subsets. Then, for every $c \in G$,*

$$1_A * 1_B(c) \geq |A| + |B| - |G|.$$

(ii) *Let $H \leq G$, let $a, b \in G$, and let $A \subseteq aH$ and $B \subseteq bH$. Then, for every $c \in abH$,*

$$1_A * 1_B(c) \geq |A| + |B| - |H|.$$

We shall also use Kneser's theorem; see, for example, [23, Thm. 5.5].

Lemma 4.2 (Kneser's theorem). *Let G be a finite abelian group, and let $A, B \subseteq G$. Let H be the stabilizer of $A \cdot B$. Then*

$$|A \cdot B| \geq |A \cdot H| + |B \cdot H| - |H| \geq |A| + |B| - |H|.$$

4.1. Reduction from popular products to product sets. We first use a popular-products lemma to reduce the problem to two cases where either many elements of the group G are represented in $A \cdot B$, or after deleting few elements one obtains a product set $A' \cdot B'$ whose elements all have many representations. We then use Kneser's theorem to analyze the latter case.

Lemma 4.3. [16, Lem. 6.1] *Let $t \geq u \geq 1$ be integers. Let A, B be finite subsets of a finite abelian group G , with $|A|, |B| \geq t$. Then at least one of the following holds.*

(a) *For at least*

$$|A| + |B| - 2t - \frac{u|G|}{t}$$

elements $c \in G$, we have

$$(1_A * 1_B)(c) \geq u.$$

(b) *There exist subsets $A' \subseteq A$ and $B' \subseteq B$ such that*

$$|A \setminus A'| + |B \setminus B'| \leq t - 1,$$

and for every $c \in A' \cdot B'$,

$$(1_A * 1_B)(c) \geq t.$$

Proof. See [16, Lem. 6.1]. □

We now combine the dense model set A with the popular-products lemma. The following proposition packages the criteria that will be used in the proofs of the two-prime and three-prime theorems.

Proposition 4.4 (Criteria on $E_2(X; \mathfrak{q})$ and $E_3(X; \mathfrak{q})$). *Let $\kappa, \varepsilon > 0$ and $C \geq 1$ be fixed, with $\varepsilon > 0$ sufficiently small. Assume*

$$X \in [Q^{\max(1, 2\alpha, 3\alpha_0) + \kappa}, Q^C].$$

Let ϑ and ϑ_0 be as in (3.6). Then, when Q is sufficiently large, at least one of the following holds.

(a) *We have*

$$|E_2(X; \mathfrak{q})| \geq (\vartheta - 3\varepsilon)|G|.$$

If moreover $\vartheta > \frac{2}{3} + 3\varepsilon$, then

$$E_3(X; \mathfrak{q}) = G.$$

(b) *There exist subsets $A', B' \subseteq G$ such that the following hold.*

(b.i) $|A'|, |B'| \geq (\frac{\vartheta}{2} - \frac{3\varepsilon}{2})|G|.$

(b.ii) $|(A' \cdot B') \cap E_2(X; \mathfrak{q})| \geq |A' \cdot B'| - \varepsilon|G|.$

(b.iii) *For every $c \in G$,*

$$(1_{A' \cdot B'} * 1_{A'})(c) \gg |G| \implies c \in E_3(X; \mathfrak{q}).$$

(b.iv) *Let $B := \frac{\log X}{\log Q}$ and $\beta \in (2\varepsilon, B]$. For every $c \in G$,*

$$\sum_{\substack{[p] \mid d=c, \\ Q^{\beta-\varepsilon} < Np \leq Q^\beta}} \frac{1}{Np} \gg 1 \implies c \in E_3(X; \mathfrak{q}).$$

(b.v) For any subgroup $H \leq G$ of index $|Y| \leq \varepsilon^{-1/2}$, there are at least $\left[\left(\frac{\vartheta_0}{2} - 3\varepsilon^{1/2}\frac{\vartheta_0}{\vartheta}\right)Y\right]$ distinct cosets bH of H such that

$$|A' \cap B' \cap bH| \geq \frac{\varepsilon}{2}|G|.$$

(b.vi) For any subgroup $H \leq G$ of index $|Y| \leq \varepsilon^{-1/2}$ and any coset $bH \subseteq G$,

$$|A' \cap B' \cap bH| \geq \left(\frac{\vartheta}{2} \frac{\pi(X; bH)}{X/\log X} - \frac{3\varepsilon}{10}\right)|G|.$$

where $\pi(X; bH)$ is defined in (3.5).

Proof. We may assume that $\varepsilon > 0$ is sufficiently small. Note that the condition $X \in [Q^{\max(1, 2\alpha, 3\alpha_0) + \kappa}, Q^C]$ ensures that the conditions on X in each part of Proposition 3.3 are satisfied.

Let $A \subseteq G$ be the set supplied by Proposition 3.3. Thus

$$(4.1) \quad |A| \geq \left(\frac{\vartheta}{2} - \varepsilon\right)|G|.$$

Let $M_\varepsilon \geq C_0\varepsilon^{-2}$ with $C_0 > 0$ chosen sufficiently large to absorb all rounding errors when applying Lemma 4.3 later. For example, we may take $C_0 = 2000$. We first handle the case $|G| < M_\varepsilon$. Since $|G|$ is then bounded in terms of ε , the logarithmic-saving thresholds and exceptional sets in Proposition 3.3 are, for Q sufficiently large, smaller than 1. Thus every element of $A \cdot A$ belongs to $E_2(X; \mathfrak{q})$, and every element of $A \cdot A \cdot A$ belongs to $E_3(X; \mathfrak{q})$. Taking $A' = B' = A$, conclusions (b.i)–(b.vi) follow directly from Proposition 3.3.

Hence we may assume $|G| \geq M_\varepsilon$. Let

$$t = \left\lceil \frac{\varepsilon|G|}{10} \right\rceil, \quad u = \left\lfloor \frac{\varepsilon^2|G|}{1000} \right\rfloor.$$

Then $u \geq \frac{\varepsilon^2|G|}{2000} \geq 1$ and $t \geq u$. As ε is sufficiently small, (4.1) gives $|A| \geq t$. We now apply Lemma 4.3 with t, u and both sets equal to A .

Case 1: Lemma 4.3(a) holds.

Then the set

$$\mathcal{C} := \{c \in G : (1_A * 1_A)(c) \geq u\}$$

has size

$$|\mathcal{C}| \geq 2|A| - 2t - \frac{u|G|}{t} > \left(\vartheta - \frac{5}{2}\varepsilon\right)|G|.$$

By Proposition 3.3(ii), all but $O(|G|(\log Q)^{-\varepsilon/2})$ of these elements in $\mathcal{C}$ belong to $E_2(X; \mathfrak{q})$. Absorbing this negligible exceptional set into the $\varepsilon|G|$ -term, we obtain

$$|E_2(X; \mathfrak{q})| \geq (\vartheta - 3\varepsilon)|G|.$$

This proves the first assertion of (a).

Assume now that $\vartheta > \frac{2}{3} + 3\varepsilon$. Then

$$|\mathcal{C}| \geq \left(\vartheta - \frac{5}{2}\varepsilon\right)|G| > \left(\frac{2}{3} + \frac{\varepsilon}{2}\right)|G|, \quad |A| \geq \left(\frac{\vartheta}{2} - \varepsilon\right)|G| > \left(\frac{1}{3} + \frac{\varepsilon}{2}\right)|G|.$$

Hence $|\mathcal{C}| + |A| > (1 + \varepsilon)|G|$. By Lemma 4.1(i), for every $c \in G$,

$$(1_{\mathcal{C}} * 1_A)(c) \geq |\mathcal{C}| + |A| - |G| > \varepsilon|G|.$$

But

$$(1_A * 1_A * 1_A)(c) = \sum_{d \in G} (1_A * 1_A)(d) 1_A(cd^{-1}) \geq u(1_{\mathcal{C}} * 1_A)(c) \geq \frac{\varepsilon^3}{2000}|G|^2.$$

Therefore Proposition 3.3(iii) implies that $c \in E_3(X; \mathfrak{q})$ for every $c \in G$. Hence $E_3(X; \mathfrak{q}) = G$. This completes the proof of (a).

Case 2: Lemma 4.3(b) holds.

Then there exist $A', B' \subseteq A$ such that

$$(4.2) \quad |A \setminus A'| + |A \setminus B'| \leq t - 1 \leq \frac{\varepsilon|G|}{10}.$$

We now verify the six claims in (b).

Proof of (b.i). Since $A', B' \subseteq A$,

$$|A'| \geq |A| - \frac{\varepsilon|G|}{10}, \quad |B'| \geq |A| - \frac{\varepsilon|G|}{10}.$$

Using (4.1), we get

$$|A'|, |B'| \geq \left(\frac{\vartheta}{2} - \frac{3\varepsilon}{2}\right)|G|.$$

Proof of (b.ii). By Lemma 4.3(b), for every $d \in A' \cdot B'$,

$$(4.3) \quad (1_A * 1_A)(d) \geq t \gg |G|.$$

Hence Proposition 3.3(ii) implies that, apart from at most $O(|G|(\log Q)^{-\varepsilon/2})$ exceptions, every element of $A' \cdot B'$ belongs to $E_2(X; \mathfrak{q})$. Since the exceptional set is $o(\varepsilon|G|)$, we get

$$|(A' \cdot B') \cap E_2(X; \mathfrak{q})| \geq |A' \cdot B'| - \varepsilon|G|.$$

Proof of (b.iii). For any $c \in G$, $(1_{A' \cdot B'} * 1_{A'})(c) = \sum_{d \in A' \cdot B'} 1_{A'}(cd^{-1})$. Using (4.3),

$$(1_A * 1_A * 1_A)(c) = \sum_{d \in G} (1_A * 1_A)(d) 1_A(cd^{-1}) \geq \sum_{d \in A' \cdot B'} (1_A * 1_A)(d) 1_{A'}(cd^{-1}) \geq t(1_{A' \cdot B'} * 1_{A'})(c).$$

Therefore, $(1_{A' \cdot B'} * 1_{A'})(c) \gg |G|$ implies $(1_A * 1_A * 1_A)(c) \geq t|G| \gg |G|^2$. By Proposition 3.3(iii), $c \in E_3(X; \mathfrak{q})$.

Proof of (b.iv). Let $\beta \in (2\varepsilon, B]$ and $L = \lfloor \frac{B}{\beta} \rfloor$. By (4.3), for every $c \in G$,

$$\sum_{\substack{[p]a_1a_2=c, \ a_1, a_2 \in A \\ Q^{\beta-\varepsilon} < Np \leq Q^\beta}} \frac{1}{Np} \geq \sum_{\substack{[p]d=c, \ d \in A' \cdot B' \\ Q^{\beta-\varepsilon} < Np \leq Q^\beta}} \frac{(1_A * 1_A)(d)}{Np} \gg |G| \sum_{\substack{[p]d=c, \ d \in A' \cdot B' \\ Q^{\beta-\varepsilon} < Np \leq Q^\beta}} \frac{1}{Np}.$$

Hence the hypothesis in (b.iv) implies the hypothesis of Proposition 3.3(iv), and thus $c \in E_3(X; \mathfrak{q})$.

Proof of (b.v). Let $H \leq G$ be a subgroup of index $Y \leq \varepsilon^{-1/2}$. By Proposition 3.3(v), Then there exist at least $\left\lceil \left(\frac{\vartheta_0}{2} - 3\varepsilon^{1/2} \frac{\vartheta_0}{\vartheta}\right)Y \right\rceil$ distinct cosets bH of H such that $|A \cap bH| > \varepsilon|G|$. Take one such coset bH . Using (4.2),

$$|A' \cap B' \cap bH| \geq |A \cap bH| - |A \setminus A'| - |A \setminus B'| > \varepsilon|G| - \frac{\varepsilon|G|}{10} > \frac{\varepsilon}{2}|G|.$$

Proof of (b.vi). By Proposition 3.3(vi), $|A \cap bH| \geq \left(\frac{\vartheta}{2} \frac{|\pi(X; bH)|}{X/\log X} - \frac{\varepsilon}{5Y}\right)|G|$. Using (4.2),

$$\begin{aligned} |A' \cap B' \cap bH| &\geq |A \cap bH| - |A \setminus A'| - |A \setminus B'| \\ &\geq \left(\frac{\vartheta}{2} \frac{|\pi(X; bH)|}{X/\log X} - \frac{\varepsilon}{5Y} - \frac{\varepsilon}{10}\right)|G| \\ &\geq \left(\frac{\vartheta}{2} \frac{|\pi(X; bH)|}{X/\log X} - \frac{3\varepsilon}{10}\right)|G|. \end{aligned}$$

This completes the proof. $\square$

4.2. Structure of sets with small doubling. The remaining lemma is a structural statement for sets with small product set. It is the finite-abelian-group form of the corresponding lemma of Matomäki–Teräväinen and follows from Kneser’s theorem.

Lemma 4.5. *Let G be a finite abelian group, and let $\alpha, \alpha', \beta \in (0, 1]$ satisfy $\beta < 2\alpha \leq 2\alpha'$. Let $A, B \subseteq G$ with $|A|, |B| \geq \alpha|G|$. Assume that each of A and B meets at least proportion α' of the cosets of every subgroup $H_0 \leq G$ with $[G : H_0] < \frac{1}{2\alpha-\beta}$. Then at least one of the following holds.*

- (a) $|A \cdot B| \geq \beta|G|$.
- (b) Let $H \leq G$ be the stabilizer of $A \cdot B$, and write $Y = [G : H]$. Then:
 - (b.i) $1 < Y < \frac{1}{2\alpha'-\beta}$.

- (b.ii) If $\alpha' > \frac{1}{3}, \beta \leq \frac{2}{3}$, then $Y = 3k + 2$ for some integer $k \geq 0$ satisfying $\frac{k+1}{3k+2} \geq \alpha'$, each of A and B meets exactly $k + 1$ cosets of H , and $A \cdot B$ is the union of exactly $2k + 1$ cosets of H .
- (b.iii) If, for some sufficiently small $\varepsilon > 0$, $\alpha' \geq \frac{3}{8} - \varepsilon$, $\beta < \frac{11}{16} - 2\varepsilon$, then $Y = 3k + 2$ for some $k \in \{0, 1, 2\}$, each of A and B meets exactly $k + 1$ cosets of H , and $A \cdot B$ is the union of exactly $2k + 1$ cosets of H .

Proof. This is the finite-abelian-group form of [16, Lemma 6.3]. Their proof uses only Kneser's theorem for finite abelian groups and elementary counting of cosets of the stabilizer of $A \cdot B$. It applies verbatim to any finite abelian group G . $\square$

5. PRIME IDEALS IN COSETS OF INDICES 5 AND 8

Let $G = \text{Cl}_{\mathfrak{q}}^{(\infty)}$, $Q = N\mathfrak{q}$. In order to deal with the case arising from Lemma 4.5, we need information on the distribution of prime ideals in cosets of subgroups of G of index $3k + 2$, for $k \in \{1, 2\}$.

We first record the Hecke-character analogue of the weighted prime-sum input used by Matomäki–Teräväinen.

Lemma 5.1. *Let $A_0 > 0$ be fixed, and define*

$$f_{A_0}(u) := \begin{cases} A_0 - u, & 0 \leq u \leq A_0, \\ 0, & u > A_0. \end{cases}$$

Let χ be a non-principal Hecke character modulo $\mathfrak{q}$ induced by the primitive character χ^ modulo $\mathfrak{f}_{\chi}$, suppose that χ^* satisfies*

$$(5.1) \quad L\left(\frac{1}{2} + it, \chi^*\right) \ll_{K, \varepsilon} C(\chi^*, t)^{\alpha_0/2 + \varepsilon},$$

where $C(\chi^, t)$ is the analytic conductor of χ^* , then*

$$(5.2) \quad \Re \sum_{\mathfrak{p} \nmid \mathfrak{q}} \chi(\mathfrak{p}) \frac{\log N\mathfrak{p}}{N\mathfrak{p}} f_{A_0}\left(\frac{\log N\mathfrak{p}}{\log Q}\right) \leq \left(\frac{A_0 \alpha_0}{2} + o_K(1)\right) \log Q.$$

For the principal character,

$$(5.3) \quad \sum_{\mathfrak{p} \nmid \mathfrak{q}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} f_{A_0}\left(\frac{\log N\mathfrak{p}}{\log Q}\right) = \left(\frac{A_0^2}{2} + o_K(1)\right) \log Q.$$

Proof. We use the number-field version of the weighted explicit-formula inequality due to Zaman [27, Prop. 6.2], which is analogous to Heath-Brown [11, Lem. 5.2]. Take $\vartheta = 1$ in Zaman's notation [27, (3.2), (3.3)], so that

$$\mathcal{L} = \log d_K + \log N\mathfrak{q} + n_K \nu(n_K) = \log Q + O_K(1), \quad \mathcal{L}_{\chi} = \log d_K + \log N\mathfrak{f}_{\chi} + n_K \nu(n_K) \leq \mathcal{L},$$

where $\nu : [1, \infty) \rightarrow [4, \infty)$ is any fixed increasing function with $\nu(x) \gg \log(x + 4)$.

We first check that [27, Lem. 4.2] is satisfied with $\phi = \alpha_0/2$. Let $\eta = (\log \mathcal{L})/\mathcal{L}$. Since χ^* is primitive and non-principal, $L(s, \chi^*)$ is entire of finite order. Let $s = \sigma + it$, where $\frac{1}{2} \leq \sigma \leq 1 + \eta$. By the assumed bound on the central line and the standard estimate $C(\chi^*, t) \ll_K d_K N\mathfrak{f}_{\chi} (1 + |t|)^{n_K}$, we have for any $\varepsilon_0 > 0$,

$$\left| L\left(\frac{1}{2} + it, \chi^*\right) \right| \ll_{K, \varepsilon} (d_K N\mathfrak{f}_{\chi})^{\alpha_0/2 + \varepsilon_0} (1 + |t|)^{n_K(\alpha_0/2 + \varepsilon_0)}.$$

On the other hand, on the line $\sigma = 1 + \eta$, the Euler product gives

$$|L(1 + \eta + it, \chi^*)| \leq \zeta_K(1 + \eta) \ll_K \eta^{-n_K}.$$

Writing $\ell(\sigma) = \frac{1 + \eta - \sigma}{1/2 + \eta}$ and applying the Phragmén–Lindelöf principle [20, Thm. 2] in the strip $\frac{1}{2} \leq \sigma \leq 1 + \eta$ to $L(s, \chi^*)$ gives

$$(5.4) \quad \log |L(s, \chi^*)| \leq \ell(\sigma) \left(\frac{\alpha_0}{2} + \varepsilon_0\right) (\log d_K + \log N\mathfrak{f}_{\chi}) + O_{K, \varepsilon_0}(\log(1 + |t|)) + O_{K, \varepsilon_0}(\log \mathcal{L}).$$

Take $\eta(x) = \log x/2 + 2$ in the notation of [27, (3.2)], we have

$$\mathcal{T} = (\mathcal{L} - n_K \nu(n_K))^{\frac{2 \log(n_K+1)}{\log n_K+4}} + \nu(n_K).$$

Now assume that $|t| \leq \mathcal{T}$, then $\log(1 + |t|) \ll_K \log \mathcal{L} = o_K(\mathcal{L})$.

Since

$$\ell(\sigma) = \frac{1 + \eta - \sigma}{1/2 + \eta} = 2(1 - \sigma) + O(\eta) \quad \text{and} \quad \log N \mathfrak{f}_\chi \leq \mathcal{L}_\chi,$$

by (5.4) and taking ε_0 sufficiently small in terms of ε , uniformly for $\frac{1}{2} \leq \sigma \leq 1 + \frac{\log \mathcal{L}}{\mathcal{L}}$, we have $|t| \leq \mathcal{T}$,

$$\log |L(s, \chi^*)| \leq \alpha_0 \mathcal{L}_\chi (1 - \sigma + \varepsilon) + o_{K, \varepsilon}(\mathcal{L}).$$

This is the input of [27, Lem. 4.2] with $\phi = \frac{\alpha_0}{2}$.

The weight f_{A_0} satisfies Condition 1 of [27, p. 335] with $x_0 = A_0$: it is continuous on $[0, \infty)$, vanishes for $u \geq A_0$, and is twice differentiable on $(0, A_0)$, with $f''_{A_0} = 0$. Applying [27, Prop. 6.2] to $L(s, \chi)$ at $s = 1$ with weight f_{A_0} and $\phi = \alpha_0/2$ gives that for any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that

$$(5.5) \quad \Re \sum_{\mathfrak{a}} \frac{\Lambda_K(\mathfrak{a}) \chi(\mathfrak{a})}{N \mathfrak{a}} f_{A_0} \left(\frac{\log N \mathfrak{a}}{\mathcal{L}} \right) \leq -\mathcal{L} \sum_{|1-\rho| \leq \delta} \Re F_{A_0}((1-\rho)\mathcal{L}) + A_0 \frac{\alpha_0}{2} \mathcal{L}_\chi + \varepsilon \mathcal{L},$$

where the sum is over the non-trivial zeros ρ of the primitive Hecke L -function $L(s, \chi^*)$ counted with multiplicity, and

$$F_{A_0}(z) := \int_0^{A_0} (A_0 - u) e^{-zu} du$$

is the Laplace transform of f_{A_0} .

We now check that the zero term in (5.5) is non-positive. First observe that F_{A_0} extends holomorphically to $z = 0$, with $F_{A_0}(0) = A_0^2/2$. Thus $\Re F_{A_0}$ is harmonic in the half-plane $\Re z > 0$ and continuous on its closure $\Re z \geq 0$. On the boundary $\Re z = 0$, say $z = iy$, we have

$$\Re F_{A_0}(iy) = \int_0^{A_0} (A_0 - u) \cos(yu) du = \frac{1 - \cos(A_0 y)}{y^2} \geq 0,$$

with the value at $y = 0$ understood by continuity.

Moreover, $F_{A_0}(z) = \frac{A_0 z - 1 + e^{-A_0 z}}{z^2}$, so $F_{A_0}(z) \rightarrow 0$ as $|z| \rightarrow \infty$ uniformly in the half-plane $\Re z \geq 0$. Applying the minimum principle [1, Cor. 1.9] to $\Re F_{A_0}(z)$ in the half-discs $\{z : \Re z \geq 0, |z| \leq R\}$, and then letting $R \rightarrow \infty$, gives

$$\Re F_{A_0}(z) \geq 0 \quad (\Re z \geq 0).$$

Since every non-trivial zero satisfies $\Re \rho \leq 1$, we have $\Re((1-\rho)\mathcal{L}) \geq 0$. Therefore

$$\Re F_{A_0}((1-\rho)\mathcal{L}) \geq 0,$$

and hence the zero term in (5.5) is non-positive and may be discarded. Therefore, since $\mathcal{L}_\chi \leq \mathcal{L}$, we have

$$\Re \sum_{\mathfrak{a}} \frac{\Lambda_K(\mathfrak{a}) \chi(\mathfrak{a})}{N \mathfrak{a}} f_{A_0} \left(\frac{\log N \mathfrak{a}}{\mathcal{L}} \right) \leq \left(\frac{A_0 \alpha_0}{2} + \varepsilon \right) \mathcal{L}.$$

As $\mathcal{L} = \log Q + O_K(1)$, and $\varepsilon > 0$ is arbitrary, it follows that

$$(5.6) \quad \Re \sum_{\mathfrak{a}} \frac{\Lambda_K(\mathfrak{a}) \chi(\mathfrak{a})}{N \mathfrak{a}} f_{A_0} \left(\frac{\log N \mathfrak{a}}{\log Q} \right) \leq \left(\frac{A_0 \alpha_0}{2} + o(1) \right) \log Q.$$

Moreover, the contribution of prime powers $\mathfrak{p}^m$, $m \geq 2$, is $O_K(1)$. Thus (5.6) yields (5.2).

We now prove the principal-character asymptotic. By Mertens' theorem for prime ideals [21, Lem. 2.3],

$$\sum_{N \mathfrak{p} \leq y} \frac{\log N \mathfrak{p}}{N \mathfrak{p}} = \log y + O_K(1).$$

Partial summation gives

$$\sum_{\mathfrak{p}} \frac{\log N \mathfrak{p}}{N \mathfrak{p}} f_{A_0} \left(\frac{\log N \mathfrak{p}}{\log Q} \right) = \log Q \int_0^{A_0} (A_0 - u) du + O_{A_0, K}(1) = \frac{A_0^2}{2} \log Q + O_{A_0, K}(1).$$

Moreover, the prime ideals dividing $\mathfrak{q}$ contribute only $o_K(\log Q)$ since $\sum_{\mathfrak{p}|\mathfrak{q}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} \ll_K (\log Q)^{1/2}$; see for example [27, Lem. 2.4]. This proves (5.3). $\square$

We now assume a bounded-order subconvexity input in order to apply Lemma 5.1. For every fixed integer $\ell \geq 2$ and every $\varepsilon > 0$, every primitive non-principal Hecke character χ^* of order at most ℓ satisfies

$$(L^b(\alpha_0)) \quad L\left(\frac{1}{2} + it, \chi^*\right) \ll_{K, \ell, \varepsilon} C(\chi^*, t)^{\alpha_0/2 + \varepsilon} \quad (t \in \mathbb{R}),$$

where $C(\chi^*, t)$ denotes the analytic conductor of χ^* .

Note that the same proof of Proposition 2.4 shows that $(L^b(\alpha_0))$ implies $(CS^b(\alpha_0))$.

Lemma 5.2. *Assume $(L^b(\alpha_0))$. Let $k \in \{1, 2\}$, and let $H \leq G$ be a subgroup of index $3k + 2$. Suppose that G/H is cyclic and generated by gH . Put $A_0 = 4\alpha_0$. Then*

$$(5.7) \quad \sum_{\substack{N\mathfrak{p} \leq Q^{A_0}, \mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \notin \bigcup_{j=1}^{k+1} g^{k+j}H}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} \gg_{K, \alpha_0} \log Q.$$

Proof. Put $A_0 = 4\alpha_0$, and define

$$w(\mathfrak{p}) := \frac{\log N\mathfrak{p}}{N\mathfrak{p}} f_{A_0}\left(\frac{\log N\mathfrak{p}}{\log Q}\right), \quad W := \sum_{\mathfrak{p} \nmid \mathfrak{q}} w(\mathfrak{p}),$$

where f_{A_0} is as defined in Lemma 5.1.

Lemma 5.1 applied to the principal character gives

$$(5.8) \quad W = \left(\frac{A_0^2}{2} + o_K(1)\right) \log Q.$$

For every non-principal character χ of G of bounded order, using $(L^b(\alpha_0))$ in Lemma 5.1 gives

$$\Re \sum_{\mathfrak{p} \nmid \mathfrak{q}} \chi(\mathfrak{p}) w(\mathfrak{p}) \leq \left(\frac{A_0 \alpha_0}{2} + o_K(1)\right) \log Q.$$

Since $A_0 = 4\alpha_0$, this may be rewritten using (5.8) as

$$(5.9) \quad \Re \sum_{\mathfrak{p} \nmid \mathfrak{q}} \chi(\mathfrak{p}) w(\mathfrak{p}) \leq \left(\frac{1}{4} + o_K(1)\right) W.$$

Since $0 \leq f_{A_0} \leq A_0$, we have $w(\mathfrak{p}) \leq A_0 \frac{\log N\mathfrak{p}}{N\mathfrak{p}}$. It is enough to show that the weighted sum outside the exceptional block of cosets is at least a fixed positive proportion of W , say $10^{-3}W$. That is, it suffices to show that

$$(5.10) \quad \sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \notin \bigcup_{j=1}^{k+1} g^{k+j}H}} w(\mathfrak{p}) \geq 10^{-3}W.$$

We now prove this estimate by contradiction.

First suppose that G/H is cyclic of order 5. Let gH generate G/H , and suppose for contradiction that

$$\sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \notin g^2H \cup g^3H}} w(\mathfrak{p}) < 10^{-3}W.$$

Define $\beta_0, \beta_1, \beta_2 \geq 0$ by

$$\beta_0 W := \sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \in H}} w(\mathfrak{p}), \quad \beta_1 W := \sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \in gH \cup g^4H}} w(\mathfrak{p}), \quad \beta_2 W := \sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \in g^2H \cup g^3H}} w(\mathfrak{p}).$$

Then $\beta_0 + \beta_1 + \beta_2 = 1$, $\beta_0 + \beta_1 < 10^{-3}$, and hence $\beta_2 > 1 - 10^{-3}$.

Let χ be the character of G induced from G/H by

$$\chi(g) = e^{2\pi i \cdot 2/5}, \quad \chi(h) = 1 \quad (h \in H).$$

Then

$$\Re \sum_{\mathfrak{p} \nmid \mathfrak{q}} \chi(\mathfrak{p}) w(\mathfrak{p}) = \left(\beta_0 + \beta_1 \cos \frac{4\pi}{5} + \beta_2 \cos \frac{2\pi}{5} \right) W \geq \left(10^{-3} \cos \frac{4\pi}{5} + (1 - 10^{-3}) \cos \frac{2\pi}{5} \right) W.$$

The constant on the right is ≈ 0.3079 which is strictly larger than $1/4$. This contradicts (5.9) for Q sufficiently large. Therefore

$$(5.11) \quad \sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \notin g^2 H \cup g^3 H}} w(\mathfrak{p}) \geq 10^{-3} W.$$

We now suppose that G/H is cyclic of order 8. Let gH generate G/H , and suppose for contradiction that

$$\sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \notin g^3 H \cup g^4 H \cup g^5 H}} w(\mathfrak{p}) < 10^{-3} W.$$

Define $\beta_0, \dots, \beta_4 \geq 0$ by

$$\beta_0 W := \sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \in H}} w(\mathfrak{p}), \quad \beta_4 W := \sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \in g^4 H}} w(\mathfrak{p}), \quad \beta_i W := \sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \in g^i H \cup g^{8-i} H}} w(\mathfrak{p}), \quad (i = 1, 2, 3).$$

Then

$$(5.12) \quad \sum_{j=0}^4 \beta_j = 1, \quad \beta_0 + \beta_1 + \beta_2 < 10^{-3}.$$

Let χ be the character induced from G/H by

$$\chi(g) = e^{2\pi i/8}, \quad \chi(h) = 1 \quad (h \in H).$$

Applying (5.9) to the non-principal character χ^2 and χ^3 , we get

$$(5.13) \quad \Re \sum_{\mathfrak{p} \nmid \mathfrak{q}} \chi(\mathfrak{p})^2 w(\mathfrak{p}) \leq \left(\frac{1}{4} + o_K(1) \right) W, \quad \Re \sum_{\mathfrak{p} \nmid \mathfrak{q}} \chi(\mathfrak{p})^3 w(\mathfrak{p}) \leq \left(\frac{1}{4} + o_K(1) \right) W$$

Since $\Re \chi(g^j)^2 = \cos \frac{\pi j}{2}$, we have

$$\Re \sum_{\mathfrak{p} \nmid \mathfrak{q}} \chi(\mathfrak{p})^2 w(\mathfrak{p}) = (\beta_0 - \beta_2 + \beta_4) W.$$

Therefore, since $\beta_2 < 10^{-3}$,

$$(5.14) \quad \beta_4 \leq \frac{1}{4} + \beta_2 + o_K(1) \leq \frac{1}{4} + 10^{-3} + o_K(1).$$

Combining (5.12) and (5.14), we obtain

$$(5.15) \quad \beta_3 = 1 - \beta_0 - \beta_1 - \beta_2 - \beta_4 \geq \frac{3}{4} - 2 \cdot 10^{-3} - o_K(1).$$

Since $\Re \chi(g^j)^3 = \cos \frac{3\pi j}{4}$, we have

$$\Re \sum_{\mathfrak{p} \nmid \mathfrak{q}} \chi(\mathfrak{p})^3 w(\mathfrak{p}) = \left(\beta_0 - \frac{\beta_1}{\sqrt{2}} + \frac{\beta_3}{\sqrt{2}} - \beta_4 \right) W.$$

Using (5.12), (5.14), and (5.15), this is at least

$$\left(-\frac{10^{-3}}{\sqrt{2}} + \frac{3/4 - 2 \cdot 10^{-3}}{\sqrt{2}} - \frac{1}{4} - 10^{-3} - o_K(1) \right) W.$$

The constant in the parentheses is ≈ 0.2772 which is strictly larger than $1/4$. This contradicts (5.13) for Q sufficiently large. Therefore

$$(5.16) \quad \sum_{\substack{\mathfrak{p} \nmid \mathfrak{q} \\ [\mathfrak{p}] \notin g^3 H \cup g^4 H \cup g^5 H}} w(\mathfrak{p}) \geq 10^{-3} W.$$

This finishes the proof of (5.10) by (5.11) and (5.16). $\square$

Corollary 5.3. *Under the hypotheses of Lemma 5.2, let $\varepsilon > 0$ be fixed and sufficiently small. Then there exists $\beta \in (2\varepsilon, 4\alpha_0]$ such that*

$$\sum_{\substack{Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta \\ [p] \notin \bigcup_{j=1}^{k+1} g^{k+j}H}} \frac{1}{N\mathfrak{p}} \gg_{K, \alpha_0, \varepsilon} 1.$$

Proof. The contribution in the range $N\mathfrak{p} \leq Q^{2\varepsilon}$ in (5.7) is $O_K(\varepsilon \log Q)$. Choosing $\varepsilon > 0$ sufficiently small, Lemma 5.2 gives

$$\sum_{\substack{Q^{2\varepsilon} < N\mathfrak{p} \leq Q^{4\alpha_0} \\ [p] \notin \bigcup_{j=1}^{k+1} g^{k+j}H}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} \gg_{K, \alpha_0, \varepsilon} \log Q.$$

Cover the interval $(Q^{2\varepsilon}, Q^{4\alpha_0}]$ by $O_{\alpha_0, \varepsilon}(1)$ logarithmic intervals of the form $(Q^{\beta-\varepsilon}, Q^\beta]$. Hence there exists at least one $\beta \in (2\varepsilon, 4\alpha_0]$ such that

$$\sum_{\substack{Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta \\ [p] \notin \bigcup_{j=1}^{k+1} g^{k+j}H}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} \gg_{K, \alpha_0, \varepsilon} \log Q.$$

Since $\log N\mathfrak{p} \asymp_{\alpha_0, \varepsilon} \log Q$ throughout this interval, the desired estimate follows. $\square$

6. REPRESENTATION BY THREE PRIME IDEALS

In this section, we prove Theorem 1.1, using exceptional quadratic estimates that will be established in the next section. Throughout, we write $G = \text{Cl}_{\mathfrak{q}}^{(\infty)}$, $Q = N\mathfrak{q}$, and assume that $(\text{CS}(\alpha))$ and $(\text{L}^b(\alpha_0))$ hold for fixed parameters $0 < \alpha_0 \leq \alpha < 1$.

Proof of Theorem 1.1. Let $\kappa > 0$ be fixed. We want to prove that if $X \geq Q^{\max(1, 3\alpha, 4\alpha_0) + \kappa}$, then $E_3(X; \mathfrak{q}) = G$. Let $\varepsilon > 0$ be fixed that is taken to be sufficiently small. Set

$$\vartheta = 1 - \varepsilon - \alpha \frac{\log Q}{\log X}, \quad \vartheta_0 = 1 - \varepsilon - \alpha_0 \frac{\log Q}{\log X}.$$

It suffices to take $X = Q^{\max(1, 3\alpha, 4\alpha_0) + \kappa}$ and we will apply Proposition 4.4.

After decreasing ε if necessary, we may fix a small constant $\eta > 0$, say $0 < \eta < 10^{-3}$, such that

$$(6.1) \quad \frac{\vartheta}{2} - \frac{3\varepsilon}{2} > \frac{1}{3} + \eta$$

and

$$(6.2) \quad \frac{\vartheta_0}{2} - 3\varepsilon^{1/2} \frac{\vartheta_0}{\vartheta} > \frac{3}{8} - \eta.$$

If Proposition 4.4(a) holds, then by (6.1) we have $\vartheta > \frac{2}{3} + 3\varepsilon$, so we immediately obtain $E_3(X; \mathfrak{q}) = G$.

We may therefore assume that Proposition 4.4(b) holds. Let $A', B' \subseteq G$ be the sets supplied there. By Proposition 4.4(b.i) and (6.1),

$$(6.3) \quad |A'|, |B'| \geq \left(\frac{1}{3} + \eta\right)|G|.$$

We apply Lemma 4.5 with

$$\alpha_* = \frac{1}{3} + \eta, \quad \alpha'_* = \frac{3}{8} - \eta, \quad \beta_* = \frac{2}{3}.$$

These parameters satisfy $\beta_* < 2\alpha_* \leq 2\alpha'_*$. Moreover, for ε sufficiently small, $\frac{1}{2\alpha_* - \beta_*} = (2\eta)^{-1} \leq \varepsilon^{-1/2}$. Hence, Proposition 4.4(b.v) together with (6.2) shows that A' and B' each meet at least proportion $\alpha'_* = \frac{3}{8} - \eta$ of the cosets of every subgroup $H_0 \leq G$ with $[G : H_0] < \frac{1}{2\alpha_* - \beta_*}$. Indeed, it gives at least α'_* proportion of cosets bH_0 satisfying $|A' \cap B' \cap bH_0| \geq \frac{\varepsilon}{2}|G|$, and each such coset is met by both A' and B' .

If Lemma 4.5(a) holds, then $|A' \cdot B'| \geq \frac{2}{3}|G|$. Together with (6.3), this gives

$$|A' \cdot B'| + |A'| \geq (1 + \eta)|G|.$$

Hence, by the convolution lower bound from Lemma 4.1, for any $c \in G$,

$$(1_{A' \cdot B'} * 1_{A'})(c) \geq \eta |G|.$$

Proposition 4.4(b.iii) then gives $c \in E_3(X; \mathfrak{q})$ and therefore $E_3(X; \mathfrak{q}) = G$.

We may therefore assume that Lemma 4.5(b) holds. Let H be the stabilizer of $A' \cdot B'$. By Lemma 4.5(b.i)(b.ii), $[G : H] < \frac{1}{2\alpha'_* - \beta_*}$, and there exists an integer $k \geq 0$ with $\frac{k+1}{3k+2} \geq \alpha'_*$ such that $[G : H] = 3k + 2$, each of A' and B' meets exactly $k + 1$ cosets of H , and $A' \cdot B'$ is the union of exactly $2k + 1$ cosets of H .

We next show that in fact $k \in \{0, 1, 2\}$. By Proposition 4.4(b.v) and (6.2), the sets A' and B' each intersects more than $(\frac{3}{8} - \eta)[G : H]$ cosets of H . Since each of A' and B' meets exactly $k + 1$ cosets of H , we get $k + 1 \geq (\frac{3}{8} - \eta)(3k + 2)$, which forces $k \in \{0, 1, 2\}$.

Moreover, the same estimate shows that the $k + 1$ cosets met by A' and the $k + 1$ cosets met by B' are the same. Indeed, the number of cosets in which $A' \cap B'$ is non-empty is at least

$$\left\lceil \left(\frac{3}{8} - \eta \right) (3k + 2) \right\rceil = k + 1 \quad (k = 0, 1, 2).$$

Let $a_1H, \dots, a_{k+1}H$ be the H -cosets met by A' and B' . Then, $A', B' \subseteq S := \bigcup_{j=1}^{k+1} a_jH$ and $A' \cdot B' \subseteq S^2$. Moreover, since A' and B' meet every H -coset contained in S , the product $A' \cdot B'$ meets every H -coset contained in S^2 . Since H is the stabilizer of $A' \cdot B'$, the set $A' \cdot B'$ is a union of H -cosets. Therefore every H -coset contained in S^2 is contained in $A' \cdot B'$, and so $A' \cdot B' = S^2$. Since $A' \cdot B' = S^2$ is the union of exactly $2k + 1$ H -cosets, denote these cosets by $b_1H, \dots, b_{2k+1}H$. Then, $A' \cdot B' = \bigcup_{j=1}^{2k+1} b_jH = \left(\bigcup_{j=1}^{k+1} a_jH \right)^2$. Furthermore, Proposition 4.4(b.v) gives

$$|A' \cap a_iH| \geq |A' \cap B' \cap a_iH| \geq \frac{\varepsilon}{2} |G| \quad (1 \leq i \leq k + 1).$$

We claim that

$$(6.4) \quad \left(\bigcup_{i=1}^{k+1} a_iH \right) \cdot \left(\bigcup_{j=1}^{2k+1} b_jH \right) \subseteq E_3(X; \mathfrak{q}).$$

Indeed, let $c \in a_iH \cdot b_jH$. For every $y \in A' \cap a_iH$, $d := y^{-1}c \in b_jH \subseteq A' \cdot B'$, and $cd^{-1} = y \in A'$. Therefore

$$(1_{A' \cdot B'} * 1_{A'})(c) = \sum_{d \in A' \cdot B'} 1_{A'}(cd^{-1}) \geq |A' \cap a_iH| \geq \frac{\varepsilon}{2} |G|.$$

By Proposition 4.4(b.iii), we obtain $c \in E_3(X; \mathfrak{q})$. This proves (6.4).

We now split into two cases.

Case 1: $k \in \{1, 2\}$, or $k = 0$ and $a_1 \in H$.

Assume first that $\bigcup_{j=1}^{k+1} a_jH$ and $\bigcup_{j=1}^{2k+1} b_jH$ are not complements of each other. Since there are exactly $3k + 2$ cosets of H in G , there exists a coset b_0H which intersects neither union.

Since $2 < 3\vartheta_0 < 3$, the lower bound in 3.1 with $\gamma = 1/3$ applies and gives a positive lower bound

$$\begin{aligned} \mathcal{N}_{1/3}(X; b_0, H) &:= \# \left\{ \mathfrak{n} \subset \mathcal{O}_K : \begin{array}{l} N\mathfrak{n} \leq X, [\mathfrak{n}] \in b_0H, \\ N\mathfrak{p} > X^{1/3} \text{ for every prime ideal } \mathfrak{p} \mid \mathfrak{n} \end{array} \right\} \\ &\geq (1 + o_K(1)) \frac{2 \log(3\vartheta_0 - 1)}{Y\vartheta_0} \frac{X}{\log X}. \end{aligned}$$

Since b_0H lies outside $\bigcup_{j=1}^{k+1} a_jH$, Proposition 4.4(b.vi) implies that

$$\pi(X; b_0H) := \# \{ \mathfrak{p} \subset \mathcal{O}_K : \mathfrak{p} \nmid \mathfrak{q}, N\mathfrak{p} \leq X, [\mathfrak{p}] \in b_0H \} = O(\varepsilon X / \log X).$$

Taking $\varepsilon > 0$ sufficiently small, we conclude that a positive proportion of the ideals $\mathfrak{n}$ counted in $\mathcal{N}_{1/3}(X; b_0, H)$ are products of exactly two prime ideals. Each such ideal $\mathfrak{n}$ may be written as $\mathfrak{n} = \mathfrak{p}\mathfrak{q}$ with $X^{1/3} < N\mathfrak{p}$, $N\mathfrak{q} \leq X^{2/3}$. Since b_0H is disjoint from $\left(\bigcup_{j=1}^{k+1} a_jH \right)^2$, the two prime factors

cannot both have classes in $\bigcup_{j=1}^{k+1} a_j H$. Hence at least one of them, say $\mathfrak{p}$, satisfies $[\mathfrak{p}] \notin \bigcup_{j=1}^{k+1} a_j H$. Thus the number of such ideals is at most the number of pairs $(\mathfrak{p}, \mathfrak{q})$ with

$$X^{1/3} < N\mathfrak{p} \leq X^{2/3}, \quad [\mathfrak{p}] \notin \bigcup_{j=1}^{k+1} a_j H, \quad N\mathfrak{q} \leq \frac{X}{N\mathfrak{p}}.$$

For fixed $\mathfrak{p}$, the number of possible $\mathfrak{q}$ is $\ll_K \frac{X}{N\mathfrak{p} \log X}$, since $X/N\mathfrak{p} \in [X^{1/3}, X^{2/3}]$. Therefore

$$\frac{X}{\log X} \ll_{K,\varepsilon} \sum_{\substack{X^{1/3} < N\mathfrak{p} \leq X^{2/3} \\ [\mathfrak{p}] \notin \bigcup_{j=1}^{k+1} a_j H}} \frac{X}{N\mathfrak{p} \log X}.$$

Hence

$$\sum_{\substack{X^{1/3} < N\mathfrak{p} \leq X^{2/3} \\ [\mathfrak{p}] \notin \bigcup_{j=1}^{k+1} a_j H}} \frac{1}{N\mathfrak{p}} \gg_{K,\varepsilon} 1.$$

Let $B := \frac{\log X}{\log Q}$. By dyadic decomposition, there exists $\beta \in [\frac{B}{3}, \frac{2B}{3}]$ and a coset $a_0 H$ outside $\bigcup_{j=1}^{k+1} a_j H$ such that

$$(6.5) \quad \sum_{\substack{Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta \\ [\mathfrak{p}] \in a_0 H}} \frac{1}{N\mathfrak{p}} \gg_{K,\varepsilon} 1.$$

Since $B \geq 1 + \kappa$, this β lies in the admissible range $(2\varepsilon, B]$ of Proposition 4.4(b.iv).

Next we consider the case that the two unions are complements. If $k = 0$ and $a_1 \in H$, this cannot happen. Thus we may assume $k \in \{1, 2\}$, corresponding to $[G : H] = 5$ and $[G : H] = 8$. The quotient G/H is clearly cyclic when $[G : H] = 5$. In the case $[G : H] = 8$, the quotient G/H is also cyclic in the complementary situation.¹ A direct check in the cyclic groups of orders 5 and 8 shows that $S = \bigcup_{j=1}^{k+1} g^{k+j} H$ for some generator gH of G/H . Therefore, by Corollary 5.3, there exist $\beta \in (2\varepsilon, 4\alpha_0]$ and a coset $a_0 H$ outside $\bigcup_{j=1}^{k+1} a_j H$ such that

$$(6.6) \quad \sum_{\substack{Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta \\ [\mathfrak{p}] \in a_0 H}} \frac{1}{N\mathfrak{p}} \gg_{K,\varepsilon} 1.$$

Since $B > 4\alpha_0$, this β also lies in the admissible range of Proposition 4.4(b.iv).

In either subcase, let $c \in a_0 H \cdot (\bigcup_{j=1}^{2k+1} b_j H)$. Then $c \in a_0 H \cdot b_j H$ for some j . Hence, for every prime ideal $\mathfrak{p}$ with $[\mathfrak{p}] \in a_0 H$, we have $[\mathfrak{p}]^{-1} c \in b_j H \subseteq A' \cdot B'$. Therefore, either (6.5) or (6.6) gives

$$\sum_{\substack{[\mathfrak{p}] d = c, d \in A' \cdot B' \\ Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta}} \frac{1}{N\mathfrak{p}} \gg_{K,\varepsilon} 1$$

with β in the admissible range of Proposition 4.4(b.iv). By Proposition 4.4(b.iv),

$$(6.7) \quad a_0 H \cdot \left(\bigcup_{j=1}^{2k+1} b_j H \right) \subseteq E_3(X; \mathfrak{q}).$$

Combining (6.4) and (6.7), we have

$$\left(\left(\bigcup_{i=1}^{k+1} a_i H \right) \cup a_0 H \right) \cdot \left(\bigcup_{j=1}^{2k+1} b_j H \right) \subseteq E_3(X; \mathfrak{q}).$$

¹Let $\Gamma = G/H$ and write $S = \{x, y, z\}$. Then $|S^2| = 5$ and $S \cap S^2 = \emptyset$. If $\Gamma \simeq C_2^3$, then $|S^2| \leq 4$, impossible. If $\Gamma \simeq C_4 \times C_2$, the square map has image of size 2, so two of x^2, y^2, z^2 are equal, say $x^2 = y^2$. Since $|S^2| = 5$, this is the only repetition among the six unordered products. Thus $S^2 = \{x^2, z^2, xy, xz, yz\}$. Multiplying the elements of $\Gamma = S \sqcup S^2$ gives $1 = (xyz)(x^2 z^2 xy xzyz) = xyzx^2$, where we used $x^2 = y^2$. Hence $yz = x$, contradicting $S \cap S^2 = \emptyset$.

Since a_0H is outside $\bigcup_{i=1}^{k+1} a_iH$, the two factors contain $k+2$ and $2k+1$ cosets of H , respectively. Hence

$$\left| \left(\bigcup_{i=1}^{k+1} a_iH \right) \cup a_0H \right| + \left| \bigcup_{j=1}^{2k+1} b_jH \right| = (3k+3)|H| > |G|.$$

Therefore their product is all of G . It follows that $E_3(X; \mathfrak{q}) = G$.

Case 2: $k = 0$ and $a_1 \notin H$.

In this case, H has index 2, and $A', B' \subseteq a_1H$, $A' \cdot B' = H$. Let ψ be the quadratic character of G with kernel H . Thus $\psi(\mathfrak{p}) = 1$ if and only if $[\mathfrak{p}] \in H$.

By (6.4), we already know that $a_1H \subseteq E_3(X; \mathfrak{q})$. It remains to prove that $H \subseteq E_3(X; \mathfrak{q})$. We will need Lemma 7.6 and Corollary 7.10 from Section 7.

Let

$$B_0 := \max(1, 3\alpha, 4\alpha_0) + \frac{\kappa}{2}.$$

Then $Q^{B_0} < X$, and $B_0 > 2\alpha + \alpha_0$, $B_0 > 2\alpha_0$. Thus, we may choose

$$\beta_* \in (\max\{2\alpha, B_0/2\}, B_0 - \alpha_0)$$

and then choose $c_* > 0$ sufficiently small so that $2\alpha + 2c_*\beta_* < B_0$. After decreasing ε if necessary, we may assume $\beta_* > 2\alpha + 5\varepsilon$ and $\beta_* + 2\varepsilon < B_0 - \alpha_0$.

We apply Lemma 7.6 with these parameters. Suppose first that Lemma 7.6(i) holds. Then

$$\sum_{\substack{Q^{\alpha+\varepsilon} < N\mathfrak{p} \leq Q^{\beta_*} \\ \psi(\mathfrak{p})=1}} \frac{1}{N\mathfrak{p}} \geq c_*.$$

By decomposing the interval $(Q^{\alpha+\varepsilon}, Q^{\beta_*}]$ into dyadic intervals of the form $(Q^{\beta-\varepsilon}, Q^{\beta}]$, we find some $\beta \in (2\varepsilon, \beta_*]$ such that

$$(6.8) \quad \sum_{\substack{Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^{\beta} \\ \psi(\mathfrak{p})=1}} \frac{1}{N\mathfrak{p}} \gg_{\varepsilon} 1.$$

Since $\beta_* < B_0 < B$, this β lies in the admissible range $(2\varepsilon, B]$ of Proposition 4.4(b.iv). For any $c \in H$, if $\psi(\mathfrak{p}) = 1$, then $[\mathfrak{p}] \in H$ and hence $[\mathfrak{p}]^{-1}c \in H = A' \cdot B'$. Therefore (6.8) implies

$$\sum_{\substack{[\mathfrak{p}]d=c, d \in A' \cdot B' \\ Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^{\beta}}} \frac{1}{N\mathfrak{p}} \geq \sum_{\substack{Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^{\beta} \\ \psi(\mathfrak{p})=1}} \frac{1}{N\mathfrak{p}} \gg_{\varepsilon} 1.$$

By Proposition 4.4(b.iv), we have $c \in E_3(X; \mathfrak{q})$. Since c was arbitrary, $H \subseteq E_3(X; \mathfrak{q})$.

Suppose now that Lemma 7.6(ii) holds. Then there exists $M \in [Q^{\beta_*}, Q^{B_0}]$ such that

$$(6.9) \quad \#\{\mathfrak{p} : M < N\mathfrak{p} \leq 2M, \psi(\mathfrak{p}) = 1\} \gg_{K, \varepsilon} M \rho_K L(1, \psi) \mathcal{V}_{\psi}(Q).$$

By the choice of B_0 and β_* , M lies in the admissible range of Corollary 7.10.

Similarly, for every $c \in H$, if $\psi(\mathfrak{p}) = 1$, then $[\mathfrak{p}]^{-1}c \in H = A' \cdot B'$, and so (6.9) gives

$$\sum_{\substack{[\mathfrak{p}]d=c, M < N\mathfrak{p} \leq 2M \\ \psi(\mathfrak{p})=1, d \in A' \cdot B'}} 1 \gg_{K, \varepsilon} M \rho_K L(1, \psi) \mathcal{V}_{\psi}(Q).$$

Corollary 7.10 then gives $c \in E_3(X; \mathfrak{q})$. Again $c \in H$ was arbitrary, so $H \subseteq E_3(X; \mathfrak{q})$. $\square$

7. THE EXCEPTIONAL QUADRATIC CASE

Let ψ be a quadratic Hecke character modulo $\mathfrak{q}$. We write

$$(1 * \psi)(\mathfrak{a}) := \sum_{\mathfrak{d}|\mathfrak{a}} \psi(\mathfrak{d}),$$

which is nonnegative on integral ideals.

As in Section 6, we assume throughout $(\mathbf{CS}(\alpha))$ and $(\mathbf{L}^b(\alpha_0))$ for fixed parameters $0 < \alpha_0 \leq \alpha < 1$. Note that $(\mathbf{L}^b(\alpha_0))$ implies $(\mathbf{CS}^b(\alpha_0))$.

We shall use the ideal-theoretic von Mangoldt function

$$(7.1) \quad \Lambda_K(\mathfrak{a}) := \begin{cases} \log N\mathfrak{p}, & \mathfrak{a} = \mathfrak{p}^m, \\ 0, & \text{otherwise,} \end{cases}$$

so that

$$\sum_{\mathfrak{d}|\mathfrak{a}} \Lambda_K(\mathfrak{d}) = \log N\mathfrak{a}.$$

Finally, define

$$\mathcal{V}_\psi(Q) := \prod_{\mathfrak{p}|\mathfrak{q}} \left(1 - \frac{1}{N\mathfrak{p}}\right) \prod_{\substack{2 < N\mathfrak{p} \leq Q \\ \psi(\mathfrak{p})=1}} \left(1 - \frac{2}{N\mathfrak{p}}\right).$$

We shall use the standard beta-sieve fundamental lemma in sieve dimension $\varkappa$; see [7, Lem. 6.8]. Although it is often stated for rational integers, its proof is purely combinatorial and depends only on unique factorization into primes. Since the nonzero integral ideals of K form a free commutative monoid generated by the prime ideals, the same argument applies verbatim to ideals, with the norm $N\mathfrak{d}$ replacing the integer d .

Lemma 7.1 (Fundamental lemma of the sieve over ideals). *Let $\varkappa \geq 1$ be fixed. Let $z \geq 2$, put $\mathcal{P}(z) := \prod_{N\mathfrak{p} < z} \mathfrak{p}$, and let $D = z^s$, where $s \geq 9\varkappa + 1$. There exist coefficients $\lambda_{\mathfrak{d}}^\pm$, supported on squarefree ideals $\mathfrak{d} \mid \mathcal{P}(z)$ with $N\mathfrak{d} \leq D$, such that $|\lambda_{\mathfrak{d}}^\pm| \leq 1$ and the following hold.*

(i) *For every integral ideal $\mathfrak{n}$,*

$$\sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^- \leq 1_{(\mathfrak{n}, \mathcal{P}(z))=1} \leq \sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+.$$

(ii) *Let h be a multiplicative function on squarefree ideals, with $0 \leq h(\mathfrak{p}) < 1$. Assume that for some $K_0 \geq 1$,*

$$\prod_{w \leq N\mathfrak{p} < y} (1 - h(\mathfrak{p}))^{-1} \leq K_0 \left(\frac{\log y}{\log w} \right)^\varkappa$$

for all $2 \leq w \leq y$. Then

$$\sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_{\mathfrak{d}}^+ h(\mathfrak{d}) \leq (1 + O_{\varkappa, K_0}(e^{-s})) \prod_{N\mathfrak{p} < z} (1 - h(\mathfrak{p})),$$

and

$$\sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_{\mathfrak{d}}^- h(\mathfrak{d}) \geq (1 - O_{\varkappa, K_0}(e^{-s})) \prod_{N\mathfrak{p} < z} (1 - h(\mathfrak{p})).$$

Lemma 7.2. *Let ψ be a non-principal quadratic Hecke character modulo $\mathfrak{q}$. For every $\varepsilon > 0$, there exists $\eta = \eta(\varepsilon) > 0$ such that, whenever $x \geq Q^{\alpha_0 + \varepsilon}$, we have*

$$\sum_{N\mathfrak{a} \leq x} (1 * \psi)(\mathfrak{a}) = \rho_K L(1, \psi) x + O_{K, \varepsilon}(x^{1-\eta}).$$

Proof. Write

$$S(x) := \sum_{N\mathfrak{a} \leq x} (1 * \psi)(\mathfrak{a}) = \sum_{N\mathfrak{b} N\mathfrak{c} \leq x} \psi(\mathfrak{c}).$$

Let $\theta := \frac{\alpha_0 + \varepsilon/2}{\alpha_0 + \varepsilon}$, $T_0 := x^\theta$. We split the sum according to whether $N\mathfrak{c} \leq T_0$ or $N\mathfrak{c} > T_0$. This gives

$$(7.2) \quad S(x) = \sum_{N\mathfrak{c} \leq T_0} \psi(\mathfrak{c}) \#\{\mathfrak{b} : N\mathfrak{b} \leq x/N\mathfrak{c}\} + \sum_{N\mathfrak{b} \leq x/T_0} \sum_{T_0 < N\mathfrak{c} \leq x/N\mathfrak{b}} \psi(\mathfrak{c}).$$

Recall the ideal-counting estimate [14, Thm. VI.3.3] $\#\{\mathfrak{a} : N\mathfrak{a} \leq T\} = \rho_K T + O_K(T^{1-1/n_K})$. Let $\delta_K := 1/n_K$. Then

$$\sum_{N\mathfrak{c} \leq T_0} \psi(\mathfrak{c}) \#\{\mathfrak{b} : N\mathfrak{b} \leq x/N\mathfrak{c}\} = \rho_K x \sum_{N\mathfrak{c} \leq T_0} \frac{\psi(\mathfrak{c})}{N\mathfrak{c}} + O_K\left(x^{1-\delta_K} \sum_{N\mathfrak{c} \leq T_0} (N\mathfrak{c})^{-1+\delta_K}\right),$$

where the error term is $\ll_K x^{1-\delta_K} T_0^{\delta_K} = x^{1-\delta_K(1-\theta)} = x^{1-\eta_1}$ with $\eta_1 = \delta_K(1-\theta) > 0$.

Since $x \geq Q^{\alpha_0+\varepsilon}$, we have $T_0 \geq Q^{\alpha_0+\varepsilon/2}$. Hence, by (CS^b(α_0)) applied with $\varepsilon/2$, there exists $\eta_2 = \eta_2(\varepsilon) > 0$ such that

$$(7.3) \quad A(T) := \sum_{N\mathfrak{a} \leq T} \psi(\mathfrak{a}) \ll_{K,\varepsilon} T^{1-\eta_2} \quad (T \geq T_0).$$

Partial summation gives, for $Y > T_0$,

$$\sum_{T_0 < N\mathfrak{c} \leq Y} \frac{\psi(\mathfrak{c})}{N\mathfrak{c}} = \frac{A(Y)}{Y} - \frac{A(T_0)}{T_0} + \int_{T_0}^Y \frac{A(t)}{t^2} dt \ll_{K,\varepsilon} T_0^{-\eta_2}.$$

Letting $Y \rightarrow \infty$, we obtain $\sum_{N\mathfrak{c} > T_0} \frac{\psi(\mathfrak{c})}{N\mathfrak{c}} \ll_{K,\varepsilon} T_0^{-\eta_2}$. Hence

$$\sum_{N\mathfrak{c} \leq T_0} \frac{\psi(\mathfrak{c})}{N\mathfrak{c}} = L(1, \psi) + O_{K,\varepsilon}(x^{-\theta\eta_2}).$$

Therefore the first sum in (7.2) is

$$(7.4) \quad \sum_{N\mathfrak{c} \leq T_0} \psi(\mathfrak{c}) \# \{ \mathfrak{b} : N\mathfrak{b} \leq x/N\mathfrak{c} \} = \rho_K L(1, \psi)x + O_{K,\varepsilon}(x^{1-\min(\eta_1, \theta\eta_2)}).$$

For the second part, since $x/N\mathfrak{b} \geq T_0$, by (7.3), we have

$$\sum_{T_0 < N\mathfrak{c} \leq x/N\mathfrak{b}} \psi(\mathfrak{c}) = A(x/N\mathfrak{b}) - A(T_0) \ll_{K,\varepsilon} \left(\frac{x}{N\mathfrak{b}} \right)^{1-\eta_2} + T_0^{1-\eta_2}.$$

Therefore, the second sum in (7.2) is

$$(7.5) \quad \begin{aligned} \sum_{N\mathfrak{b} \leq x/T_0} \sum_{T_0 < N\mathfrak{c} \leq x/N\mathfrak{b}} \psi(\mathfrak{c}) &\ll_{K,\varepsilon} x^{1-\eta_2} \sum_{N\mathfrak{b} \leq x/T_0} (N\mathfrak{b})^{-1+\eta_2} + T_0^{1-\eta_2} \# \{ N\mathfrak{b} \leq x/T_0 \} \\ &\ll_{K,\varepsilon} x^{1-\eta_2} \left(\frac{x}{T_0} \right)^{\eta_2} + T_0^{1-\eta_2} \frac{x}{T_0} \ll x^{1-\theta\eta_2}. \end{aligned}$$

Let $\eta = \min\{\eta_1, \theta\eta_2\} > 0$. Combining (7.4) and (7.5), we have

$$S(x) = \rho_K L(1, \psi)x + O_{K,\varepsilon}(x^{1-\eta}).$$

□

Lemma 7.3. *Let ψ be a non-principal quadratic Hecke character modulo $\mathfrak{q}$. For every $\varepsilon > 0$, there exists $\eta = \eta(\varepsilon) > 0$ such that, uniformly for every nonzero integral ideal $\mathfrak{d}$, whenever $\frac{x}{N\mathfrak{d}} \geq Q^{\alpha_0+\varepsilon}$, we have*

$$\sum_{\substack{N\mathfrak{a} \leq x \\ \mathfrak{d}|\mathfrak{a}}} (1 * \psi)(\mathfrak{a}) = \rho_K L(1, \psi) h(\mathfrak{d}) x + O_{K,\varepsilon} \left(\tau_3(\mathfrak{d}) \left(\frac{x}{N\mathfrak{d}} \right)^{1-\eta} \right),$$

where h is defined on nonzero integral ideals by

$$h(\mathfrak{d}) := \sum_{\mathfrak{d} = \mathfrak{d}_1 \mathfrak{f} \mathfrak{g}} \frac{\mu(\mathfrak{f}) \psi(\mathfrak{f} \mathfrak{g})}{N\mathfrak{d}_1 (N\mathfrak{f})^2 N\mathfrak{g}}.$$

Moreover, h is multiplicative and for every prime ideal $\mathfrak{p}$,

$$(7.6) \quad h(\mathfrak{p}) = \frac{1 + \psi(\mathfrak{p})}{N\mathfrak{p}} - \frac{\psi(\mathfrak{p})}{(N\mathfrak{p})^2},$$

$$(7.7) \quad |h(\mathfrak{p}^k)| \leq \frac{k+1}{(N\mathfrak{p})^k} \quad (k \geq 1).$$

Proof. Write

$$S_{\mathfrak{d}}(x) := \sum_{\substack{N\mathfrak{a} \leq x \\ \mathfrak{d}|\mathfrak{a}}} (1 * \psi)(\mathfrak{a}) = \sum_{\substack{N\mathfrak{m}_1 N\mathfrak{m}_2 \leq x \\ \mathfrak{d}|\mathfrak{m}_1 \mathfrak{m}_2}} \psi(\mathfrak{m}_2).$$

For a pair $(\mathfrak{m}_1, \mathfrak{m}_2)$ with $\mathfrak{d} \mid \mathfrak{m}_1 \mathfrak{m}_2$, put $\mathfrak{d}_1 = (\mathfrak{d}, \mathfrak{m}_1)$, $\mathfrak{d}_2 = \mathfrak{d}/\mathfrak{d}_1$. Then $\mathfrak{d}_2 \mid \mathfrak{m}_2$, and writing $\mathfrak{m}_1 = \mathfrak{d}_1 \mathfrak{n}_1$, $\mathfrak{m}_2 = \mathfrak{d}_2 \mathfrak{n}_2$, we have $(\mathfrak{n}_1, \mathfrak{d}_2) = 1$. Conversely, this condition gives a unique such pair $(\mathfrak{m}_1, \mathfrak{m}_2)$. Hence

$$S_{\mathfrak{d}}(x) = \sum_{\mathfrak{d}=\mathfrak{d}_1 \mathfrak{d}_2} \psi(\mathfrak{d}_2) \sum_{\substack{N \mathfrak{n}_1 N \mathfrak{n}_2 \leq x/N\mathfrak{d} \\ (\mathfrak{n}_1, \mathfrak{d}_2)=1}} \psi(\mathfrak{n}_2).$$

Writing $\mathfrak{d}_2 = \mathfrak{f}\mathfrak{g}$ and $\mathfrak{n}_1 = \mathfrak{f}\mathfrak{r}$, by Möbius inversion $1_{(\mathfrak{n}_1, \mathfrak{d}_2)=1} = \sum_{\mathfrak{f} \mid (\mathfrak{n}_1, \mathfrak{d}_2)} \mu(\mathfrak{f})$, we have

$$(7.8) \quad S_{\mathfrak{d}}(x) = \sum_{\mathfrak{d}=\mathfrak{d}_1 \mathfrak{f}\mathfrak{g}} \mu(\mathfrak{f}) \psi(\mathfrak{f}\mathfrak{g}) \sum_{N\mathfrak{m} \leq x/(N\mathfrak{d} N\mathfrak{f})} (1 * \psi)(\mathfrak{m}).$$

Let

$$X := \frac{x}{N\mathfrak{d}}, \quad \gamma := \frac{\varepsilon}{4(\alpha_0 + \varepsilon)}.$$

For the terms with $N\mathfrak{f} \geq X^\gamma$, we use the elementary bound $\sum_{N\mathfrak{m} \leq y} (1 * \psi)(\mathfrak{m}) \ll_K y \log(2y)$. To see this, using $|(1 * \psi)(\mathfrak{m})| \leq \tau_K(\mathfrak{m})$, then

$$\sum_{N\mathfrak{m} \leq y} |(1 * \psi)(\mathfrak{m})| \leq \sum_{N\mathfrak{m} \leq y} \tau_K(\mathfrak{m}) = \sum_{N\mathfrak{r} \leq y} \#\{\mathfrak{s} : N\mathfrak{s} \leq y/N\mathfrak{r}\} \ll_K \sum_{N\mathfrak{r} \leq y} \frac{y}{N\mathfrak{r}} \ll_K y \log(2y).$$

Therefore, the total contribution of the terms with $N\mathfrak{f} \geq X^\gamma$ to (7.8) is

$$\sum_{\substack{\mathfrak{d}=\mathfrak{d}_1 \mathfrak{f}\mathfrak{g} \\ N\mathfrak{f} \geq X^\gamma}} \left| \sum_{N\mathfrak{m} \leq X/N\mathfrak{f}} (1 * \psi)(\mathfrak{m}) \right| \ll_K \sum_{\substack{\mathfrak{d}=\mathfrak{d}_1 \mathfrak{f}\mathfrak{g} \\ N\mathfrak{f} \geq X^\gamma}} \frac{X}{N\mathfrak{f}} \log(2X) \ll_K \tau_3(\mathfrak{d}) X^{1-\gamma} \log(2X) \ll_{K,\varepsilon} \tau_3(\mathfrak{d}) X^{1-\gamma/2}.$$

Now suppose $N\mathfrak{f} < X^\gamma$. Since $X \geq Q^{\alpha_0+\varepsilon}$, we have $\frac{X}{N\mathfrak{f}} > X^{1-\gamma} \geq Q^{\alpha_0+3\varepsilon/4}$. Applying Lemma 7.2 with $3\varepsilon/4$, we have, for some $\eta_0 = \eta(\varepsilon) > 0$,

$$\sum_{N\mathfrak{m} \leq X/N\mathfrak{f}} (1 * \psi)(\mathfrak{m}) = \rho_K L(1, \psi) \frac{X}{N\mathfrak{f}} + O_{K,\varepsilon} \left(\left(\frac{X}{N\mathfrak{f}} \right)^{1-\eta_0} \right).$$

Therefore the contribution of the terms with $N\mathfrak{f} < X^\gamma$ is

$$\rho_K L(1, \psi) x \sum_{\substack{\mathfrak{d}=\mathfrak{d}_1 \mathfrak{f}\mathfrak{g} \\ N\mathfrak{f} < X^\gamma}} \frac{\mu(\mathfrak{f}) \psi(\mathfrak{f}\mathfrak{g})}{N\mathfrak{d}_1 (N\mathfrak{f})^2 N\mathfrak{g}} + O_{K,\varepsilon} (\tau_3(\mathfrak{d}) X^{1-\eta_0}).$$

We may extend the main term to all factorizations $\mathfrak{d} = \mathfrak{d}_1 \mathfrak{f}\mathfrak{g}$. Indeed, the omitted part with $N\mathfrak{f} \geq X^\gamma$ is

$$(7.9) \quad \ll_K L(1, \psi) x \sum_{\substack{\mathfrak{d}=\mathfrak{d}_1 \mathfrak{f}\mathfrak{g} \\ N\mathfrak{f} \geq X^\gamma}} \frac{1}{N\mathfrak{d} N\mathfrak{f}} \ll_K L(1, \psi) \tau_3(\mathfrak{d}) X^{1-\gamma}.$$

We shall use the standard bound $L(1, \psi) \ll_{K,\varepsilon'} Q^{\varepsilon'}$ for any $\varepsilon' > 0$. Indeed, for the primitive character ψ^* inducing ψ with conductor $\mathfrak{f}_\psi$, [27, Lem. 2.1] applied at $s = 1$, gives $L(1, \psi^*) \ll_{K,\varepsilon'} (N\mathfrak{f}_\psi)^{\varepsilon'} \ll_{K,\varepsilon'} Q^{\varepsilon'}$. The missing Euler factors between $L(s, \psi)$ and $L(s, \psi^*)$ contribute only $Q^{o_K(1)}$, by [27, Lem. 2.4]. After relabeling ε' , this gives the stated bound.

Using this bound with $\varepsilon' > 0$ sufficiently small, since $X \geq Q^{\alpha_0+\varepsilon}$, (7.9) is $\ll_{K,\varepsilon} \tau_3(\mathfrak{d}) X^{1-\gamma/2}$.

Combining the above estimates, we have

$$S_{\mathfrak{d}}(x) = \rho_K L(1, \psi) h(\mathfrak{d}) x + O_{K,\varepsilon} (\tau_3(\mathfrak{d}) X^{1-\eta}),$$

for some $\eta > 0$, where

$$h(\mathfrak{d}) := \sum_{\mathfrak{d}=\mathfrak{d}_1 \mathfrak{f}\mathfrak{g}} \frac{\mu(\mathfrak{f}) \psi(\mathfrak{f}\mathfrak{g})}{N\mathfrak{d}_1 (N\mathfrak{f})^2 N\mathfrak{g}}.$$

The function h is multiplicative, since it is defined by a finite Dirichlet convolution of multiplicative functions on ideals.

For $\mathfrak{d} = \mathfrak{p}^k$, since $\mu(\mathfrak{p}^e) = 0$ for $e \geq 2$, we have

$$h(\mathfrak{p}^k) = \sum_{j=0}^k \frac{\psi(\mathfrak{p}^j)}{(N\mathfrak{p})^k} - \sum_{j=0}^{k-1} \frac{\psi(\mathfrak{p}^{j+1})}{(N\mathfrak{p})^{k+1}}.$$

In particular,

$$h(\mathfrak{p}) = \frac{1 + \psi(\mathfrak{p})}{N\mathfrak{p}} - \frac{\psi(\mathfrak{p})}{(N\mathfrak{p})^2},$$

We have $\psi(\mathfrak{p}) \in \{0, \pm 1\}$. If $\psi(\mathfrak{p}) = 1$, then $h(\mathfrak{p}^k) = \frac{k+1}{(N\mathfrak{p})^k} - \frac{k}{(N\mathfrak{p})^{k+1}}$, while if $\psi(\mathfrak{p}) = 0$, then $h(\mathfrak{p}^k) = \frac{1}{(N\mathfrak{p})^k}$. Finally, if $\psi(\mathfrak{p}) = -1$, then $|\sum_{j=0}^k \psi(\mathfrak{p}^j)|, |\sum_{j=1}^k \psi(\mathfrak{p}^j)| \leq 1$, and hence $|h(\mathfrak{p}^k)| \leq \frac{1}{(N\mathfrak{p})^k} + \frac{1}{(N\mathfrak{p})^{k+1}}$. Therefore, in any case,

$$|h(\mathfrak{p}^k)| \leq \frac{k+1}{(N\mathfrak{p})^k}.$$

□

Lemma 7.4. *Let h be as in Lemma 7.3. Then $0 \leq h(\mathfrak{p}) < 1$ for every prime ideal $\mathfrak{p}$, and h satisfies the sieve dimension condition with $\varkappa = 2$; that is, there exists $K_0 > 1$ such that for all $2 \leq w \leq z$,*

$$\prod_{w \leq N\mathfrak{p} < z} (1 - h(\mathfrak{p}))^{-1} \leq K_0 \left(\frac{\log z}{\log w} \right)^2.$$

And for any fixed $\varepsilon > 0$,

$$\prod_{N\mathfrak{p} < Q^\varepsilon} (1 - h(\mathfrak{p})) \asymp_{K, \varepsilon} \mathcal{V}_\psi(Q) := \prod_{\mathfrak{p} | \mathfrak{q}} \left(1 - \frac{1}{N\mathfrak{p}} \right) \prod_{\substack{2 < N\mathfrak{p} \leq Q \\ \psi(\mathfrak{p}) = 1}} \left(1 - \frac{2}{N\mathfrak{p}} \right).$$

Proof. If $\mathfrak{p} \mid \mathfrak{q}$, then $\psi(\mathfrak{p}) = 0$, and $h(\mathfrak{p}) = \frac{1}{N\mathfrak{p}}$. If $\psi(\mathfrak{p}) = -1$, then $h(\mathfrak{p}) = \frac{1}{(N\mathfrak{p})^2}$. Finally, if $\psi(\mathfrak{p}) = 1$, then $h(\mathfrak{p}) = \frac{2}{N\mathfrak{p}} - \frac{1}{N\mathfrak{p}^2}$, and $1 - h(\mathfrak{p}) = (1 - \frac{1}{N\mathfrak{p}})^2$. In any case, $0 \leq h(\mathfrak{p}) < 1$ and $(1 - h(\mathfrak{p}))^{-1} \leq (1 - \frac{1}{N\mathfrak{p}})^{-2}$. Therefore, by Mertens' theorem [21, Thm. 2] (see (3.2)),

$$\prod_{w \leq N\mathfrak{p} < z} (1 - h(\mathfrak{p}))^{-1} \leq \prod_{w \leq N\mathfrak{p} < z} \left(1 - \frac{1}{N\mathfrak{p}} \right)^{-2} \ll_K \left(\frac{\log z}{\log w} \right)^2.$$

This proves the sieve dimension assertion with $\varkappa = 2$.

We now compare the Euler product with $\mathcal{V}_\psi(Q)$. The factors corresponding to prime ideals with $\psi(\mathfrak{p}) = -1$ satisfy

$$1 \geq \prod_{\substack{N\mathfrak{p} < Q^\varepsilon \\ \psi(\mathfrak{p}) = -1}} \left(1 - \frac{1}{(N\mathfrak{p})^2} \right) \gg_K 1,$$

since $\sum_{\mathfrak{p}} (N\mathfrak{p})^{-2} \ll_K 1$ and hence the product is bounded below by a positive constant depending only on K . For prime ideals with $\psi(\mathfrak{p}) = 1$ and $N\mathfrak{p} > 2$, we have

$$1 - h(\mathfrak{p}) = \left(1 - \frac{1}{N\mathfrak{p}} \right)^2 = \left(1 - \frac{2}{N\mathfrak{p}} \right) \left(1 + O\left(\frac{1}{(N\mathfrak{p})^2} \right) \right)$$

Therefore, by the absolute convergence of $\sum_{\mathfrak{p}} (N\mathfrak{p})^{-2}$,

$$\prod_{\substack{N\mathfrak{p} < Q^\varepsilon \\ \psi(\mathfrak{p}) = 1}} (1 - h(\mathfrak{p})) \asymp_K \prod_{\substack{2 < N\mathfrak{p} < Q^\varepsilon \\ \psi(\mathfrak{p}) = 1}} \left(1 - \frac{2}{N\mathfrak{p}} \right).$$

The finitely many prime ideals with $N\mathfrak{p} = 2$ are absorbed into the implied constants.

Finally, we compare the cutoffs Q^ε and Q . By Mertens theorem in the form [21, Lem. 2.4], there is a constant $B_K > 0$ such that

$$(7.10) \quad \sum_{N\mathfrak{p} \leq z} \frac{1}{N\mathfrak{p}} = \log \log z + B_K + O_K\left(\frac{1}{\log z} \right).$$

Thus $\sum_{Q^\varepsilon \leq N\mathfrak{p} \leq Q} \frac{1}{N\mathfrak{p}} \ll_{K, \varepsilon} 1$, and therefore

$$\prod_{\substack{Q^\varepsilon \leq N\mathfrak{p} \leq Q \\ \psi(\mathfrak{p}) = 1}} \left(1 - \frac{2}{N\mathfrak{p}} \right) \asymp_{K, \varepsilon} 1.$$

Next, consider the factors corresponding to prime ideals dividing $\mathfrak{q}$. Since $N\mathfrak{p} \leq Q$, we have

$$\prod_{\substack{\mathfrak{p}|\mathfrak{q} \\ N\mathfrak{p} \leq Q^\varepsilon}} \left(1 - \frac{1}{N\mathfrak{p}}\right) \geq \prod_{Q^\varepsilon \leq N\mathfrak{p} \leq Q} \left(1 - \frac{1}{N\mathfrak{p}}\right) \gg_{K,\varepsilon} 1.$$

Combining these estimates gives

$$\prod_{N\mathfrak{p} < Q^\varepsilon} (1 - h(\mathfrak{p})) \asymp_{K,\varepsilon} \prod_{\mathfrak{p}|\mathfrak{q}} \left(1 - \frac{1}{N\mathfrak{p}}\right) \prod_{\substack{2 < N\mathfrak{p} \leq Q \\ \psi(\mathfrak{p})=1}} \left(1 - \frac{2}{N\mathfrak{p}}\right) = \mathcal{V}_\psi(Q).$$

□

Lemma 7.5. *Let ψ be a non-principal quadratic Hecke character modulo $\mathfrak{q}$. For every sufficiently small constant $\varepsilon > 0$, there exists an ineffective constant $C_{K,\varepsilon} > 0$ such that, whenever $y \geq Q^{\alpha_0+\varepsilon}$, we have*

$$\#\{\mathfrak{p} : N\mathfrak{p} \leq y, \psi(\mathfrak{p}) = 1\} \leq C_{K,\varepsilon} y \rho_K L(1, \psi) \mathcal{V}_\psi(Q).$$

Proof. Let $s_0 \geq 19$ be fixed and put

$$z = Q^{\varepsilon/(2s_0)}, \quad D = z^{s_0} = Q^{\varepsilon/2}.$$

Let $\lambda_\mathfrak{d}^+$ be the upper-bound beta-sieve weights from Lemma 7.1.

Since ψ is quadratic, $(1 * \psi)(\mathfrak{a}) \geq 0$ for every $\mathfrak{a}$. Moreover, if $\psi(\mathfrak{p}) = 1$, then $(1 * \psi)(\mathfrak{p}) = 1 + \psi(\mathfrak{p}) = 2$. The prime ideals with $N\mathfrak{p} < z$ contribute $O_K(z)$ and the remaining primes satisfies $(\mathfrak{p}, \mathcal{P}(z)) = 1$, hence

$$\begin{aligned} \#\{\mathfrak{p} : N\mathfrak{p} \leq y, \psi(\mathfrak{p}) = 1\} &\ll_K z + \sum_{N\mathfrak{a} \leq y} (1 * \psi)(\mathfrak{a}) 1_{(\mathfrak{a}, \mathcal{P}(z))=1} \\ &\leq O_K(z) + \sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_\mathfrak{d}^+ \sum_{\substack{N\mathfrak{a} \leq y \\ \mathfrak{d}|\mathfrak{a}}} (1 * \psi)(\mathfrak{a}). \end{aligned}$$

Since $\lambda_\mathfrak{d}^+$ is supported on $N\mathfrak{d} \leq D = Q^{\varepsilon/2}$, we have $\frac{y}{N\mathfrak{d}} \geq Q^{\alpha_0+\varepsilon/2}$. Applying Lemma 7.3 with $\varepsilon/2$ gives

$$\sum_{\substack{N\mathfrak{a} \leq y \\ \mathfrak{d}|\mathfrak{a}}} (1 * \psi)(\mathfrak{a}) = \rho_K L(1, \psi) h(\mathfrak{d}) y + O_{K,\varepsilon} \left(\tau_3(\mathfrak{d}) \left(\frac{y}{N\mathfrak{d}} \right)^{1-\eta} \right)$$

for some $\eta > 0$. Therefore

$$\sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_\mathfrak{d}^+ \sum_{\substack{N\mathfrak{a} \leq y \\ \mathfrak{d}|\mathfrak{a}}} (1 * \psi)(\mathfrak{a}) = \rho_K L(1, \psi) y \sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_\mathfrak{d}^+ h(\mathfrak{d}) + O_{K,\varepsilon} (y^{1-\eta/2}),$$

where we use that $\sum_{N\mathfrak{d} \leq D} \frac{\tau_3(\mathfrak{d})}{(N\mathfrak{d})^{1-\eta}} \ll_{K,\varepsilon} D^\eta (\log D)^{O_K(1)} \ll_{K,\varepsilon} y^{\eta/2}$ for the error term.

We now apply the fundamental lemma Lemma 7.1 with sieve dimension $\varkappa = 2$. Since $s_0 \geq 19$, by Lemma 7.4 and Lemma 7.1, we have

$$\sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_\mathfrak{d}^+ h(\mathfrak{d}) \ll_K \prod_{N\mathfrak{p} < z} (1 - h(\mathfrak{p})) \asymp_{K,\varepsilon} \mathcal{V}_\psi(Q).$$

Hence

$$\#\{\mathfrak{p} : N\mathfrak{p} \leq y, \psi(\mathfrak{p}) = 1\} \ll_{K,\varepsilon} y \rho_K L(1, \psi) \mathcal{V}_\psi(Q) + y^{1-\eta/2} + z.$$

By the (ineffective) Siegel-type lower bound for real nontrivial Hecke characters [17, Lem. 10], for every $\delta > 0$ one has

$$(7.11) \quad L(1, \psi) \gg_{K,\delta} Q^{-\delta}.$$

Also, by Mertens' theorem in the form [21, Lem. 2.4] (see (7.10)),

$$(7.12) \quad \mathcal{V}_\psi(Q) \gg_K (\log Q)^{-3} \gg_{K,\delta} Q^{-\delta}.$$

Choosing $\delta > 0$ sufficiently small so that $2\delta < \min\{\frac{(\alpha_0+\varepsilon)\eta}{2}, \alpha_0 + \varepsilon - \frac{\varepsilon}{2s_0}\}$, and since $y \geq Q^{\alpha_0+\varepsilon}$, $z = Q^{\varepsilon/(2s_0)}$, the terms $y^{1-\eta/2}$ and z are absorbed into the main term. This proves the lemma. □

Lemma 7.6. *Let $B_0 > 0$ be fixed and suppose $B_0 > \max\{2\alpha + \alpha_0, 2\alpha_0\}$. Choose*

$$\beta_* \in (\max\{2\alpha, B_0/2\}, B_0 - \alpha_0), \quad c_* > 0,$$

such that

$$2\alpha + 2c_*\beta_* < B_0.$$

Let $\varepsilon > 0$ be sufficiently small so that

$$\beta_* > 2\alpha + 5\varepsilon, \quad \beta_* + 2\varepsilon < B_0 - \alpha_0.$$

Let ψ be a non-principal quadratic Hecke character modulo $\mathfrak{q}$. Then at least one of the following holds.

(i)

$$\sum_{\substack{Q^{\alpha+\varepsilon} < N\mathfrak{p} \leq Q^{\beta_*} \\ \psi(\mathfrak{p})=1}} \frac{1}{N\mathfrak{p}} \geq c_*.$$

(ii) *There exists $M \in [Q^{\beta_*}, Q^{B_0}]$ such that*

$$\#\{\mathfrak{p} : M < N\mathfrak{p} \leq 2M, \psi(\mathfrak{p}) = 1\} \gg_{K,\varepsilon} M \rho_K L(1, \psi) \mathcal{V}_\psi(Q).$$

Proof. Put $Y := Q^{B_0}$. Let $s_0 \geq 19$ be a constant to be chosen sufficiently large later, and put

$$w := Q^{\varepsilon/s_0}, \quad D := w^{s_0} = Q^\varepsilon.$$

Let $\lambda_\mathfrak{e}^\pm$ be the beta-sieve weights from Lemma 7.1, with sifting range $z = w$ and level D .

Define

$$a_n := (1 * \psi)(n) 1_{(n, \mathcal{P}(w))=1}.$$

Let Λ_K denote the von Mangoldt function on ideals as defined in (7.1). Using $\sum_{\mathfrak{d}|n} \Lambda_K(\mathfrak{d}) = \log Nn$, we have

$$(7.13) \quad S(Y) := \sum_{N\mathfrak{d} \leq Y} \Lambda_K(\mathfrak{d}) \sum_{\substack{Nn \leq Y \\ \mathfrak{d}|n}} a_n = \sum_{Nn \leq Y} a_n \log Nn.$$

We first lower-bound the right-hand side of (7.13). Since ψ is quadratic, $1 * \psi \geq 0$, and hence $a_n \geq 0$. Using $1 * \psi \geq 0$ and $1_{(n, \mathcal{P}(w))=1} \geq \sum_{\mathfrak{e}|n} \lambda_\mathfrak{e}^-$, we have

$$\sum_{Nn \leq Y} a_n \log Nn \geq \sum_{\mathfrak{e}|\mathcal{P}(w)} \lambda_\mathfrak{e}^- \sum_{\substack{Nn \leq Y \\ \mathfrak{e}|n}} (1 * \psi)(n) \log Nn.$$

For $N\mathfrak{e} \leq D = Q^\varepsilon$, we have $\frac{Y}{N\mathfrak{e}} \geq Q^{B_0-\varepsilon} \geq Q^{\alpha_0+\varepsilon}$ after decreasing ε , since $B_0 > 2\alpha_0 \geq \alpha_0$. Applying Lemma 7.3 and partial summation, there is $\eta' > 0$ such that uniformly for all such $\mathfrak{e}$,

$$\sum_{\substack{Nn \leq Y \\ \mathfrak{e}|n}} (1 * \psi)(n) \log Nn = \rho_K L(1, \psi) h(\mathfrak{e}) (Y \log Y - Y) + O_{K,\varepsilon}(\tau_3(\mathfrak{e}) Y^{1-\eta'}).$$

By Lemma 7.4, the multiplicative function h satisfies the sieve-dimension hypothesis of Lemma 7.1 with dimension 2. Since $s_0 \geq 19 = 9 \cdot 2 + 1$, the fundamental lemma gives

$$\sum_{\mathfrak{e}|\mathcal{P}(w)} \lambda_\mathfrak{e}^- h(\mathfrak{e}) \geq (1 - O_K(e^{-s_0})) V(w), \quad \sum_{\mathfrak{e}|\mathcal{P}(w)} \lambda_\mathfrak{e}^+ h(\mathfrak{e}) \leq (1 + O_K(e^{-s_0})) V(w),$$

where $V(w) := \prod_{N\mathfrak{p} < w} (1 - h(\mathfrak{p}))$.

Summing over $\mathfrak{e} \mid \mathcal{P}(w)$, the total contribution of the error terms is

$$\ll_{K,\varepsilon} Y^{1-\eta'} \sum_{N\mathfrak{e} \leq D} \tau_3(\mathfrak{e}) \ll_{K,\varepsilon} Y^{1-\eta'} D^{1+o(1)}.$$

Since $D = Q^\varepsilon$, $Y = Q^{B_0}$, after decreasing ε once more, and then choosing the exponent δ in the Siegel lower bound (7.11) sufficiently small, this error is absorbed into the main term. Moreover, since $Y \log Y - Y = (1 - o(1)) B_0 Y \log Q$, we obtain

$$(7.14) \quad \sum_{Nn \leq Y} a_n \log Nn \geq (1 - O_K(e^{-s_0}) - o_{K,\varepsilon,s_0}(1)) B_0 Y \log Q \rho_K L(1, \psi) V(w).$$

We now prove an upper bound for $S(Y)$. First we estimate the contribution of the terms with $N\mathfrak{d} \leq Q^{\beta_*}$ in (7.13).

Using the upper-bound sieve $1_{(n, \mathcal{P}(w))=1} \leq \sum_{\mathfrak{e}|\mathfrak{n}} \lambda_{\mathfrak{e}}^+$, and that $(\mathfrak{d}, \mathfrak{e}) = 1$ when $\mathfrak{e} \mid \mathcal{P}(w)$, we have

$$\sum_{\substack{N\mathfrak{d} \leq Q^{\beta_*} \\ (\mathfrak{d}, \mathcal{P}(w))=1}} \Lambda_K(\mathfrak{d}) \sum_{\substack{N\mathfrak{n} \leq Y \\ \mathfrak{d}|\mathfrak{n}}} a_{\mathfrak{n}} \leq \sum_{\substack{N\mathfrak{d} \leq Q^{\beta_*} \\ (\mathfrak{d}, \mathcal{P}(w))=1}} \Lambda_K(\mathfrak{d}) \sum_{\mathfrak{e}|\mathcal{P}(w)} \lambda_{\mathfrak{e}}^+ \sum_{\substack{N\mathfrak{n} \leq Y \\ \mathfrak{de}|\mathfrak{n}}} (1 * \psi)(\mathfrak{n}).$$

For $N\mathfrak{d} \leq Q^{\beta_*}$ and $N\mathfrak{e} \leq D = Q^\varepsilon$, $\frac{Y}{N\mathfrak{d}N\mathfrak{e}} \geq Q^{B_0 - \beta_* - \varepsilon} > Q^{\alpha_0 + \varepsilon}$, by our choice of ε . Applying Lemma 7.3, the total error is again absorbed into the main term as before. Therefore, the contribution of the terms with $N\mathfrak{d} \leq Q^{\beta_*}$ in $S(Y)$ is

$$(7.15) \quad \leq (1 + o_{K, \varepsilon, s_0}(1)) Y \rho_K L(1, \psi) \left(\sum_{\mathfrak{e}|\mathcal{P}(w)} \lambda_{\mathfrak{e}}^+ h(\mathfrak{e}) \right) \left(\sum_{\substack{N\mathfrak{d} \leq Q^{\beta_*} \\ (\mathfrak{d}, \mathcal{P}(w))=1}} \Lambda_K(\mathfrak{d}) h(\mathfrak{d}) \right).$$

We next bound the inner sum over $\mathfrak{d}$ in (7.15). The contribution of prime powers $\mathfrak{d} = \mathfrak{p}^k$, $k \geq 2$, is $O_K(1)$, since $|h(\mathfrak{p}^k)| \leq \frac{k+1}{(N\mathfrak{p})^k}$ by (7.7). For primes, if $\psi(\mathfrak{p}) = 1$, then $h(\mathfrak{p}) = \frac{2}{N\mathfrak{p}} - \frac{1}{N\mathfrak{p}^2} \leq \frac{2}{N\mathfrak{p}}$. If $\psi(\mathfrak{p}) = -1$, then $h(\mathfrak{p}) = \frac{1}{(N\mathfrak{p})^2}$. And if $\mathfrak{p} \mid \mathfrak{q}$, then $h(\mathfrak{p}) = \frac{1}{N\mathfrak{p}}$. Moreover, the primes $\mathfrak{p}$ with $(\mathfrak{p}, \mathcal{P}(w)) = 1$ have $N\mathfrak{p} \geq w$. Therefore, by (7.10), the primes with $\psi(\mathfrak{p}) = -1$ contribute $O_{K, \varepsilon, s_0}(1)$. For primes $\mathfrak{p}$ with $\mathfrak{p} \mid \mathfrak{q}$, the contribution is at most

$$\sum_{\substack{\mathfrak{p}|\mathfrak{q} \\ N\mathfrak{p} \geq w}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} \leq \frac{1}{w} \sum_{\mathfrak{p}|\mathfrak{q}} \log N\mathfrak{p} \leq \frac{\log Q}{w} = o_{\varepsilon, s_0}(\log Q).$$

We now split the sum at $Q^{\alpha+\varepsilon}$. In the range $N\mathfrak{d} \leq Q^{\alpha+\varepsilon}$, by the above estimates and (7.10), we have

$$\begin{aligned} \sum_{\substack{N\mathfrak{d} \leq Q^{\alpha+\varepsilon} \\ (\mathfrak{d}, \mathcal{P}(w))=1}} \Lambda_K(\mathfrak{d}) h(\mathfrak{d}) &\leq \sum_{\substack{N\mathfrak{p} \leq Q^{\alpha+\varepsilon} \\ \psi(\mathfrak{p})=1}} \Lambda_K(\mathfrak{p}) h(\mathfrak{p}) + o_{K, \varepsilon, s_0}(\log Q) \\ &\leq 2 \sum_{N\mathfrak{p} \leq Q^{\alpha+\varepsilon}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} + o_{K, \varepsilon, s_0}(\log Q) = (2\alpha + 2\varepsilon + o_{K, \varepsilon, s_0}(1)) \log Q. \end{aligned}$$

If (i) fails, then

$$\sum_{\substack{Q^{\alpha+\varepsilon} < N\mathfrak{p} \leq Q^{\beta_*} \\ \psi(\mathfrak{p})=1}} \frac{1}{N\mathfrak{p}} < c_*.$$

Hence

$$\sum_{\substack{Q^{\alpha+\varepsilon} < N\mathfrak{d} \leq Q^{\beta_*} \\ (\mathfrak{d}, \mathcal{P}(w))=1}} \Lambda_K(\mathfrak{d}) h(\mathfrak{d}) \leq 2 \sum_{\substack{Q^{\alpha+\varepsilon} < N\mathfrak{p} \leq Q^{\beta_*} \\ \psi(\mathfrak{p})=1}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} + o_{K, \varepsilon, s_0}(\log Q) \leq (2c_*\beta_* + o_{K, \varepsilon, s_0}(1)) \log Q.$$

Summing over these two ranges together with (7.15), we have

$$\sum_{\substack{N\mathfrak{d} \leq Q^{\beta_*} \\ (\mathfrak{d}, \mathcal{P}(w))=1}} \Lambda_K(\mathfrak{d}) \sum_{\substack{N\mathfrak{n} \leq Y \\ \mathfrak{d}|\mathfrak{n}}} a_{\mathfrak{n}} \leq (1 + O_K(e^{-s_0}) + o_{K, \varepsilon, s_0}(1)) (2\alpha + 2\varepsilon + 2c_*\beta_*) Y \log Q \rho_K L(1, \psi) V(w).$$

Set $\Delta := B_0 - (2\alpha + 2c_*\beta_*) > 0$. Taking ε sufficiently small, we may assume $B_0 - (2\alpha + 2\varepsilon + 2c_*\beta_*) > \Delta/2$. We then choose s_0 sufficiently large so that the $O_K(e^{-s_0})$ terms are small compared with Δ , and then take Q sufficiently large so that the $o_{K, \varepsilon, s_0}(1)$ terms are also small so that from (7.14), we obtain

$$\sum_{Q^{\beta_*} < N\mathfrak{d} \leq Y} \Lambda_K(\mathfrak{d}) \sum_{\substack{N\mathfrak{n} \leq Y \\ \mathfrak{d}|\mathfrak{n}}} a_{\mathfrak{n}} \gg_{K, \varepsilon, s_0} Y \log Q \rho_K L(1, \psi) V(w).$$

Since $w = Q^{\varepsilon/s_0}$ and s_0 is now fixed, Lemma 7.4 gives $V(w) \asymp_{K, \varepsilon} \mathcal{V}_\psi(Q)$, hence we have

$$(7.16) \quad \sum_{Q^{\beta_*} < N\mathfrak{d} \leq Y} \Lambda_K(\mathfrak{d}) \sum_{\substack{N\mathfrak{n} \leq Y \\ \mathfrak{d}|\mathfrak{n}}} a_{\mathfrak{n}} \gg_{K, \varepsilon} Y \log Q \rho_K L(1, \psi) \mathcal{V}_\psi(Q).$$

Note that since $0 \leq a_{\mathfrak{n}} \leq \tau(\mathfrak{n})$, for any ideal $\mathfrak{b}$, we have

$$(7.17) \quad \sum_{\substack{N\mathfrak{n} \leq Y \\ \mathfrak{b} | \mathfrak{n}}} a_{\mathfrak{n}} \leq \sum_{N\mathfrak{m} \leq Y/N\mathfrak{b}} \tau(\mathfrak{bm}) \ll_K \tau(\mathfrak{b}) \frac{Y}{N\mathfrak{b}} \log Y.$$

Therefore, the contribution to the sum in (7.16) from prime powers $\mathfrak{d} = \mathfrak{p}^m$ with $m \geq 2$, is

$$\ll_K Y \log Y \sum_{\substack{m \geq 2 \\ (N\mathfrak{p})^m > Q^{\beta_*}}} \frac{(m+1) \log N\mathfrak{p}}{(N\mathfrak{p})^m} \ll_K Y (\log Q)^2 Q^{-\beta_*/2},$$

which is negligible compared with $Y \log Q \rho_K L(1, \psi) \mathcal{V}_{\psi}(Q)$, using (7.11) and (7.12).

Now we consider $\mathfrak{d} = \mathfrak{p}$. Since $\beta_* > B_0/2$, any prime ideal with $N\mathfrak{p} > Q^{\beta_*}$ satisfies $(N\mathfrak{p})^2 > Y$. Thus, if $N\mathfrak{n} \leq Y$ and $\mathfrak{p} | \mathfrak{n}$, then $\mathfrak{p}$ divides $\mathfrak{n}$ exactly once. Hence, writing $\mathfrak{n} = \mathfrak{p}\mathfrak{m}$, then $(\mathfrak{p}, \mathfrak{m}) = 1$ and $(1 * \psi)(\mathfrak{n}) = (1 + \psi(\mathfrak{p}))(1 * \psi)(\mathfrak{m})$. Therefore the inner sum vanishes whenever $\psi(\mathfrak{p}) = -1$.

The remaining primes with $\psi(\mathfrak{p}) = 0$ divide $\mathfrak{q}$. By (7.17), their total contribution is

$$\ll_K Y \log Y \sum_{\substack{\mathfrak{p} | \mathfrak{q} \\ N\mathfrak{p} > Q^{\beta_*}}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} \leq Y \log Y Q^{-\beta_*} \log Q,$$

which is negligible compared with $Y \log Q \rho_K L(1, \psi) \mathcal{V}_{\psi}(Q)$, again using (7.11) and (7.12).

Therefore, in (7.16) we may restrict the outer sum to primes $\mathfrak{d} = \mathfrak{p}$ with $\psi(\mathfrak{p}) = 1$. Thus

$$(7.18) \quad Y \log Q \rho_K L(1, \psi) \mathcal{V}_{\psi}(Q) \ll_{K, \varepsilon} \sum_{\substack{Q^{\beta_*} < N\mathfrak{p} \leq Y \\ \psi(\mathfrak{p}) = 1}} \log N\mathfrak{p} \sum_{\substack{N\mathfrak{n} \leq Y \\ \mathfrak{p} | \mathfrak{n}}} a_{\mathfrak{n}}.$$

Moreover, as noted above, if $N\mathfrak{p} > Q^{\beta_*}$, then $\mathfrak{p}$ divides $\mathfrak{n}$ exactly once. Writing $\mathfrak{n} = \mathfrak{p}\mathfrak{m}$, we therefore have $(\mathfrak{p}, \mathfrak{m}) = 1$ and $a_{\mathfrak{n}} \leq 2\tau(\mathfrak{m})$. Since $N\mathfrak{p} > w$, the condition $(\mathfrak{n}, \mathcal{P}(w)) = 1$ is equivalent to $(\mathfrak{m}, \mathcal{P}(w)) = 1$.

Hence we have

$$(7.19) \quad \begin{aligned} \sum_{\substack{N\mathfrak{n} \leq Y \\ \mathfrak{p} | \mathfrak{n}}} a_{\mathfrak{n}} &\ll \sum_{\substack{N\mathfrak{m} \leq Y/N\mathfrak{p} \\ (\mathfrak{m}, \mathcal{P}(w)) = 1}} \tau(\mathfrak{m}) \leq \sum_{\substack{N\mathfrak{s} \leq Y/N\mathfrak{p} \\ (\mathfrak{s}, \mathcal{P}(w)) = 1}} \#\{\mathfrak{r} : N\mathfrak{r} \leq \frac{Y}{N\mathfrak{p}N\mathfrak{s}}, (\mathfrak{r}, \mathcal{P}(w)) = 1\} \\ &\ll_K \sum_{\substack{N\mathfrak{s} \leq Y/N\mathfrak{p} \\ (\mathfrak{s}, \mathcal{P}(w)) = 1}} \left(\frac{Y}{N\mathfrak{p}N\mathfrak{s} \log w} + 1 \right) \\ &\ll_{K, \varepsilon} \frac{Y}{N\mathfrak{p} \log w} + 1. \end{aligned}$$

Here we use the standard upper-bound linear sieve estimate

$$\#\{\mathfrak{a} : N\mathfrak{a} \leq X, (\mathfrak{a}, \mathcal{P}(w)) = 1\} \ll_K \frac{X}{\log w} + 1,$$

which can be obtained by applying Lemma 2.1 to the sequence of all integral ideals $N\mathfrak{a} \leq X$ with density $g(\mathfrak{p}) = 1/N\mathfrak{p}$, together with Mertens' theorem for prime ideals. By partial summation this also gives

$$\sum_{\substack{N\mathfrak{a} \leq X \\ (\mathfrak{a}, \mathcal{P}(w)) = 1}} \frac{1}{N\mathfrak{a}} \ll_K 1 + \frac{\log X}{\log w}.$$

Applying (7.19) to (7.18), we obtain

$$(7.20) \quad Y \log Q \rho_K L(1, \psi) \mathcal{V}_{\psi}(Q) \ll_{K, \varepsilon} \frac{Y}{\log w} \sum_{\substack{Q^{\beta_*} < N\mathfrak{p} \leq Y \\ \psi(\mathfrak{p}) = 1}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} + (\log Y) \#\{\mathfrak{p} : Q^{\beta_*} < N\mathfrak{p} \leq Y, \psi(\mathfrak{p}) = 1\}.$$

If the second term in (7.20) dominates, then

$$\#\{\mathfrak{p} : Q^{\beta_*} < N\mathfrak{p} \leq Y, \psi(\mathfrak{p}) = 1\} \gg_{K, \varepsilon} Y \rho_K L(1, \psi) \mathcal{V}_{\psi}(Q).$$

A dyadic decomposition then gives some $M \in [Q^{\beta_*}, Y]$ such that

$$\#\{\mathfrak{p} : M < N\mathfrak{p} \leq 2M, \psi(\mathfrak{p}) = 1\} \gg_{K,\varepsilon} M \rho_K L(1, \psi) \mathcal{V}_\psi(Q).$$

Otherwise the first term in (7.20) dominates. Since $\log w \asymp_\varepsilon \log Q$, we get

$$\sum_{\substack{Q^{\beta_*} < N\mathfrak{p} \leq Y \\ \psi(\mathfrak{p})=1}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} \gg_{K,\varepsilon} (\log Q)^2 \rho_K L(1, \psi) \mathcal{V}_\psi(Q).$$

Again by dyadic decomposition, for some $M \in [Q^{\beta_*}, Y]$,

$$\sum_{\substack{M < N\mathfrak{p} \leq 2M \\ \psi(\mathfrak{p})=1}} \frac{\log N\mathfrak{p}}{N\mathfrak{p}} \gg_{K,\varepsilon} \log Q \rho_K L(1, \psi) \mathcal{V}_\psi(Q).$$

Since $\log N\mathfrak{p} \ll_{B_0} \log Q$ on this range, this implies

$$\#\{\mathfrak{p} : M < N\mathfrak{p} \leq 2M, \psi(\mathfrak{p}) = 1\} \gg_{K,\varepsilon} M \rho_K L(1, \psi) \mathcal{V}_\psi(Q).$$

This proves (ii). $\square$

Lemma 7.7. *Let ψ and ξ be non-principal Hecke characters modulo $\mathfrak{q}$, with ψ quadratic. Suppose $\psi \neq \xi$. For every $\varepsilon > 0$, there exists $\eta = \eta(\varepsilon) > 0$ such that whenever $T \geq Q^{2\alpha+4\varepsilon}$, uniformly for nonzero integral ideals $\mathfrak{d}$ with $N\mathfrak{d} \leq Q^\varepsilon$, we have*

$$\sum_{\substack{N\mathfrak{a} \leq T \\ \mathfrak{d}|\mathfrak{a}}} \xi(\mathfrak{a})(1 * \psi)(\mathfrak{a}) \ll_{K,\varepsilon} \frac{T}{N\mathfrak{d}} Q^{-\eta}.$$

Proof. We use

$$\xi(\mathfrak{a})(1 * \psi)(\mathfrak{a}) = (\xi * \xi\psi)(\mathfrak{a}).$$

Since $\xi \notin \{\chi_0, \psi\}$, both ξ and $\xi\psi$ are non-principal.

First we record a hyperbola estimate. For $U \geq Q^{2\alpha+2\varepsilon}$, put

$$H(U) := \sum_{N\mathfrak{b} N\mathfrak{c} \leq U} \xi(\mathfrak{b})(\xi\psi)(\mathfrak{c}).$$

By Dirichlet's hyperbola decomposition,

$$H(U) = \sum_{N\mathfrak{b} \leq U^{1/2}} \xi(\mathfrak{b}) \sum_{N\mathfrak{c} \leq U/N\mathfrak{b}} (\xi\psi)(\mathfrak{c}) + \sum_{N\mathfrak{c} \leq U^{1/2}} (\xi\psi)(\mathfrak{c}) \sum_{N\mathfrak{b} \leq U/N\mathfrak{c}} \xi(\mathfrak{b}) - \left(\sum_{N\mathfrak{b} \leq U^{1/2}} \xi(\mathfrak{b}) \right) \left(\sum_{N\mathfrak{c} \leq U^{1/2}} (\xi\psi)(\mathfrak{c}) \right).$$

By (CS(α)), for some $\theta = \theta(\varepsilon) > 0$, whenever $X \geq Q^{\alpha+\varepsilon}$, we have

$$\sum_{N\mathfrak{a} \leq X} \xi(\mathfrak{a}) \ll_{K,\varepsilon} X^{1-\theta}, \quad \sum_{N\mathfrak{a} \leq X} (\xi\psi)(\mathfrak{a}) \ll_{K,\varepsilon} X^{1-\theta}.$$

Since every inner sum in $H(U)$ has length at least $U^{1/2} \geq Q^{\alpha+\varepsilon}$, we have

$$H(U) \ll_{K,\varepsilon} U^{1-\theta} \sum_{N\mathfrak{b} \leq U^{1/2}} \frac{1}{(N\mathfrak{b})^{1-\theta}} + U^{1-\theta} \sum_{N\mathfrak{c} \leq U^{1/2}} \frac{1}{(N\mathfrak{c})^{1-\theta}} + U^{1-\theta} \ll_{K,\varepsilon} U^{1-\theta/2}.$$

Since $U \geq Q^{2\alpha+2\varepsilon}$, this gives

$$(7.21) \quad H(U) \ll_{K,\varepsilon} U Q^{-\eta_0}$$

for some $\eta_0 = \eta_0(\varepsilon) > 0$.

We now impose the divisibility condition. Expanding the convolution,

$$\sum_{\substack{N\mathfrak{a} \leq T \\ \mathfrak{d}|\mathfrak{a}}} \xi(\mathfrak{a})(1 * \psi)(\mathfrak{a}) = \sum_{\substack{N\mathfrak{b} N\mathfrak{c} \leq T \\ \mathfrak{d}|\mathfrak{bc}}} \xi(\mathfrak{b})(\xi\psi)(\mathfrak{c}).$$

For a given pair $(\mathfrak{b}, \mathfrak{c})$, put $\mathfrak{d}_1 = (\mathfrak{d}, \mathfrak{b})$, $\mathfrak{d}_2 = \mathfrak{d}/\mathfrak{d}_1$. Then $\mathfrak{d}_2 \mid \mathfrak{c}$, and writing $\mathfrak{b} = \mathfrak{d}_1 \mathfrak{b}_0$, $\mathfrak{c} = \mathfrak{d}_2 \mathfrak{c}_0$, we have $(\mathfrak{b}_0, \mathfrak{d}_2) = 1$. Conversely, these conditions recover the condition $(\mathfrak{d}, \mathfrak{b}) = \mathfrak{d}_1$ and $\mathfrak{d} \mid \mathfrak{bc}$. Thus the sum can be written as

$$\sum_{\mathfrak{d}_1 \mid \mathfrak{d}} \xi(\mathfrak{d}_1)(\xi\psi)(\mathfrak{d}_2) \sum_{\substack{N\mathfrak{b}_0 N\mathfrak{c}_0 \leq T/N\mathfrak{d} \\ (\mathfrak{b}_0, \mathfrak{d}_2)=1}} \xi(\mathfrak{b}_0)(\xi\psi)(\mathfrak{c}_0).$$

By Möbius inversion, $1_{(\mathfrak{b}_0, \mathfrak{d}_2)=1} = \sum_{\mathfrak{r} | (\mathfrak{b}_0, \mathfrak{d}_2)} \mu(\mathfrak{r})$. Writing $\mathfrak{b}_0 = \mathfrak{r}\mathfrak{b}$, this becomes

$$\sum_{\mathfrak{d}_1 | \mathfrak{d}} \sum_{\mathfrak{r} | \mathfrak{d}_2} \mu(\mathfrak{r}) \xi(\mathfrak{d}_1 \mathfrak{r}) (\xi \psi)(\mathfrak{d}_2) \sum_{N\mathfrak{b} N\mathfrak{c} \leq T/(N\mathfrak{d} N\mathfrak{r})} \xi(\mathfrak{b}) (\xi \psi)(\mathfrak{c}).$$

For the inner sum, since $N\mathfrak{d} \leq Q^\varepsilon$ and $N\mathfrak{r} \leq Q^\varepsilon$, we have $T/(N\mathfrak{d} N\mathfrak{r}) \geq Q^{2\alpha+4\varepsilon-\varepsilon-\varepsilon} = Q^{2\alpha+2\varepsilon}$. Thus, the hyperbola estimate (7.21) applies to the inner sum and we have

$$\sum_{\substack{N\mathfrak{a} \leq T \\ \mathfrak{d} | \mathfrak{a}}} \xi(\mathfrak{a}) (1 * \psi)(\mathfrak{a}) \ll_{K,\varepsilon} Q^{-\eta_0} \sum_{\mathfrak{d}_1 | \mathfrak{d}} \sum_{\mathfrak{r} | \mathfrak{d}/\mathfrak{d}_1} \frac{T}{N\mathfrak{d} N\mathfrak{r}} \ll_{K,\varepsilon} \frac{T}{N\mathfrak{d}} Q^{-\eta_0} \tau(\mathfrak{d})^2.$$

Finally, since $N\mathfrak{d} \leq Q^\varepsilon$, the divisor bound for ideals gives $\tau(\mathfrak{d})^2 \ll_{K,\varepsilon,\eta_0} Q^{\eta_0/2}$. Therefore, we have

$$\sum_{\substack{N\mathfrak{a} \leq T \\ \mathfrak{d} | \mathfrak{a}}} \xi(\mathfrak{a}) (1 * \psi)(\mathfrak{a}) \ll_{K,\varepsilon} \frac{T}{N\mathfrak{d}} Q^{-\frac{\eta_0}{2}}.$$

□

Lemma 7.8 (Sharp Halász–Montgomery with weight $1 * \psi$). *Let $\varepsilon > 0$ and $C \geq 1$ be fixed. Suppose*

$$Q^{2\alpha+4\varepsilon} \leq X \leq Q^C.$$

Let $\chi_1, \dots, \chi_R$ be distinct Hecke characters modulo $\mathfrak{q}$, and let ψ be a non-principal quadratic Hecke character modulo $\mathfrak{q}$. Then there exists $\eta = \eta(\varepsilon) > 0$ such that, for any complex coefficients $a_{\mathfrak{n}}$,

$$\sum_{j=1}^R \left| \sum_{\substack{N\mathfrak{n} \leq X \\ (\mathfrak{n}, \mathcal{P}(Q^\varepsilon))=1}} (1 * \psi)(\mathfrak{n}) a_{\mathfrak{n}} \chi_j(\mathfrak{n}) \right|^2 \ll_{K,\varepsilon,C} (X \rho_K L(1, \psi) \mathcal{V}_\psi(Q) + X Q^{-\eta} R) \sum_{\substack{N\mathfrak{n} \leq X \\ (\mathfrak{n}, \mathcal{P}(Q^\varepsilon))=1}} (1 * \psi)(\mathfrak{n}) |a_{\mathfrak{n}}|^2.$$

Proof. Define

$$W(\mathfrak{n}) := (1 * \psi)(\mathfrak{n}).$$

Since ψ is quadratic, $W(\mathfrak{n}) \geq 0$.

By the duality principle [13, Sec. 7.1, p. 170], it suffices to prove that for arbitrary complex numbers $c_1, \dots, c_R$,

$$(7.22) \quad \sum_{\substack{N\mathfrak{n} \leq X \\ (\mathfrak{n}, \mathcal{P}(Q^\varepsilon))=1}} W(\mathfrak{n}) \left| \sum_{j=1}^R c_j \chi_j(\mathfrak{n}) \right|^2 \ll_{K,\varepsilon,C} (X \rho_K L(1, \psi) \mathcal{V}_\psi(Q) + X Q^{-\eta} R) \sum_{j=1}^R |c_j|^2.$$

Fix $s_0 \geq 19$, and let

$$z := Q^{\varepsilon/s_0}, \quad D := z^{s_0} = Q^\varepsilon.$$

Let $\lambda_{\mathfrak{d}}^+$ be the upper-bound sieve weights with sifting range z and level D . Applying the sieve inequality $1_{(\mathfrak{n}, \mathcal{P}(Q^\varepsilon))=1} \leq 1_{(\mathfrak{n}, \mathcal{P}(z))=1} \leq \sum_{\mathfrak{d} | \mathfrak{n}} \lambda_{\mathfrak{d}}^+$, the left-hand side of (7.22) is bounded by

$$(7.23) \quad \sum_{N\mathfrak{n} \leq X} \left(\sum_{\mathfrak{d} | \mathfrak{n}} \lambda_{\mathfrak{d}}^+ \right) W(\mathfrak{n}) \left| \sum_{j=1}^R c_j \chi_j(\mathfrak{n}) \right|^2 = \sum_{j,k=1}^R c_j \overline{c_k} \sum_{\mathfrak{d} | \mathcal{P}(z)} \lambda_{\mathfrak{d}}^+ \sum_{\substack{N\mathfrak{n} \leq X \\ \mathfrak{d} | \mathfrak{n}}} \chi_j(\mathfrak{n}) \overline{\chi_k(\mathfrak{n})} W(\mathfrak{n}).$$

Write $\xi_{j,k} := \chi_j \overline{\chi_k}$. First suppose $\xi_{j,k} \notin \{\chi_0, \psi\}$. For $N\mathfrak{d} \leq D = Q^\varepsilon$ and $X \geq Q^{2\alpha+4\varepsilon}$, Lemma 7.7 gives

$$\sum_{\substack{N\mathfrak{n} \leq X \\ \mathfrak{d} | \mathfrak{n}}} \xi_{j,k}(\mathfrak{n}) W(\mathfrak{n}) \ll_{K,\varepsilon} \frac{X}{N\mathfrak{d}} Q^{-\eta_0}$$

for some $\eta_0 = \eta_0(\varepsilon) > 0$. Using $|\lambda_{\mathfrak{d}}^+| \leq 1$ and that

$$\sum_{\substack{\mathfrak{d} | \mathcal{P}(z) \\ N\mathfrak{d} \leq D}} \frac{1}{N\mathfrak{d}} \leq \prod_{N\mathfrak{p} < z} \left(1 + \frac{1}{N\mathfrak{p}} \right) \ll_K \log z \ll_{K,\varepsilon} \log Q,$$

we have, for some $\eta_1 = \eta_1(\varepsilon) > 0$,

$$\sum_{\mathfrak{d} | \mathcal{P}(z)} \lambda_{\mathfrak{d}}^+ \sum_{\substack{N\mathfrak{n} \leq X \\ \mathfrak{d} | \mathfrak{n}}} \xi_{j,k}(\mathfrak{n}) W(\mathfrak{n}) \ll_{K,\varepsilon} X Q^{-\eta_1}.$$

Therefore the total contribution of these pairs with $\xi_{j,k} \notin \{\chi_0, \psi\}$ to (7.23) is

$$(7.24) \quad \ll_{K,\varepsilon} XQ^{-\eta_1} \sum_{j,k=1}^R |c_j||c_k| \ll XQ^{-\eta_1} R \sum_{j=1}^R |c_j|^2.$$

It remains to treat the pairs for which $\xi_{j,k} \in \{\chi_0, \psi\}$. If $\xi_{j,k} = \chi_0$, then $j = k$. If $\xi_{j,k} = \psi$, then $\chi_j = \psi\chi_k$, for each j there is at most one such k , and these corresponding k 's are distinct as j varies. Hence

$$\sum_{\substack{j,k \\ \xi_{j,k} \in \{\chi_0, \psi\}}} |c_j||c_k| \leq 2 \sum_{j=1}^R |c_j|^2.$$

Note that $\sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+ \geq 0$ for any $\mathfrak{n}$, the exceptional contribution in (7.23) is bounded in absolute value by

$$\sum_{\substack{j,k \\ \xi_{j,k} \in \{\chi_0, \psi\}}} |c_j||c_k| \sum_{N\mathfrak{n} \leq X} \left(\sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+ \right) W(\mathfrak{n}) \leq 2 \sum_{j=1}^R |c_j|^2 \sum_{N\mathfrak{n} \leq X} \left(\sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+ \right) W(\mathfrak{n}) = 2 \sum_{j=1}^R |c_j|^2 \sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_{\mathfrak{d}}^+ \sum_{\substack{N\mathfrak{n} \leq X \\ \mathfrak{d}|\mathfrak{n}}} W(\mathfrak{n}).$$

For $N\mathfrak{d} \leq D = Q^\varepsilon$, by Lemma 7.3, there is some $\eta_2 = \eta_2(\varepsilon) > 0$, such that

$$\sum_{\substack{N\mathfrak{n} \leq X \\ \mathfrak{d}|\mathfrak{n}}} W(\mathfrak{n}) = X\rho_K L(1, \psi) h(\mathfrak{d}) + O_{K,\varepsilon} \left(\tau_3(\mathfrak{d}) \left(\frac{X}{N\mathfrak{d}} \right)^{1-\eta_2} \right).$$

Summing over $\mathfrak{d}$, the error term is bounded by

$$\sum_{N\mathfrak{d} \leq D} \tau_3(\mathfrak{d}) \left(\frac{X}{N\mathfrak{d}} \right)^{1-\eta_2} \leq X^{1-\eta_2} \sum_{N\mathfrak{d} \leq D} \frac{\tau_3(\mathfrak{d})}{(N\mathfrak{d})^{1-\eta_2}} \ll_{K,\varepsilon} X^{1-\eta_2} D^{\eta_2} (\log D)^{O_K(1)}.$$

Hence, for some $\eta_3 = \eta_3(\eta) > 0$,

$$\sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_{\mathfrak{d}}^+ \sum_{\substack{N\mathfrak{n} \leq X \\ \mathfrak{d}|\mathfrak{n}}} W(\mathfrak{n}) \ll_{K,\varepsilon} X\rho_K L(1, \psi) \sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_{\mathfrak{d}}^+ h(\mathfrak{d}) + XQ^{-\eta_3}.$$

By Lemma 7.1 and Lemma 7.4,

$$\sum_{\mathfrak{d}|\mathcal{P}(z)} \lambda_{\mathfrak{d}}^+ h(\mathfrak{d}) \ll_K \prod_{N\mathfrak{p} < z} (1 - h(\mathfrak{p})) \asymp_{K,\varepsilon} \mathcal{V}_\psi(Q).$$

Thus the exceptional contribution from pairs with $\xi_{j,k} \in \{\chi_0, \psi\}$ in (7.23) is

$$(7.25) \quad \ll_{K,\varepsilon} (X\rho_K L(1, \psi) \mathcal{V}_\psi(Q) + XQ^{-\eta_3}) \sum_{j=1}^R |c_j|^2.$$

Combining (7.24), (7.25) and taking $\eta = \min\{\eta_1, \eta_3\}$ gives (7.22). $\square$

Proposition 7.9 (Exceptional quadratic transference). *Assume the setup of Proposition 3.3. Let $A \subseteq G$ be the dense-model set constructed there.*

Let ψ be a non-principal quadratic Hecke character modulo $\mathfrak{q}$. Suppose that, for some $c \in G$ and some M with

$$Q^{2\alpha+5\varepsilon} \leq M \leq X/2,$$

$$(7.26) \quad \sum_{\substack{[\mathfrak{p}]a_1 a_2 = c, \ a_1, a_2 \in A \\ M < N\mathfrak{p} \leq 2M, \ \psi(\mathfrak{p})=1}} 1 \gg \frac{|G| M \rho_K L(1, \psi) \mathcal{V}_\psi(Q)}{(\log Q)^{1/2-\varepsilon}}.$$

Then $c \in E_3(X; \mathfrak{q})$.

Proof. Let f and g be the functions appearing in the proof of Proposition 3.3. Thus $f : \mathcal{A}(X; \mathfrak{q}) \rightarrow \mathbb{R}_{\geq 0}$ is given by

$$f(\mathfrak{a}) := \frac{\vartheta}{2} V_{\mathfrak{q}} \log X \, 1_{\mathfrak{a}=\mathfrak{p}} 1_{N\mathfrak{p} \geq z},$$

and $g : G \rightarrow [0, 1 + o(1)]$ is its dense model.

Define $f_0 : \mathcal{A}(X; \mathfrak{q}) \rightarrow \mathbb{R}_{\geq 0}$ by

$$f_0(\mathfrak{a}) := 1_{\mathfrak{a}=\mathfrak{p}} 1_{M < N\mathfrak{p} \leq 2M} 1_{\psi(\mathfrak{p})=1}.$$

Note that the function f_0 is supported on prime ideals of norm at most X since $2M \leq X$.

For $c \in G$, define

$$T_{f,f_0}(c) := \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} \sum_{\substack{\mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3 \in \mathcal{A}(X; \mathfrak{q}) \\ [\mathfrak{a}_1 \mathfrak{a}_2 \mathfrak{a}_3] = c}} f(\mathfrak{a}_1) f(\mathfrak{a}_2) f_0(\mathfrak{a}_3),$$

$$T_{g,f_0}(c) := \frac{1}{|G|^2} \sum_{\substack{a_1, a_2 \in G, \mathfrak{a}_3 \in \mathcal{A}(X; \mathfrak{q}) \\ a_1 a_2 [\mathfrak{a}_3] = c}} g(a_1) g(a_2) f_0(\mathfrak{a}_3).$$

If $T_{f,f_0}(c) > 0$, then by definition, $c \in E_3(X; \mathfrak{q})$.

For a character χ of G , define

$$\widehat{f}(\chi) := \frac{1}{|\mathcal{A}(X; \mathfrak{q})|} \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a}) \chi(\mathfrak{a}), \quad \widehat{g}(\chi) := \frac{1}{|G|} \sum_{a \in G} g(a) \chi(a),$$

and

$$\widehat{f_0}(\chi) := \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f_0(\mathfrak{a}) \chi(\mathfrak{a}) = \sum_{\substack{M < N\mathfrak{p} \leq 2M \\ \psi(\mathfrak{p})=1}} \chi(\mathfrak{p}).$$

Here we do not include the normalization factor $|\mathcal{A}(X; \mathfrak{q})|^{-1}$ in the definition of $\widehat{f_0}(\chi)$. This avoids introducing a different normalization scale and leads to cleaner definition for T_{g,f_0} .

By orthogonality of characters,

$$T_{f,f_0}(c) - T_{g,f_0}(c) = \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \left(\widehat{f}(\chi)^2 - \widehat{g}(\chi)^2 \right) \widehat{f_0}(\chi) \overline{\chi(c)}.$$

Therefore

$$(7.27) \quad |T_{f,f_0}(c) - T_{g,f_0}(c)| \leq \frac{1}{|G|} \sum_{\chi \in \widehat{G}} (|\widehat{f}(\chi)| + |\widehat{g}(\chi)|) |\widehat{f_0}(\chi)| |\widehat{f}(\chi) - \widehat{g}(\chi)|.$$

Define

$$A_\psi(M) := M \rho_K L(1, \psi) \mathcal{V}_\psi(Q), \quad \mathcal{L} := n_K + 10.$$

We split the characters according to the size of $\widehat{f_0}(\chi)$. Define

$$\mathcal{X}_1 := \left\{ \chi \in \widehat{G} : |\widehat{f_0}(\chi)| < (\log Q)^{-\mathcal{L}} A_\psi(M) \right\}, \quad \mathcal{X}_2 := \widehat{G} \setminus \mathcal{X}_1.$$

For $\chi \in \mathcal{X}_1$, using $|\widehat{g}(\chi)| \leq |\widehat{f}(\chi)|$, $|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq |\widehat{f}(\chi)|$, and the mean-square estimate $\sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 \ll_K (\log Q)^{n_K+1}$ from (3.17), we have

$$(7.28) \quad \frac{1}{|G|} \sum_{\chi \in \mathcal{X}_1} (|\widehat{f}(\chi)| + |\widehat{g}(\chi)|) |\widehat{f_0}(\chi)| |\widehat{f}(\chi) - \widehat{g}(\chi)| \ll_K \frac{A_\psi(M)}{|G|(\log Q)^\mathcal{L}} \sum_{\chi \in \widehat{G}} |\widehat{f}(\chi)|^2 \ll_K \frac{A_\psi(M)}{|G|(\log Q)^9}.$$

For $\chi \in \mathcal{X}_2$, we will apply Lemma 7.8 at length $2M$ with coefficients

$$a_{\mathfrak{n}} = \frac{1}{2} 1_{\mathfrak{n}=\mathfrak{p}} 1_{M < N\mathfrak{p} \leq 2M} 1_{\psi(\mathfrak{p})=1}.$$

Since $(1 * \psi)(\mathfrak{p}) = 2$ when $\psi(\mathfrak{p}) = 1$ and $M > Q^\varepsilon$, we have

$$\sum_{\substack{N\mathfrak{n} \leq 2M \\ (\mathfrak{n}, \mathcal{P}(Q^\varepsilon))=1}} (1 * \psi)(\mathfrak{n}) a_{\mathfrak{n}} \chi(\mathfrak{n}) = \widehat{f_0}(\chi).$$

Moreover, by Lemma 7.5,

$$\sum_{\substack{N\mathfrak{n} \leq 2M \\ (\mathfrak{n}, \mathcal{P}(Q^\varepsilon))=1}} (1 * \psi)(\mathfrak{n}) |a_{\mathfrak{n}}|^2 = \frac{1}{2} \# \{ \mathfrak{p} : M < N\mathfrak{p} \leq 2M, \psi(\mathfrak{p}) = 1 \} \ll_{K,\varepsilon} A_\psi(M).$$

Therefore Lemma 7.8 gives, for some $\eta_0 = \eta_0(\varepsilon) > 0$,

$$(7.29) \quad \sum_{\chi \in \mathcal{X}_2} |\widehat{f}_0(\chi)|^2 \ll_{K,\varepsilon} (A_\psi(M) + MQ^{-\eta_0} |\mathcal{X}_2|) A_\psi(M).$$

We now bound $|\mathcal{X}_2|$. By the definition of $\mathcal{X}_2$, $|\widehat{f}_0(\chi)| \geq (\log Q)^{-\mathcal{L}} A_\psi(M)$ for any $\chi \in \mathcal{X}_2$. Using this lower bound in (7.29), we have

$$|\mathcal{X}_2| (\log Q)^{-2\mathcal{L}} A_\psi(M)^2 \ll_{K,\varepsilon} A_\psi(M)^2 + MQ^{-\eta_0} |\mathcal{X}_2| A_\psi(M).$$

The second term on the right is absorbed into the left. Indeed, by the Siegel lower bound (7.11) and Lemma 7.4, for any $\delta > 0$,

$$(7.30) \quad \frac{A_\psi(M)}{M} = \rho_K L(1, \psi) \mathcal{V}_\psi(Q) \gg_{K,\delta} Q^{-2\delta}.$$

Choosing $\delta > 0$ sufficiently small, we have $MQ^{-\eta_0} A_\psi(M) \leq \frac{1}{2} (\log Q)^{-2\mathcal{L}} A_\psi(M)^2$ for Q sufficiently large. Therefore

$$(7.31) \quad |\mathcal{X}_2| \ll_{K,\varepsilon} (\log Q)^{2\mathcal{L}}.$$

Applying (7.30) and (7.31) to (7.29), we have

$$(7.32) \quad \sum_{\chi \in \mathcal{X}_2} |\widehat{f}_0(\chi)|^2 \ll_{K,\varepsilon} A_\psi(M)^2.$$

We now prove the mean-square bound for f restricted to $\mathcal{X}_2$. This follows similarly as in the proof of Proposition 3.3. By Lemma 2.9(i), for some $\eta_1 = \eta_1(\varepsilon) > 0$,

$$\sum_{\chi \in \mathcal{X}_2} |\widehat{f}(\chi)|^2 \ll_{K,\varepsilon} \left(\frac{X}{\log Q} + |\mathcal{X}_2| X^{1-\eta_1} Q^{\varepsilon\eta_1} \right) \frac{1}{|\mathcal{A}(X; \mathfrak{q})|^2} \sum_{\mathfrak{a} \in \mathcal{A}(X; \mathfrak{q})} f(\mathfrak{a})^2.$$

Using (3.14), this gives

$$\sum_{\chi \in \mathcal{X}_2} |\widehat{f}(\chi)|^2 \ll_{K,\varepsilon} \frac{\log X}{\log Q} + |\mathcal{X}_2| X^{-\eta_1} Q^{\varepsilon\eta_1} \log X.$$

Since $Q^{2\alpha+5\varepsilon} \leq X \leq Q^C$, $|\mathcal{X}_2| \ll_{K,\varepsilon} (\log Q)^{2\mathcal{L}}$, taking ε sufficiently small and fixed, the second term is $o_{K,\varepsilon}(1)$, and $\log X / \log Q \leq C$. Hence

$$\sum_{\chi \in \mathcal{X}_2} |\widehat{f}(\chi)|^2 \ll_{K,\varepsilon} 1.$$

Moreover, $|\widehat{g}(\chi)| \leq |\widehat{f}(\chi)|$, this implies

$$(7.33) \quad \sum_{\chi \in \mathcal{X}_2} (|\widehat{f}(\chi)| + |\widehat{g}(\chi)|)^2 \ll_{K,\varepsilon} 1.$$

Finally, Proposition 3.3 gives the dense-model estimate

$$|\widehat{f}(\chi) - \widehat{g}(\chi)| \leq \delta, \quad \delta = (\log Q)^{-1/2+\varepsilon/2}.$$

Combining this with (7.32), (7.33), and the Cauchy–Schwarz inequality, we obtain

$$(7.34) \quad \begin{aligned} & \frac{1}{|G|} \sum_{\chi \in \mathcal{X}_2} (|\widehat{f}(\chi)| + |\widehat{g}(\chi)|) |\widehat{f}_0(\chi)| |\widehat{f}(\chi) - \widehat{g}(\chi)| \\ & \leq \frac{\delta}{|G|} \left(\sum_{\chi \in \mathcal{X}_2} (|\widehat{f}(\chi)| + |\widehat{g}(\chi)|)^2 \right)^{1/2} \left(\sum_{\chi \in \mathcal{X}_2} |\widehat{f}_0(\chi)|^2 \right)^{1/2} \ll_{K,\varepsilon} \frac{\delta}{|G|} A_\psi(M). \end{aligned}$$

Combining (7.27), (7.28), and (7.34), we obtain

$$(7.35) \quad T_{f,f_0}(c) = T_{g,f_0}(c) + O_{K,\varepsilon} \left(\frac{\delta}{|G|} A_\psi(M) \right).$$

By the definition of A in (3.15), $g(a) \geq \varepsilon/10$ for $a \in A$. Thus, by hypothesis (7.26),

$$T_{g,f_0}(c) \gg_\varepsilon \frac{1}{|G|^2} \sum_{\substack{[\mathfrak{p}] a_1 a_2 = c, \ a_1, a_2 \in A \\ M < N\mathfrak{p} \leq 2M, \ \psi(\mathfrak{p})=1}} 1 \gg \frac{A_\psi(M)}{|G|(\log Q)^{1/2-\varepsilon}}.$$

This dominates the error term in (7.35) as $\delta = (\log Q)^{-1/2+\varepsilon/2}$. Hence $T_{f,f_0}(c) > 0$, and $c \in E_3(X; \mathfrak{q})$. $\square$

Corollary 7.10. *Assume the setup of Proposition 4.4(b), and let $A', B' \subseteq A$ be the sets there. Let ψ be a non-principal quadratic Hecke character modulo $\mathfrak{q}$. Suppose that, for some $c \in G$ and some M with*

$$(7.36) \quad Q^{2\alpha+5\varepsilon} \leq M \leq X/2, \\ \sum_{\substack{[\mathfrak{p}]d=c, \ d \in A' \cdot B' \\ M < N\mathfrak{p} \leq 2M, \ \psi(\mathfrak{p})=1}} 1 \gg M \rho_K L(1, \psi) \mathcal{V}_\psi(Q).$$

Then $c \in E_3(X; \mathfrak{q})$.

Proof. By (4.3) in the proof of Proposition 4.4(b.ii), we have that for every $d \in A' \cdot B'$,

$$(1_A * 1_A)(d) = \#\{(a_1, a_2) \in A^2 : a_1 a_2 = d\} \gg |G|.$$

Consequently,

$$\sum_{\substack{[\mathfrak{p}]a_1 a_2=c, \ a_1, a_2 \in A \\ M < N\mathfrak{p} \leq 2M, \ \psi(\mathfrak{p})=1}} 1 \geq \sum_{\substack{[\mathfrak{p}]d=c, \ d \in A' \cdot B' \\ M < N\mathfrak{p} \leq 2M, \ \psi(\mathfrak{p})=1}} (1_A * 1_A)(d) \gg |G| \sum_{\substack{[\mathfrak{p}]d=c, \ d \in A' \cdot B' \\ M < N\mathfrak{p} \leq 2M, \ \psi(\mathfrak{p})=1}} 1.$$

Hence (7.36) implies condition (7.26) of Proposition 7.9. Applying Proposition 7.9, we conclude that $c \in E_3(X; \mathfrak{q})$. $\square$

8. PRODUCTS OF TWO PRIME IDEALS

In this section, we will prove Theorem 1.2. Before the proof we need some lemmas. As before, we write $G = \text{Cl}_{\mathfrak{q}}^{(\infty)}$, $Q = N\mathfrak{q}$, and assume that $(\text{CS}(\alpha))$ and $(\text{L}^b(\alpha_0))$ hold for fixed parameters $0 < \alpha_0 \leq \alpha < 1$. Note that $(\text{L}^b(\alpha_0))$ implies $(\text{CS}^b(\alpha_0))$.

Lemma 8.1. *Let $\kappa > 0$ and $B \geq \max(1, 3\alpha) + \kappa$ be fixed, and let $\varepsilon > 0$ be sufficiently small. Set $X := Q^B$, $z := X^{2/3}$, and let $\mathcal{P}_1, \mathcal{P}_2$ be finite sets of prime ideals such that for some $Q^{2\varepsilon} \leq Y_1 \leq X/2$,*

$$\mathcal{P}_1 \subseteq \{\mathfrak{p} \nmid \mathfrak{q} : Y_1 < N\mathfrak{p} \leq 2Y_1\}, \quad \mathcal{P}_2 \subseteq \{\mathfrak{p} \nmid \mathfrak{q} : z < N\mathfrak{p} \leq X\}.$$

Suppose that, for some fixed constant $C_0 \geq 1$ and for any $\nu > 0$,

$$(8.1) \quad |\mathcal{P}_1| \gg_\nu Y_1 Q^{-\nu}, \quad \frac{2X}{\log z} \leq (C_0 + o(1))|\mathcal{P}_2|.$$

Define the multiplicative energy

$$\mathcal{E}_G(\mathcal{P}_1, \mathcal{P}_2) := \#\{(\mathfrak{p}_1, \mathfrak{p}'_1, \mathfrak{p}_2, \mathfrak{p}'_2) \in \mathcal{P}_1^2 \times \mathcal{P}_2^2 : [\mathfrak{p}_1 \mathfrak{p}_2] = [\mathfrak{p}'_1 \mathfrak{p}'_2]\}.$$

Then

$$\mathcal{E}_G(\mathcal{P}_1, \mathcal{P}_2) \leq \frac{C_0 + o_{K,\varepsilon}(1)}{|G|} |\mathcal{P}_1|^2 |\mathcal{P}_2|^2.$$

Proof. Let $\lambda_{\mathfrak{d}}^+$ be the upper-bound linear sieve weights as in Lemma 2.1 with sifting range z and level $D = z$ so that $s = 1$. Since all prime ideals in $\mathcal{P}_2$ have norm $> z$, using the upper bound sieve inequality $1_{\mathfrak{n}=\mathfrak{p} \in \mathcal{P}_2} \leq 1_{(\mathfrak{n}, \mathcal{P}(z))=1} \leq \sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+$, we have

$$\mathcal{E}_G(\mathcal{P}_1, \mathcal{P}_2) \leq \sum_{\mathfrak{p}_1, \mathfrak{p}'_1 \in \mathcal{P}_1} \sum_{\mathfrak{p}_2 \in \mathcal{P}_2} \sum_{\substack{N\mathfrak{n} \leq X \\ (\mathfrak{n}, \mathfrak{q})=1}} \left(\sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+ \right) 1_{[\mathfrak{p}_1 \mathfrak{p}_2] = [\mathfrak{p}'_1 \mathfrak{n}]}.$$

By orthogonality of characters of G , this is

$$(8.2) \quad \mathcal{E}_G(\mathcal{P}_1, \mathcal{P}_2) \leq \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \left| \sum_{\mathfrak{p} \in \mathcal{P}_1} \chi(\mathfrak{p}) \right|^2 \left(\sum_{\mathfrak{p} \in \mathcal{P}_2} \chi(\mathfrak{p}) \right) \left(\sum_{\substack{N\mathfrak{n} \leq X \\ (\mathfrak{n}, \mathfrak{q})=1}} \bar{\chi}(\mathfrak{n}) \sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+ \right).$$

The contribution of the principal character $\chi = \chi_0$ in (8.2) is

$$\frac{1}{|G|} |\mathcal{P}_1|^2 |\mathcal{P}_2| \sum_{\substack{N\mathfrak{n} \leq X \\ (\mathfrak{n}, \mathfrak{q})=1}} \sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+.$$

Interchanging the order of summation and applying the ideal-counting estimate in Lemma 2.8, we get

$$\begin{aligned} \sum_{\substack{N\mathfrak{n} \leq X \\ (\mathfrak{n}, \mathfrak{q})=1}} \sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+ &= \sum_{(\mathfrak{d}, \mathfrak{q})=1} \lambda_{\mathfrak{d}}^+ \#\left\{ \mathfrak{m} : N\mathfrak{m} \leq \frac{X}{N\mathfrak{d}}, (\mathfrak{m}, \mathfrak{q})=1 \right\} \\ &= V_{\mathfrak{q}} X \sum_{(\mathfrak{d}, \mathfrak{q})=1} \frac{\lambda_{\mathfrak{d}}^+}{N\mathfrak{d}} + O_{K, \delta} \left(\sum_{N\mathfrak{d} \leq z} \left(\frac{X}{N\mathfrak{d}} \right)^{1-1/n_K} Q^{\delta} \right) \end{aligned}$$

where $V_{\mathfrak{q}} := \rho_K \prod_{\mathfrak{p}|\mathfrak{q}} (1 - \frac{1}{N\mathfrak{p}})$, Since $\sum_{N\mathfrak{d} \leq z} (N\mathfrak{d})^{-1+1/n_K} \ll_K z^{1/n_K}$, the error term is $\ll_{K, \delta} Q^{\delta} X^{1-1/n_K} z^{1/n_K} = o_K(X/\log z)$ provided $\delta > 0$ is chosen sufficiently small.

By the upper-bound linear sieve, Lemma 2.1 with $s = 1$, together with the prime-ideal Mertens theorem (3.2), we have

$$V_{\mathfrak{q}} \sum_{(\mathfrak{d}, \mathfrak{q})=1} \frac{\lambda_{\mathfrak{d}}^+}{N\mathfrak{d}} \leq (2e^{\gamma} + o_K(1)) \rho_K \prod_{N\mathfrak{p} < z} \left(1 - \frac{1}{N\mathfrak{p}} \right) = \frac{2 + o_K(1)}{\log z}.$$

Hence, using the lower bound for $|\mathcal{P}_2|$ in (8.1), the contribution of the principal character is

$$(8.3) \quad \leq \frac{(2 + o_K(1))}{|G|} |\mathcal{P}_1|^2 |\mathcal{P}_2| \frac{X}{\log z} \leq \frac{C_0 + o_K(1)}{|G|} |\mathcal{P}_1|^2 |\mathcal{P}_2|^2.$$

Write

$$S_1(\chi) := \sum_{\mathfrak{p} \in \mathcal{P}_1} \chi(\mathfrak{p}), \quad S_2(\chi) := \sum_{\mathfrak{p} \in \mathcal{P}_2} \chi(\mathfrak{p}), \quad R(\chi) := \sum_{\substack{N\mathfrak{n} \leq X \\ (\mathfrak{n}, \mathfrak{q})=1}} \chi(\mathfrak{n}) \sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+.$$

We claim that the non-principal-characters contribution in (8.2) is

$$(8.4) \quad \frac{1}{|G|} \sum_{\chi \neq \chi_0} |S_1(\chi)|^2 |S_2(\chi)| |R(\bar{\chi})| \ll_{K, \varepsilon} \frac{|\mathcal{P}_1|^2 |\mathcal{P}_2|^2}{|G| \log Q}.$$

Assuming the claim, combining it with (8.3) proves the lemma.

We now prove (8.4).

Split the non-principal characters into

$$\mathcal{X}_1 := \left\{ \chi \neq \chi_0 : |S_1(\chi)| \leq \frac{|\mathcal{P}_1|}{(\log Q)^{A_1}} \right\},$$

and

$$\mathcal{X}_2 := \left\{ \chi \neq \chi_0 : |S_1(\chi)| > \frac{|\mathcal{P}_1|}{(\log Q)^{A_1}} \right\},$$

where $A_1 > 0$ is a constant to be chosen later.

For $\mathcal{X}_1$, Cauchy–Schwarz gives

$$\frac{1}{|G|} \sum_{\chi \in \mathcal{X}_1} |S_1(\chi)|^2 |S_2(\chi)| |R(\bar{\chi})| \leq \frac{|\mathcal{P}_1|^2}{|G| (\log Q)^{2A_1}} \left(\sum_{\chi \in \hat{G}} |S_2(\chi)|^2 \right)^{1/2} \left(\sum_{\chi \in \hat{G}} |R(\chi)|^2 \right)^{1/2}.$$

Applying Lemma 2.7, with the coefficients supported on $\mathcal{P}_2$ and using that that $X \geq Q^{1+\kappa}$, we have

$$(8.5) \quad \sum_{\chi \in \hat{G}} |S_2(\chi)|^2 \ll_K (X + Q) (\log(3Q))^{n_K} |\mathcal{P}_2| \ll_K X (\log Q)^{n_K} |\mathcal{P}_2|.$$

Next write

$$R(\chi) = \sum_{\substack{N\mathfrak{n} \leq X \\ (\mathfrak{n}, \mathfrak{q})=1}} b_{\mathfrak{n}} \chi(\mathfrak{n}), \quad b_{\mathfrak{n}} := \sum_{\mathfrak{d}|\mathfrak{n}} \lambda_{\mathfrak{d}}^+.$$

Since $|\lambda_{\mathfrak{d}}^+| \leq 1$, we have $|b_{\mathfrak{n}}| \leq \tau(\mathfrak{n})$. Moreover, using $\tau(\mathfrak{n})^2 \leq \tau_4(\mathfrak{n})$, we have

$$\sum_{\substack{N\mathfrak{n} \leq X \\ (\mathfrak{n}, \mathfrak{q})=1}} |b_{\mathfrak{n}}|^2 \leq \sum_{N\mathfrak{n} \leq X} \tau(\mathfrak{n})^2 \leq \sum_{N\mathfrak{n} \leq X} \tau_4(\mathfrak{n}) \ll_K X (\log 2X)^3.$$

Applying Lemma 2.7 again gives

$$(8.6) \quad \sum_{\chi \in \widehat{G}} |R(\chi)|^2 \ll_K (X+Q)(\log(3Q))^{n_K} X(\log 2X)^3 \ll_K X^2(\log Q)^{n_K+3}.$$

From (8.1), we have $\frac{X}{\log z} \ll |\mathcal{P}_2|$. Combining (8.5) and (8.6), and using that $\log z \asymp \log Q$, we obtain

$$(8.7) \quad \frac{1}{|G|} \sum_{\chi \in \mathcal{X}_1} |S_1(\chi)|^2 |S_2(\chi)| |R(\bar{\chi})| \ll_K \frac{|\mathcal{P}_1|^2 |\mathcal{P}_2|^2}{|G|(\log Q)^{2A_1-n_K-3}}.$$

Choose A_1 so that $2A_1 - n_K - 3 \geq 1$; for example, $A_1 = n_K + 2$ suffices. Then (8.7) is $O_K\left(\frac{|\mathcal{P}_1|^2 |\mathcal{P}_2|^2}{|G|(\log Q)^{n_K+1}}\right)$.

We next consider $\mathcal{X}_2$. First we record a pointwise bound for $R(\chi)$ for χ non-principal. Write

$$R(\chi) = \sum_{\substack{\mathfrak{d}|\mathcal{P}(z), \ N\mathfrak{d} \leq D \\ (\mathfrak{d}, q)=1}} \lambda_{\mathfrak{d}}^+ \chi(\mathfrak{d}) \sum_{\substack{N\mathfrak{m} \leq X/N\mathfrak{d} \\ (\mathfrak{m}, q)=1}} \chi(\mathfrak{m}).$$

For $N\mathfrak{d} \leq D = z = X^{2/3}$, taking $\varepsilon < \kappa/3$, we have $\frac{X}{N\mathfrak{d}} \geq X^{1/3} \geq Q^{\alpha+\kappa/3} \geq Q^{\alpha+\varepsilon}$. Applying (CS(α)) to the inner sum, we have for some $\eta_0 = \eta_0(\varepsilon) > 0$,

$$\sum_{\substack{N\mathfrak{m} \leq X/N\mathfrak{d} \\ (\mathfrak{m}, q)=1}} \chi(\mathfrak{m}) \ll_{K, \varepsilon} \left(\frac{X}{N\mathfrak{d}}\right)^{1-\eta_0}.$$

Therefore,

$$(8.8) \quad R(\chi) \ll_{K, \varepsilon} X^{1-\eta_0} \sum_{N\mathfrak{d} \leq D} \frac{1}{(N\mathfrak{d})^{1-\eta_0}} \ll_K X^{1-\eta_0} D^{\eta_0} = XQ^{-\eta_1}, \quad \eta_1 := \frac{B}{3}\eta_0.$$

We now bound $|\mathcal{X}_2|$ using higher-moment estimates, similar to that in the proof of Proposition 3.3(iv). Define

$$\beta := \frac{\log Y_1}{\log Q}, \quad L := \left\lfloor \frac{B}{\beta} \right\rfloor.$$

Then $\beta \geq 2\varepsilon$, by the same elementary proof for (3.26), we have $L\beta > B/2 \geq \alpha + \kappa/2$. Taking ε sufficiently small, then $Y_1^L = Q^{L\beta} \geq Q^{\alpha+2\varepsilon}$. Expanding the L -th power, we have

$$(S_1(\chi))^L = \left(\sum_{\mathfrak{p} \in \mathcal{P}_1} \chi(\mathfrak{p}) \right)^L = \sum_{\mathfrak{n}} c_{\mathfrak{n}} \chi(\mathfrak{n}).$$

The coefficients $c_{\mathfrak{n}}$ are supported on products of L prime ideals from $\mathcal{P}_1$. Hence $Y_1^L < N\mathfrak{n} \leq (2Y_1)^L$ and $\sum_{\mathfrak{n}} |c_{\mathfrak{n}}|^2 \ll_L |\mathcal{P}_1|^L$. Hence, $\sum_{\mathfrak{n}} N\mathfrak{n} |c_{\mathfrak{n}}|^2 \ll_L Y_1^L |\mathcal{P}_1|^L$.

Moreover, since every $\mathfrak{p} \in \mathcal{P}_1$ has $N\mathfrak{p} > Y_1 \geq Q^{2\varepsilon}$, all ideals $\mathfrak{n}$ in the support of $c_{\mathfrak{n}}$ are coprime to $\mathcal{P}(Q^\varepsilon)$. Applying Lemma 2.9(ii) to the coefficients $a_{\mathfrak{n}} := N\mathfrak{n} c_{\mathfrak{n}}$ in the interval $Q^{\alpha+2\varepsilon} \leq Y_1^L < N\mathfrak{n} \leq (2Y_1)^L$, there is some $\eta_2 = \eta_2(\varepsilon) > 0$ such that

$$(8.9) \quad \begin{aligned} \sum_{\chi \in \mathcal{X}_2} |S_1(\chi)|^{2L} &\ll_{K, \varepsilon} \left(1 + |\mathcal{X}_2| \left(\frac{Q^\varepsilon}{Y_1^L}\right)^{\eta_2}\right) \sum_{\mathfrak{n}} N\mathfrak{n} |c_{\mathfrak{n}}|^2 \\ &\ll_{K, \varepsilon} (1 + |\mathcal{X}_2| Y_1^{-L\eta_2} Q^{\varepsilon\eta_2}) Y_1^L |\mathcal{P}_1|^L. \end{aligned}$$

On the other hand, by the definition of $\mathcal{X}_2$,

$$|\mathcal{X}_2| \left(\frac{|\mathcal{P}_1|}{(\log Q)^{A_1}} \right)^{2L} < \sum_{\chi \in \mathcal{X}_2} |S_1(\chi)|^{2L}.$$

Applying this in (8.9), we obtain

$$|\mathcal{X}_2| \frac{|\mathcal{P}_1|^{2L}}{(\log Q)^{2A_1L}} \ll_{K, \varepsilon} Y_1^L |\mathcal{P}_1|^L + |\mathcal{X}_2| Y_1^{L(1-\eta_2)} Q^{\varepsilon\eta_2} |\mathcal{P}_1|^L.$$

Since $|\mathcal{P}_1| \gg_\nu Y_1 Q^{-\nu}$ by (8.1) and $Y_1 \geq Q^{2\varepsilon}$. Choosing $\nu > 0$ sufficiently small, say $\nu < \frac{\varepsilon\eta_2(2L-1)}{L}$, then the second term on the right-hand side of (8.9) is absorbed into the left-hand side. Thus

$$(8.10) \quad |\mathcal{X}_2| \ll_{K,\varepsilon} \left(\frac{Y_1}{|\mathcal{P}_1|} \right)^L (\log Q)^{2A_1 L} \ll_{K,\varepsilon,\nu} Q^{\nu L} (\log Q)^{2A_1 L}.$$

Finally, using the trivial bounds $|S_1(\chi)| \leq |\mathcal{P}_1|$, $|S_2(\chi)| \leq |\mathcal{P}_2|$, together with (8.8) and (8.10), we have

$$\frac{1}{|G|} \sum_{\chi \in \mathcal{X}_2} |S_1(\chi)|^2 |S_2(\chi)| |R(\bar{\chi})| \ll_{K,\varepsilon} \frac{X Q^{-\eta_1}}{|G|} |\mathcal{P}_1|^2 |\mathcal{P}_2| |\mathcal{X}_2| \ll_{K,\varepsilon,\nu} \frac{X}{|G|} |\mathcal{P}_1|^2 |\mathcal{P}_2| Q^{\nu L - \eta_1} (\log Q)^{2A_1 L}.$$

Since $|\mathcal{P}_2| \gg \frac{X}{\log z} \gg \frac{X}{\log Q}$, taking $\nu > 0$ sufficiently small so that $\nu L < \min\{\varepsilon\eta_2(2L-1), \eta_1\}$, the above is

$$(8.11) \quad \frac{1}{|G|} \sum_{\chi \in \mathcal{X}_2} |S_1(\chi)|^2 |S_2(\chi)| |R(\bar{\chi})| \ll_{K,\varepsilon,\nu} \frac{|\mathcal{P}_1|^2 |\mathcal{P}_2|^2}{|G|} Q^{\nu L - \eta_1} (\log Q)^{2A_1 L + 1} \ll_{K,\varepsilon} \frac{|\mathcal{P}_1|^2 |\mathcal{P}_2|^2}{|G| (\log Q)}.$$

Combining (8.7) and (8.11) gives the claim (8.4), completing the proof. $\square$

Proposition 8.2. *Assume the setup of Proposition 4.4(b) and let $A', B' \subseteq G$ be the sets there. Assume additionally that $X \geq Q^{\max(1, 3\alpha, 4\alpha_0) + \kappa}$. Suppose Lemma 4.5(b) holds, and let H be the stabilizer of $A' \cdot B'$. Assume that*

$$[G : H] = 3k + 2, \quad k \in \{0, 1, 2\},$$

and that

$$A', B' \subset \bigcup_{i=1}^{k+1} a_i H, \quad A' \cdot B' = \bigcup_{j=1}^{2k+1} b_j H = \left(\bigcup_{i=1}^{k+1} a_i H \right)^2.$$

Then

$$|E_2(X; \mathfrak{q}) \setminus (A' \cdot B')| \geq (\gamma_k - O(\varepsilon^{1/2}) - o_K(1)) |G|,$$

where

$$\gamma_0 = \frac{1}{3}, \quad \gamma_1 = \frac{7}{45}, \quad \gamma_2 = \frac{1}{9}.$$

Proof. Let

$$S := \bigcup_{i=1}^{k+1} a_i H, \quad Y := [G : H] = 3k + 2, \quad B := \frac{\log X}{\log Q}.$$

We follow the discussion in the proof of Theorem 1.1 in Section 6. There are two cases.

The non-complementary case. Suppose that S and $S^2 = A' \cdot B'$ are not complementary subsets of G/H . By (6.5), there exist a coset $a_0 H \not\subset S$ and $\beta \in [B/3, 2B/3]$ such that

$$(8.12) \quad \sum_{\substack{Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta \\ [\mathfrak{p}] \in a_0 H}} \frac{1}{N\mathfrak{p}} \gg_{K,\varepsilon} 1.$$

The complementary case. Suppose that S and S^2 are complementary subsets of G/H .

If $k = 1$ or $k = 2$, then G/H has order 5 or 8. In these cases, by Corollary 5.3 or (6.6), there exist a coset $a_0 H \not\subset S$ and $\beta \in (2\varepsilon, 4\alpha_0]$ such that

$$(8.13) \quad \sum_{\substack{Q^{\beta-\varepsilon} < N\mathfrak{p} \leq Q^\beta \\ [\mathfrak{p}] \in a_0 H}} \frac{1}{N\mathfrak{p}} \gg_{K,\varepsilon} 1.$$

In either case (8.12) or (8.13), $Q^\beta \leq X/2$. A dyadic decomposition of the interval $[Q^{\beta-\varepsilon}, Q^\beta]$ therefore gives Y_1 with $2Y_1 \leq X$ such that

$$\#\{\mathfrak{p} : Y_1 < N\mathfrak{p} \leq 2Y_1, [\mathfrak{p}] \in a_0 H\} \gg_{K,\varepsilon} \frac{Y_1}{\log Q}.$$

We now consider the complementary case with $k = 0$, following Case 2 in the proof of Theorem 1.1. In this situation, $[G : H] = 2$ and $S \neq H$. Let ψ be the non-principal quadratic character of G with kernel H . Then, $\psi(\mathfrak{p}) = 1 \iff [\mathfrak{p}] \in H$.

Choose a fixed number $B_0 < B$ with $B_0 > \max\{2\alpha + \alpha_0, 2\alpha_0\}$, for example $B_0 = B - \kappa/2$. Applying Lemma 7.6 to ψ , one of the following two alternatives holds for $\varepsilon > 0$ sufficiently small. *Case (i).* Suppose that Lemma 7.6(i) holds. Then there exists $\beta_* \in (\max\{2\alpha, B_0/2\}, B_0 - \alpha_0)$ such that

$$\sum_{\substack{Q^{\alpha+\varepsilon} < N\mathfrak{p} \leq Q^{\beta_*} \\ [\mathfrak{p}] \in H}} \frac{1}{N\mathfrak{p}} \geq c_*,$$

A dyadic decomposition of the interval $[Q^{\alpha+\varepsilon}, Q^{\beta_*}]$ then gives Y_1 with $2Y_1 \leq X$ such that

$$\#\{\mathfrak{p} : Y_1 < N\mathfrak{p} \leq 2Y_1, [\mathfrak{p}] \in H\} \gg_{K,\varepsilon} \frac{Y_1}{\log Q}.$$

Case (ii). Suppose that Lemma 7.6(ii) holds. Then there exists $M \in [Q^{\beta_*}, Q^{B_0}]$ such that

$$\#\{\mathfrak{p} : M < N\mathfrak{p} \leq 2M, \psi(\mathfrak{p}) = 1\} \gg_{K,\varepsilon} M \rho_K L(1, \psi) \mathcal{V}_\psi(Q).$$

Since $\psi(\mathfrak{p}) = 1$ if and only if $[\mathfrak{p}] \in H$, and $L(1, \psi) \gg_{K,\delta} Q^{-\delta}$, $\mathcal{V}_\psi(Q) \gg_{K,\delta} Q^{-\delta}$, it follows that, for every fixed $\delta > 0$,

$$\#\{\mathfrak{p} : M < N\mathfrak{p} \leq 2M, [\mathfrak{p}] \in H\} \gg_{K,\varepsilon,\delta} M Q^{-\delta}.$$

Since $M \leq Q^{B_0}$, we also have $2M \leq X$.

Therefore, in both the complementary and non-complementary cases, we obtain a coset $a_0H \not\subset S$ and a dyadic interval $Y_1 < N\mathfrak{p} \leq 2Y_1$ with $2Y_1 \leq X$ such that writing

$$\mathcal{P}_1 := \{\mathfrak{p} : Y_1 < N\mathfrak{p} \leq 2Y_1, [\mathfrak{p}] \in a_0H\},$$

we have, for every fixed $\delta > 0$,

$$(8.14) \quad |\mathcal{P}_1| \gg_{K,\varepsilon,\delta} Y_1 Q^{-\delta}.$$

Since $a_0H \not\subset S$, the product $(S \cup a_0H)S$ contains strictly more H -cosets than S^2 . Hence there is $j_0 \in \{1, \dots, k+1\}$ such that $a_0a_{j_0}H \not\subset A' \cdot B'$.

We now lower-bound the number of primes in the coset $a_{j_0}H$. For a coset bH , write

$$\pi(T; bH) := \#\{\mathfrak{p} \subset \mathcal{O}_K : \mathfrak{p} \nmid \mathfrak{q}, N\mathfrak{p} \leq T, [\mathfrak{p}] \in bH\}.$$

If $bH \not\subset S$, then $A' \cap B' \cap bH = \emptyset$. Hence, by Proposition 4.4(b.vi),

$$\pi(X; bH) \ll \frac{\varepsilon X}{Y \log X}.$$

Summing over the $2k+1$ cosets outside S , we have

$$(8.15) \quad \sum_{bH \not\subset S} \pi(X; bH) \ll \frac{\varepsilon X}{\log X}.$$

On the other hand, for any coset bH , we apply Lemma 3.1 to bH . The contribution to $\pi(X; bH)$ from prime ideals with $N\mathfrak{p} \leq X^{1/3}$ is $o_K(X/\log X)$ by the prime ideal theorem. And the number of remaining prime ideals is bounded by $\mathcal{N}_{1/3}(X; b, H)$, hence Lemma 3.1 gives

$$(8.16) \quad \pi(X; bH) \leq \left(\frac{2}{Y\vartheta_0} + o_K(1) \right) \frac{X}{\log X}.$$

Moreover, the prime ideal theorem gives

$$\sum_{bH \in G/H} \pi(X; bH) = (1 + o_K(1)) \frac{X}{\log X}.$$

Subtracting the contributions of the k cosets $a_iH \neq a_{j_0}H$ contained in S using (8.16), together with the contribution from the cosets outside S using (8.15), we obtain

$$\pi(X; a_{j_0}H) \geq \left(1 - \frac{2k}{Y\vartheta_0} - O(\varepsilon) - o_K(1) \right) \frac{X}{\log X}.$$

Since $X \geq Q^{\max(1, 3\alpha, 4\alpha_0) + \kappa}$, by (6.2), $\vartheta_0 \geq \frac{3}{4} - O(\varepsilon^{1/2})$. Therefore,

$$1 - \frac{2k}{Y\vartheta_0} \geq 1 - \frac{8k}{3(3k+2)} - O(\varepsilon^{1/2}).$$

Taking $c_k = 1 - \frac{8k}{3(3k+2)}$, then we have

$$(8.17) \quad \pi(X; a_{j_0}H) \geq (c_k - O(\varepsilon^{1/2}) - o_K(1)) \frac{X}{\log X},$$

where

$$c_0 = 1, \quad c_1 = \frac{7}{15}, \quad c_2 = \frac{1}{3}.$$

Let

$$z := X^{2/3}, \quad \mathcal{P}_2 := \{\mathfrak{p} : z < N\mathfrak{p} \leq X, [\mathfrak{p}] \in a_{j_0}H\}.$$

Since the number of prime ideals of norm at most z is $o_K(X/\log X)$, it follows from (8.17) that

$$(8.18) \quad |\mathcal{P}_2| \geq (c_k - O(\varepsilon^{1/2}) - o_K(1)) \frac{X}{\log X}.$$

By (8.14), (8.18), and $\log z = \frac{2}{3} \log X$, Lemma 8.1 applies to $\mathcal{P}_1$ and $\mathcal{P}_2$ with $C_k = \frac{3}{c_k} + O(\varepsilon^{1/2}) + o_K(1)$.

For $u \in G$, define

$$r(u) := \#\{(\mathfrak{p}_1, \mathfrak{p}_2) \in \mathcal{P}_1 \times \mathcal{P}_2 : [\mathfrak{p}_1 \mathfrak{p}_2] = u\}.$$

Since $a_0 a_{j_0} H \not\subset A' \cdot B'$, if $r(u) > 0$, then

$$u \in E_2(X; \mathfrak{q}) \cap a_0 a_{j_0} H \subseteq E_2(X; \mathfrak{q}) \setminus (A' \cdot B').$$

Therefore, by Cauchy–Schwarz,

$$(8.19) \quad |\mathcal{P}_1|^2 |\mathcal{P}_2|^2 = \left(\sum_{u \in G} r(u) \right)^2 \leq |E_2(X; \mathfrak{q}) \cap a_0 a_{j_0} H| \sum_{u \in G} r(u)^2.$$

The second factor $\sum_{u \in G} r(u)^2$ is the multiplicative energy

$$\mathcal{E}(\mathcal{P}_1, \mathcal{P}_2) := \#\{(\mathfrak{p}_1, \mathfrak{p}'_1, \mathfrak{p}_2, \mathfrak{p}'_2) \in \mathcal{P}_1^2 \times \mathcal{P}_2^2 : [\mathfrak{p}_1 \mathfrak{p}_2] = [\mathfrak{p}'_1 \mathfrak{p}'_2]\}.$$

And by Lemma 8.1,

$$(8.20) \quad \mathcal{E}(\mathcal{P}_1, \mathcal{P}_2) \leq \frac{C_k + O(\varepsilon^{1/2}) + o_K(1)}{|G|} |\mathcal{P}_1|^2 |\mathcal{P}_2|^2.$$

Combining (8.19) and (8.20), we have

$$|E_2(X; \mathfrak{q}) \setminus (A' \cdot B')| \geq |E_2(X; \mathfrak{q}) \cap a_0 a_{j_0} H| \geq \left(\frac{1}{C_k} - O(\varepsilon^{1/2}) - o_K(1) \right) |G|.$$

Taking $\gamma_k := \frac{c_k}{3}$, we have

$$|E_2(X; \mathfrak{q}) \setminus (A' \cdot B')| \geq (\gamma_k - O(\varepsilon^{1/2}) - o_K(1)) |G|,$$

where

$$\gamma_0 = \frac{1}{3}, \quad \gamma_1 = \frac{7}{45}, \quad \gamma_2 = \frac{1}{9}.$$

□

Proof of Theorem 1.2. Let $\kappa > 0$ be fixed, and let $\varepsilon > 0$ be chosen sufficiently small. For Q sufficiently large in terms of ε , we want to show that

- (i) If $X \geq Q^{\max(1, 3\alpha, 4\alpha_0) + \kappa}$, then $|E_2(X; \mathfrak{q})| \geq (\frac{2}{3} - \varepsilon) |G|$.
- (ii) If $X \geq Q^{\max(1, 4\alpha, 4\alpha_0) + \kappa}$, then $|E_2(X; \mathfrak{q})| \geq (\frac{11}{16} - \varepsilon) |G|$.

It suffices to take $X = Q^{\max(1, 3\alpha, 4\alpha_0) + \kappa}$ in (i) and $X = Q^{\max(1, 4\alpha, 4\alpha_0) + \kappa}$ in (ii). Let $B := \frac{\log X}{\log Q}$. We apply Proposition 4.4.

If Proposition 4.4(a) holds, then

$$|E_2(X; \mathfrak{q})| \geq (\vartheta - 3\varepsilon) |G|.$$

In case (i), the assumption $B \geq 3\alpha + \kappa$ implies that

$$\vartheta \geq \frac{2}{3} + \frac{\kappa}{9\alpha + 3\kappa} - \varepsilon.$$

Hence, if $\varepsilon > 0$ is sufficiently small that $3\varepsilon < \frac{\kappa}{9\alpha+3\kappa}$, then

$$|E_2(X; \mathfrak{q})| \geq \left(\frac{2}{3} - \varepsilon\right)|G|.$$

Similarly, in case (ii), the assumption $B \geq 4\alpha + \kappa$ gives that for sufficiently small ε ,

$$|E_2(X; \mathfrak{q})| \geq \left(\frac{3}{4} - \varepsilon\right)|G|.$$

This is stronger than the desired $(11/16 - \varepsilon)|G|$ bound.

We may therefore assume that Proposition 4.4(b) holds. Let $A', B' \subseteq G$ be the sets in Proposition 4.4(b). By the same argument as in the proof of Theorem 1.1, using Proposition 4.4(b.v), the $k+1$ cosets of H met by A' and B' are the same. Thus, for $S = \bigcup_{i=1}^{k+1} a_i H$, we have $A', B' \subseteq S$, $A'B' = S^2$. And Proposition 8.2 applies.

First we prove (i). We apply Lemma 4.5 with

$$\alpha_* = \frac{1}{3} - \varepsilon^{1/2}, \quad \alpha'_* = \frac{3}{8} - \varepsilon^{1/2}, \quad \beta_* = \frac{2}{3} - 5\varepsilon^{1/2}.$$

These parameters satisfy the conditions of Lemma 4.5 for $\varepsilon > 0$ sufficiently small. The lower bounds for $|A'|$, $|B'|$, and the upper bound for indices of cosets follow from Proposition 4.4(b.i), (b.v).

If Lemma 4.5(a) holds, then

$$|A' \cdot B'| \geq \left(\frac{2}{3} - 5\varepsilon^{1/2}\right)|G|.$$

By Proposition 4.4(b.ii),

$$|E_2(X; \mathfrak{q})| \geq |(A' \cdot B') \cap E_2(X; \mathfrak{q})| \geq |A' \cdot B'| - \varepsilon|G| \geq \left(\frac{2}{3} - 6\varepsilon^{1/2}\right)|G|.$$

We may therefore assume that Lemma 4.5(b) holds. Then, for some $k \in \{0, 1, 2\}$, the set $A' \cdot B'$ is the union of $2k+1$ cosets of a subgroup H of index $3k+2$. By Proposition 4.4(b.ii) again,

$$|(A' \cdot B') \cap E_2(X; \mathfrak{q})| \geq |A' \cdot B'| - \varepsilon|G| \geq \left(\frac{2k+1}{3k+2} - \varepsilon\right)|G|.$$

By Proposition 8.2,

$$|E_2(X; \mathfrak{q}) \setminus (A' \cdot B')| \geq (\gamma_k - O(\varepsilon^{1/2}) - o_K(1))|G|.$$

Hence,

$$|E_2(X; \mathfrak{q})| \geq \left(\frac{2k+1}{3k+2} + \gamma_k - O(\varepsilon^{1/2}) - o_K(1)\right)|G|.$$

Since $\frac{2k+1}{3k+2} + \gamma_k > \frac{2}{3}$, for $k = 0, 1, 2$, renaming ε if necessary, the desired bound follows for $\varepsilon > 0$ sufficiently small and Q sufficiently large.

For (ii), we repeat the same argument by applying Lemma 4.5 with

$$\alpha_* = \alpha'_* = \frac{3}{8} - \varepsilon^{1/2}, \quad \beta_* = \frac{11}{16} - 5\varepsilon^{1/2}.$$

The assumptions $B \geq 4\alpha + \kappa$ and $B \geq 4\alpha_0 + \kappa$ ensure that Proposition 4.4(b.i), (b.v) give the required condition for the bounds of size and coset indices in Lemma 4.5. If Lemma 4.5(a) holds, then

$$|E_2(X; \mathfrak{q})| \geq \left(\frac{11}{16} - 6\varepsilon^{1/2}\right)|G|.$$

Otherwise, Lemma 4.5(b) holds, and the same argument above gives

$$|E_2(X; \mathfrak{q})| \geq \left(\frac{2k+1}{3k+2} + \gamma_k - O(\varepsilon^{1/2}) - o_K(1)\right)|G|.$$

Since $\frac{2k+1}{3k+2} + \gamma_k > \frac{11}{16}$, for $k = 0, 1, 2$, renaming ε if necessary completes the proof. $\square$

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